

Preparation and Guideline for Lab 3

Week 6

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This PowerPoint is an optional guide designed to support your understanding of the official Lab 3 instructions. It includes step-by-step guidance for selected tasks.

Please follow the official Lab instructions and assessment requirements on Canvas when completing and submitting your work.

What Are We Doing in This Lab?

We will use Wireshark to capture and analyse real network traffic generated by a ping. You will examine Ethernet frames, ARP messages and ICMP packets, identify key fields (MAC and IP addresses), and explain how the communication happens.

Today's Overall Process

1  **Check PC network information**
Use ipconfig to view your PC's IP address, MAC address and network settings.



2  **Find the default gateway**
Identify the IP address of your default gateway (router).



3  **Understand Ethernet frame fields**
Review the Ethernet II frame structure (preamble, destination MAC, source MAC, type, data, FCS).



4  **Understand ARP**
Learn how ARP is used to find the MAC address of a device on the local network.



5  **Open Wireshark**
Start Wireshark and select the correct network interface.



6  **Capture traffic**
Begin capturing network packets.



7  **Generate traffic using ping**
Use the ping command (e.g. ping 192.168.1.1 or ping www.cisco.com).



8  **Filter ARP / ICMP packets**
Use Wireshark display filters (e.g. arp or icmp, or just icmp) to view relevant packets.



9  **Inspect MAC and IP addresses**
Examine the captured packets and identify source/destination MAC and IP addresses.



10  **Explain what happened**
Interpret the packet-capture evidence and explain the communication process.

**From capture to understanding —
see what really happens on the network!**

Preparation and Guideline for Lab 3

Network Setup (Typical Setup)



PC
(Your Computer)

IP: 192.168.1.17
MAC: 00-1A-73-EA-63-8C

Default Gateway
(Router)

IP: 192.168.1.1
MAC: 80:37:73-EA-B1-7A

Key Protocols We Will Analyse



Ethernet II

- Frame structure and header fields
- Source and destination MAC addresses
- Frame type (0x0800 for IPv4, 0x0806 for ARP)



Address Resolution Protocol (ARP)

- Used to find the MAC address of a device
- ARP request (broadcast)
- ARP reply (unicast)



Internet Control Message Protocol (ICMP)

- Used by the ping command
- Echo (ping) request
- Echo (ping) reply

What You Will achieve

- ✓ Capture real network traffic using Wireshark
- ✓ Identify Ethernet frame fields and MAC addressing
- ✓ Analyse ARP request and reply messages
- ✓ Examine ICMP packets generated by ping
- ✓ Explain how encapsulation works across layers
- ✓ Interpret packet-capture evidence to justify network behaviour

Why Lab 3 Matters

Most network communication happens in the background.

Wireshark allows us to see what is really happening.

In this lab you will:

- See real network packets
- Understand how devices find each other
- See how ARP finds a MAC address
- See how ping uses ICMP
- Understand MAC addresses and IP addresses
- Learn how to investigate network problems
- Use packet evidence instead of guessing

Main Idea

Wireshark helps us see, understand and explain what is happening on a network.

Skills You Will Develop

During this lab, you will learn how to:

- Open and use Wireshark
- Select the correct network interface
- Start and stop packet captures
- Apply simple filters such as arp and icmp
- Identify Ethernet II frames
- Find source and destination MAC addresses
- Find source and destination IP addresses
- Identify ARP requests and replies

- Identify ICMP ping requests and replies
- Read packet details
- Explain what captured packets show

Practical Process

Capture → Filter → Inspect → Explain

Learning Outcomes for Lab 3

By the end of this lab, you should be able to:

- Capture network traffic using Wireshark
- Identify Ethernet II frame fields
- Identify source and destination MAC addresses
- Explain how ARP works
- Identify ARP requests and replies
- Examine ICMP ping packets

- Identify source and destination IP addresses
- Explain how Ethernet, IP and ICMP work together
- Compare local and remote communication
- Use packet evidence to explain network behaviour

In Simple Terms

You should be able to look at a packet and explain:

Who sent it? → Where is it going? → What protocol is being used? → Why was it sent?

We will complete Tasks 1 and 2 together.

These tasks teach you the main Wireshark skills needed for the lab. Once you can capture, filter, inspect and explain packets, you will continue with the remaining activities using the same process.

Three practical tasks.

Task	Main Focus	Suggested Approach
Task 1	Understand Ethernet II and ARP	We do them together
Task 2	Capture and analyse a local ping	
Task 3	Analyse remote traffic and complete reflection	Students continue more independently

Preparation and Guideline for Lab 3



What You Will Build Today

Today you will create a packet-capture investigation using Wireshark.

*Follow the steps
and discover how
it all works!*

1 Check your PC configuration

Open Command Prompt and type:

```
C:\> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    IPv4 Address. . . . . : 192.168.1.17
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . : 192.168.1.1
```

This shows your IP address and default gateway.

2 Find your default gateway

Look at the output of ipconfig.

```
Default Gateway . . . . . : 192.168.1.1
```

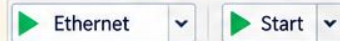
Note the default gateway address. You will use this for ping and in your analysis.

3 Open Wireshark

Start Wireshark and select your network interface (e.g. Ethernet or Wi-Fi), then click the shark fin (Start) button.



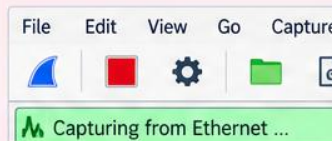
Wireshark



Choose the correct interface and start the capture.

4 Capture network traffic

Wireshark will now capture all traffic on the selected interface.



You will see packets appearing in the top pane.

5 Generate traffic using ping

Go back to Command Prompt and ping the default gateway (first) and a remote destination (later).

```
C:\> ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms

C:\> ping www.cisco.com
```

This generates ARP and ICMP traffic for you to capture.

6 Filter ARP and ICMP packets

Use the display filter to show only relevant packets.

No.	Time	Source	Destination	Protocol
1	0.000000	192.168.1.17	192.168.1.1	ARP
2	0.000512	192.168.1.1	192.168.1.17	ARP
3	0.001024	192.168.1.17	192.168.1.1	ICMP
4	0.001536	192.168.1.1	192.168.1.17	ICMP

This makes it easier to analyse the packets you need.

7 Inspect Ethernet, IP and ICMP information

Click on a packet and look at the details in the middle pane.

▼ Ethernet II
Source: 00:1A:73:EA:63:8C
Destination: 80:37:73:EA:B1:7A
▼ Internet Protocol Version 4
Source: 192.168.1.17
Destination: 192.168.1.1
▼ Internet Control Message Protocol
Type: 8 (Echo request)
Code: 0

You can see the Ethernet, IP and ICMP fields.

8 Compare request and reply packets

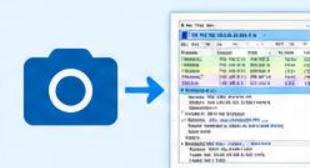
Look at both the request and the reply. Compare the addresses, types and other fields.

Packet	Source	Destination	Info
ARP Req	192.168.1.17	192.168.1.1	Who has 1
ARP Rep	192.168.1.1	192.168.1.17	192.168.1
ICMP Req	192.168.1.17	192.168.1.1	Echo (ping)
ICMP Rep	192.168.1.1	192.168.1.17	Echo (ping)

The request and reply packets should have opposite source and destination addresses.

9 Capture evidence

Take screenshots of key packets (e.g. ARP request/reply and ICMP request/reply).



Save your screenshots to include in your lab report.

10 Explain what happened

Use your observations to explain what you saw in the capture.



- Who sent the packet?
- Where is it going?
- Which protocol is being used?
- What information is inside?
- What is the purpose of each packet?

Write clear and concise explanations in your report.



Final Result

You will be able to show how:

PC → ARP → Ethernet → ICMP → Router
work together during real network communication.



PC
192.168.1.17



ARP
Finds the gateway
MAC address



Ethernet
Carries the frame
on the local network



ICMP
Carries the ping
message



Router / Default Gateway
192.168.1.1

These protocols work together to enable real network communication!

Capture
Analyse
Understand!

TASK 1 – Understand Ethernet Frames and ARP

What are we trying to learn?

Before using Wireshark, we first need to understand:

- What an Ethernet frame contains
- What a MAC address is used for
- Why ARP is needed
- Why ARP happens before some network communication

Check that Wireshark is installed

In Windows, click Start and search:

Wireshark

Open **Wireshark**.

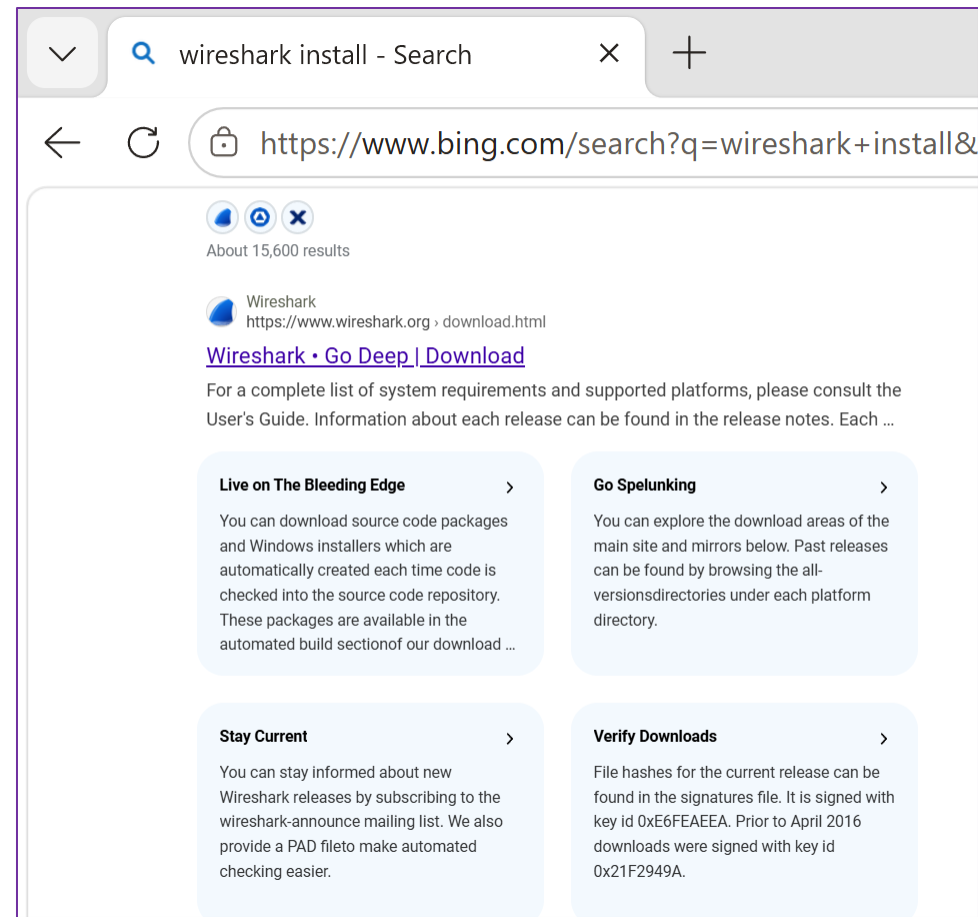
Why?

Lab 3 requires packet-capture evidence. Wireshark allows us to see the actual packets sent and received by the PC.

If Wireshark is not installed

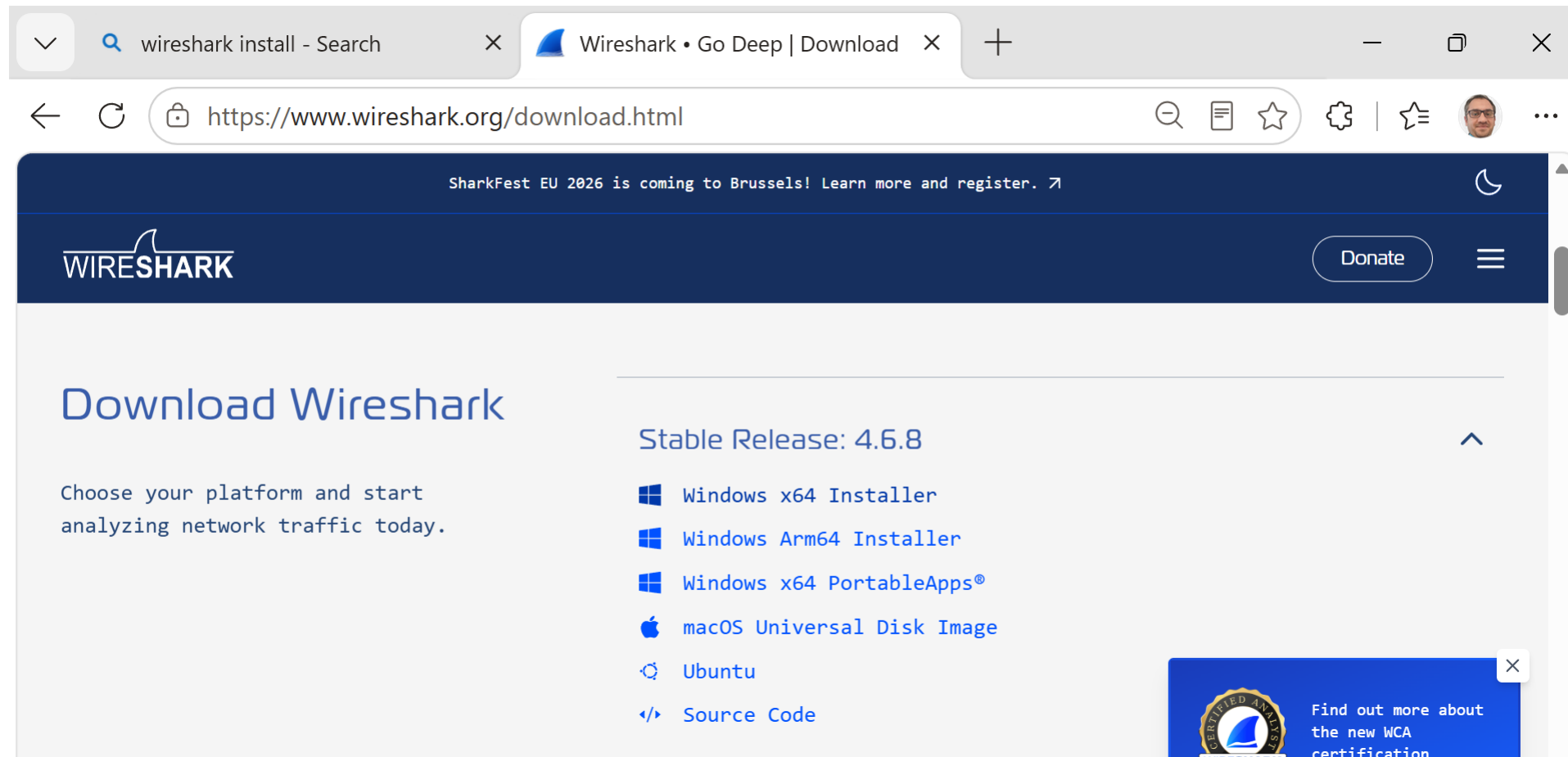
Install the current Wireshark version before the lab. On Windows, the installer normally includes Npcap, which Wireshark needs to capture network traffic.

You already have Packet Tracer, so do not reinstall Packet Tracer for this lab.



Scroll Down

Windows x64



The screenshot shows a web browser window with two tabs. The active tab is titled 'Wireshark • Go Deep | Download'. The address bar shows the URL 'https://www.wireshark.org/download.html'. The page header is dark blue with the 'WIRESHARK' logo on the left, a 'Donate' button, and a menu icon on the right. A banner at the top of the page reads 'SharkFest EU 2026 is coming to Brussels! Learn more and register. ↗'. The main content area has the heading 'Download Wireshark' followed by the text 'Choose your platform and start analyzing network traffic today.' To the right, under the heading 'Stable Release: 4.6.8', is a list of download links with icons: Windows x64 Installer, Windows Arm64 Installer, Windows x64 PortableApps®, macOS Universal Disk Image, Ubuntu, and Source Code. A small blue box in the bottom right corner contains the Wireshark logo and the text 'Find out more about the new WCA certification'.

wireshark install - Search

Wireshark • Go Deep | Download

https://www.wireshark.org/download.html

SharkFest EU 2026 is coming to Brussels! Learn more and register. ↗

WIRESHARK Donate

Download Wireshark

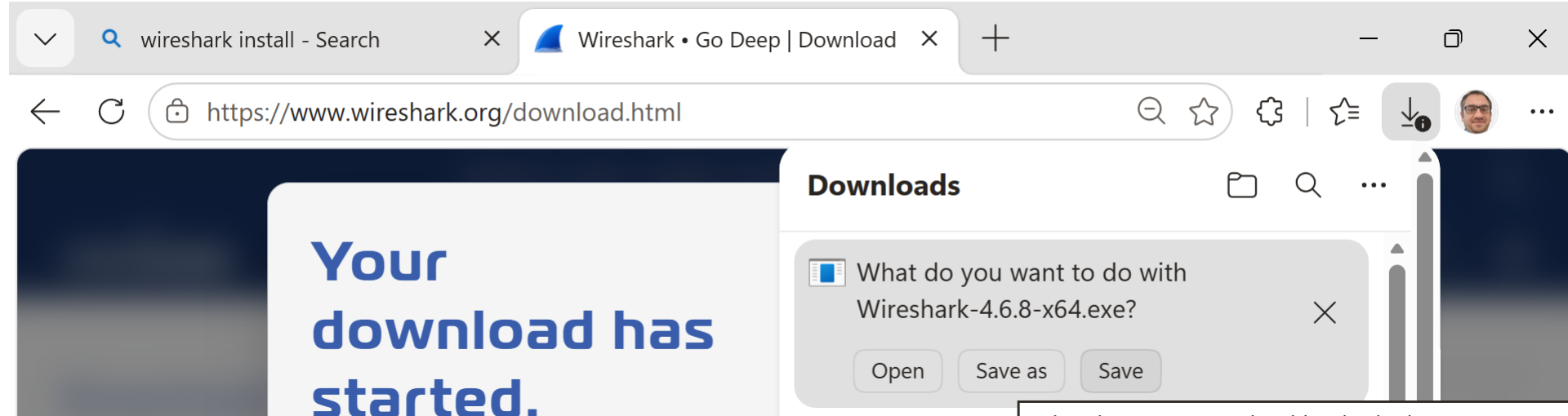
Choose your platform and start analyzing network traffic today.

Stable Release: 4.6.8

- Windows x64 Installer
- Windows Arm64 Installer
- Windows x64 PortableApps®
- macOS Universal Disk Image
- Ubuntu
- Source Code

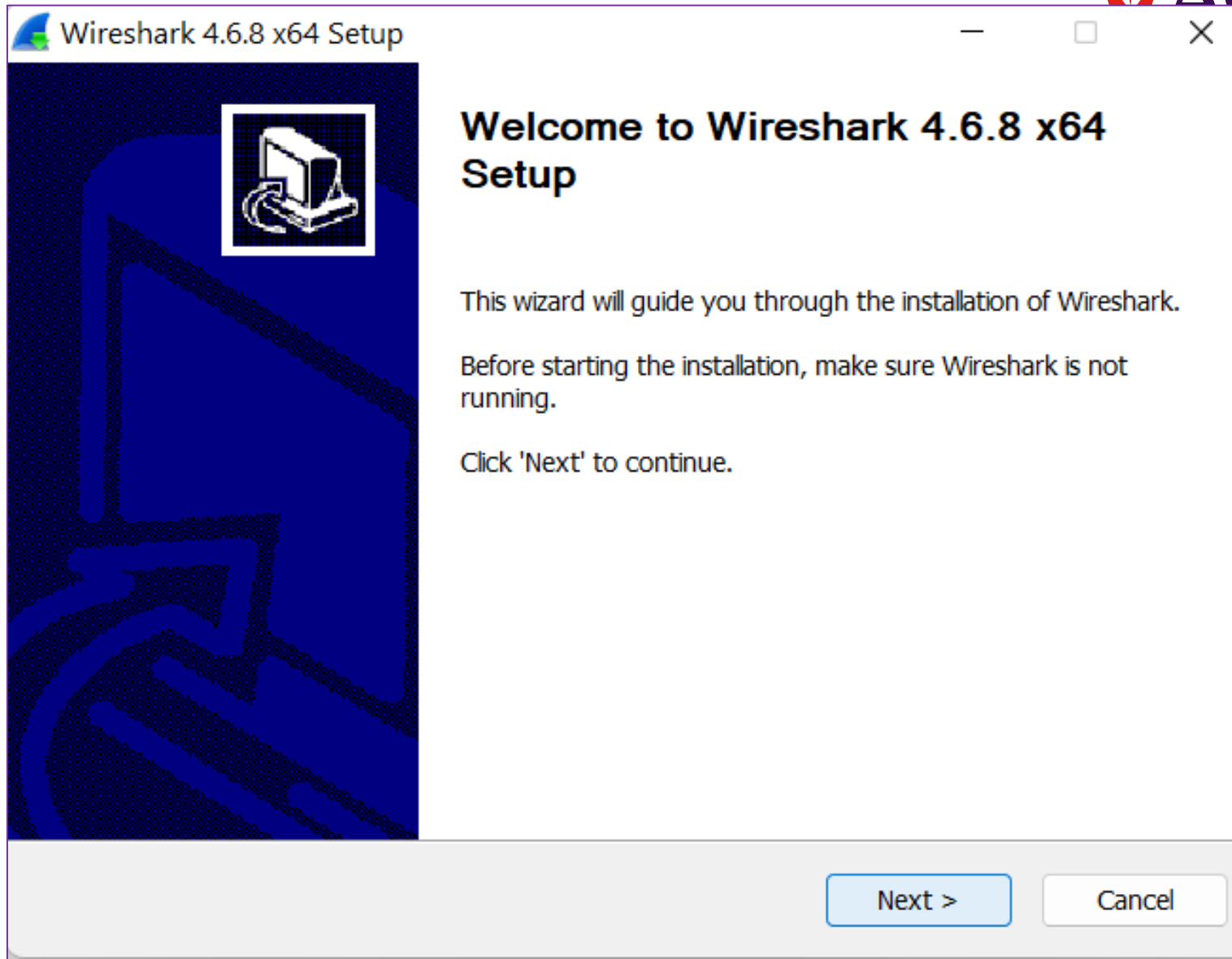
Find out more about the new WCA certification

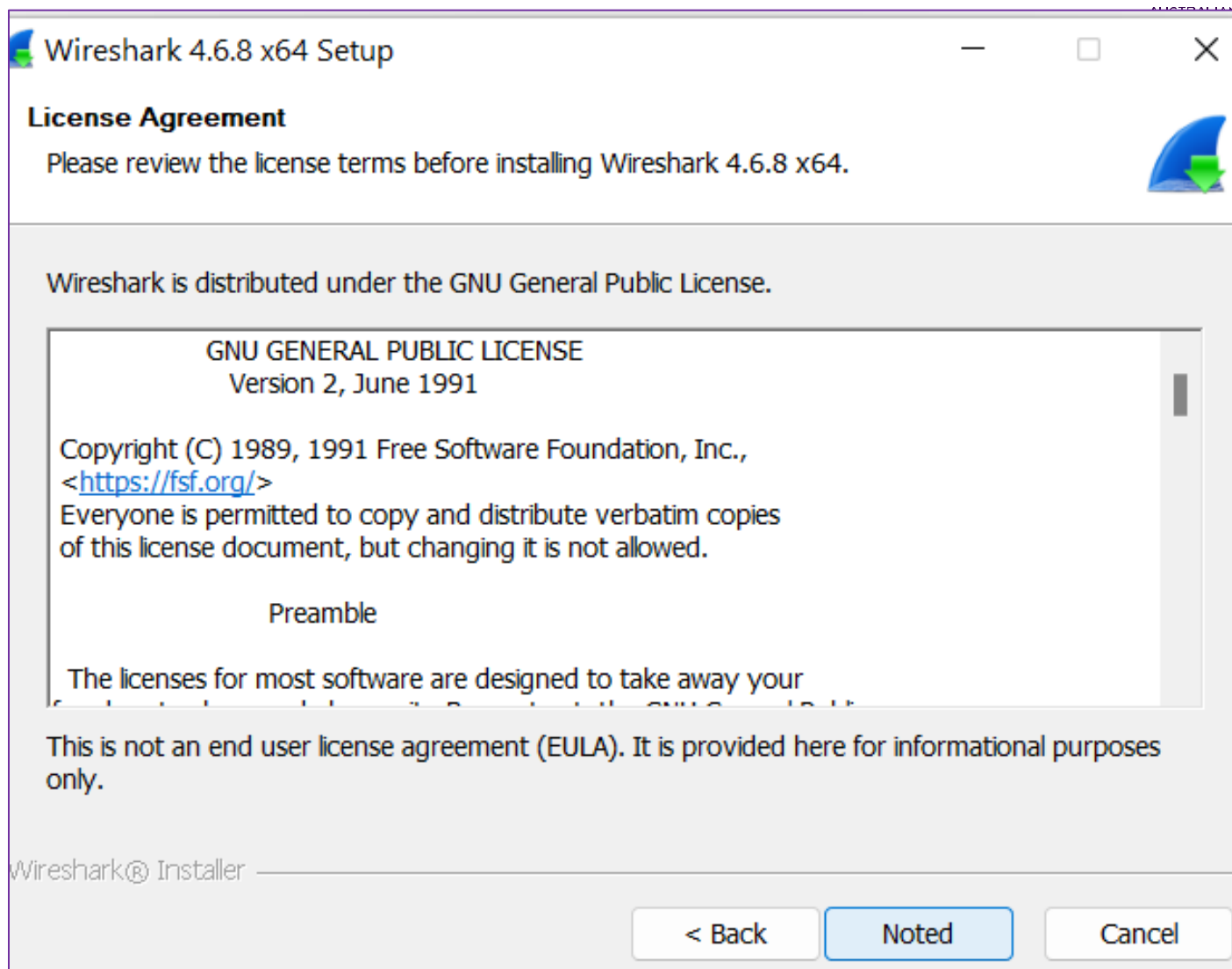
Save



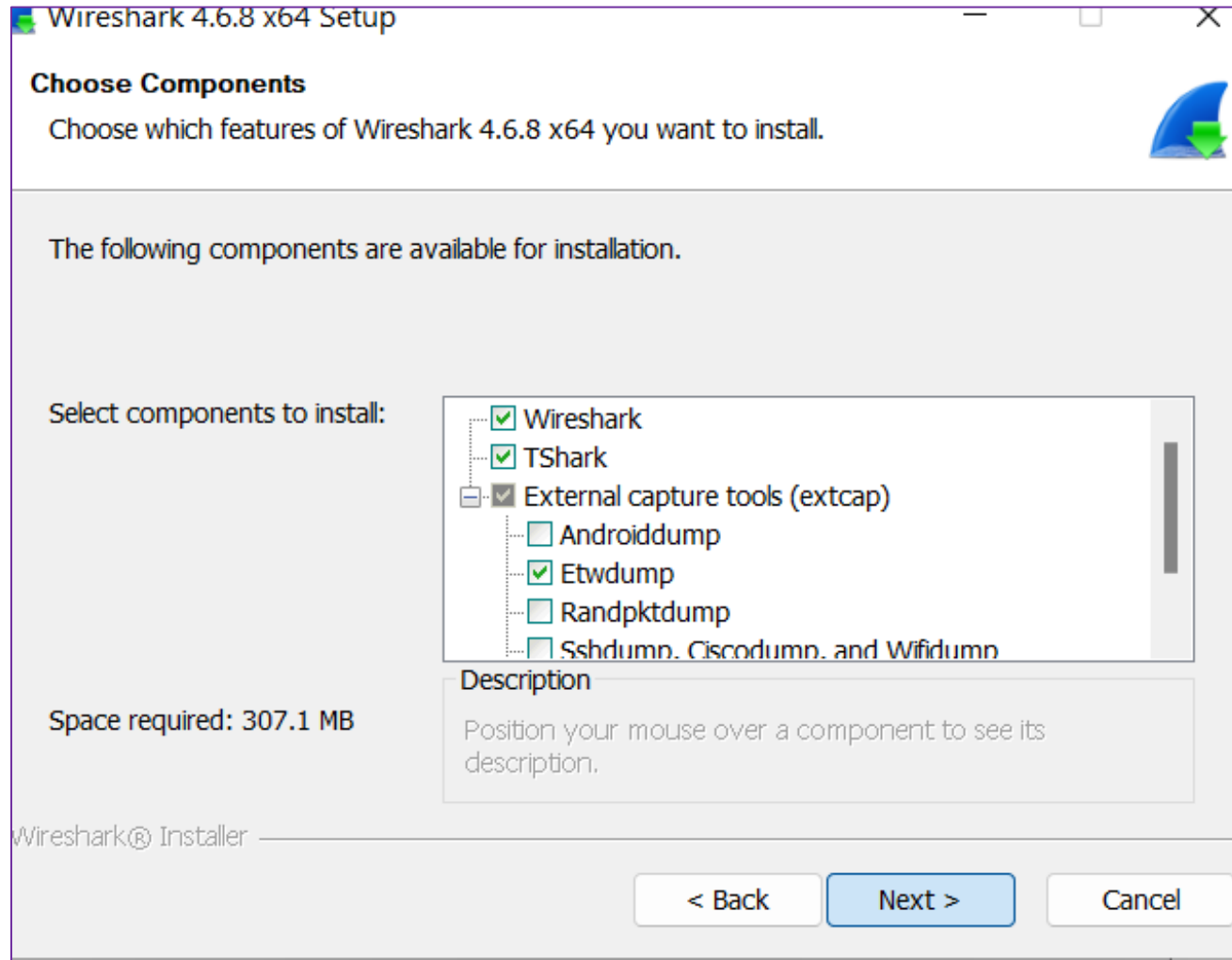
Double Click (Run)

Name	Date modified	Type	Size
 Wireshark-4.6.8-x64	05/09/2026 18:57	Application	96,617 KB





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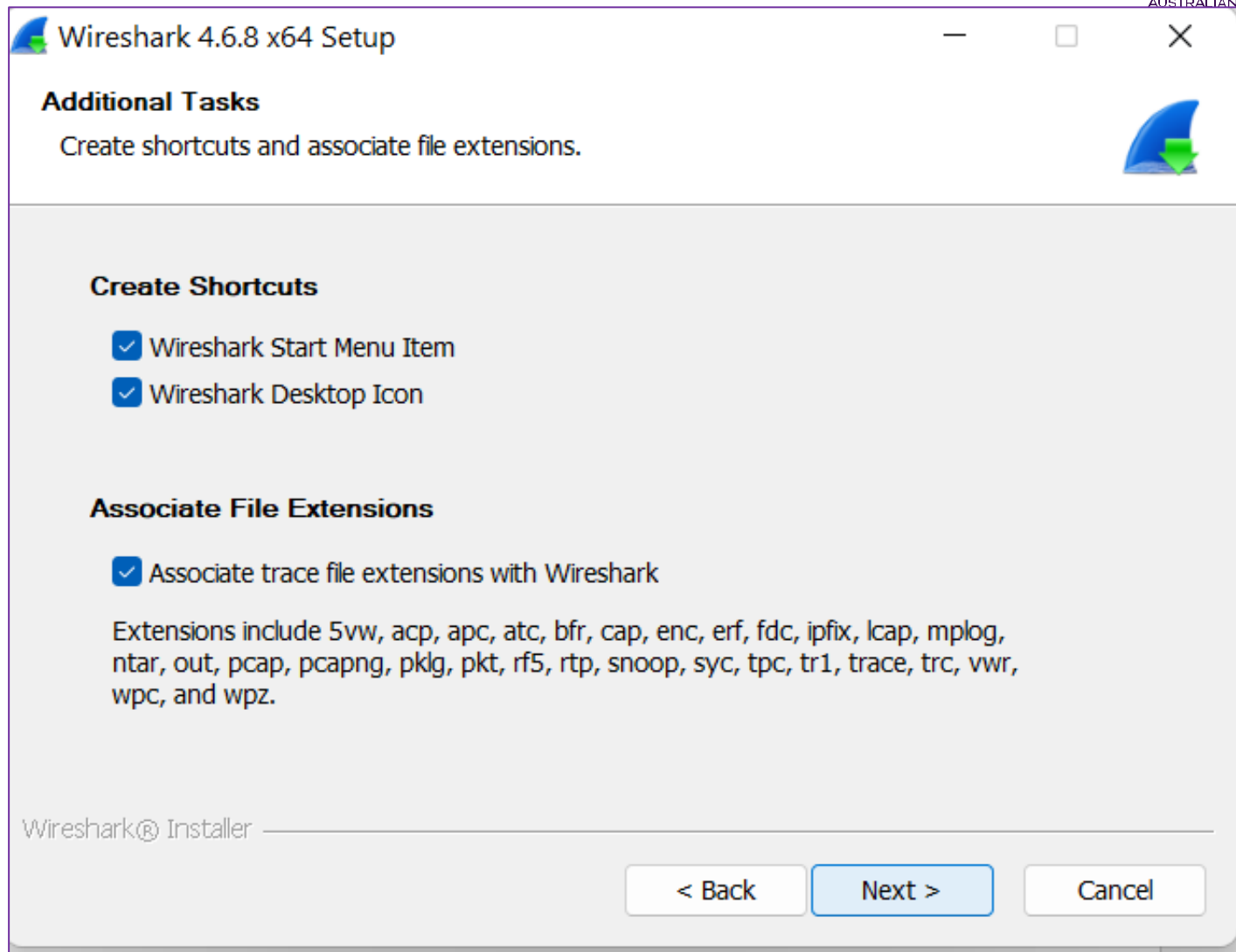


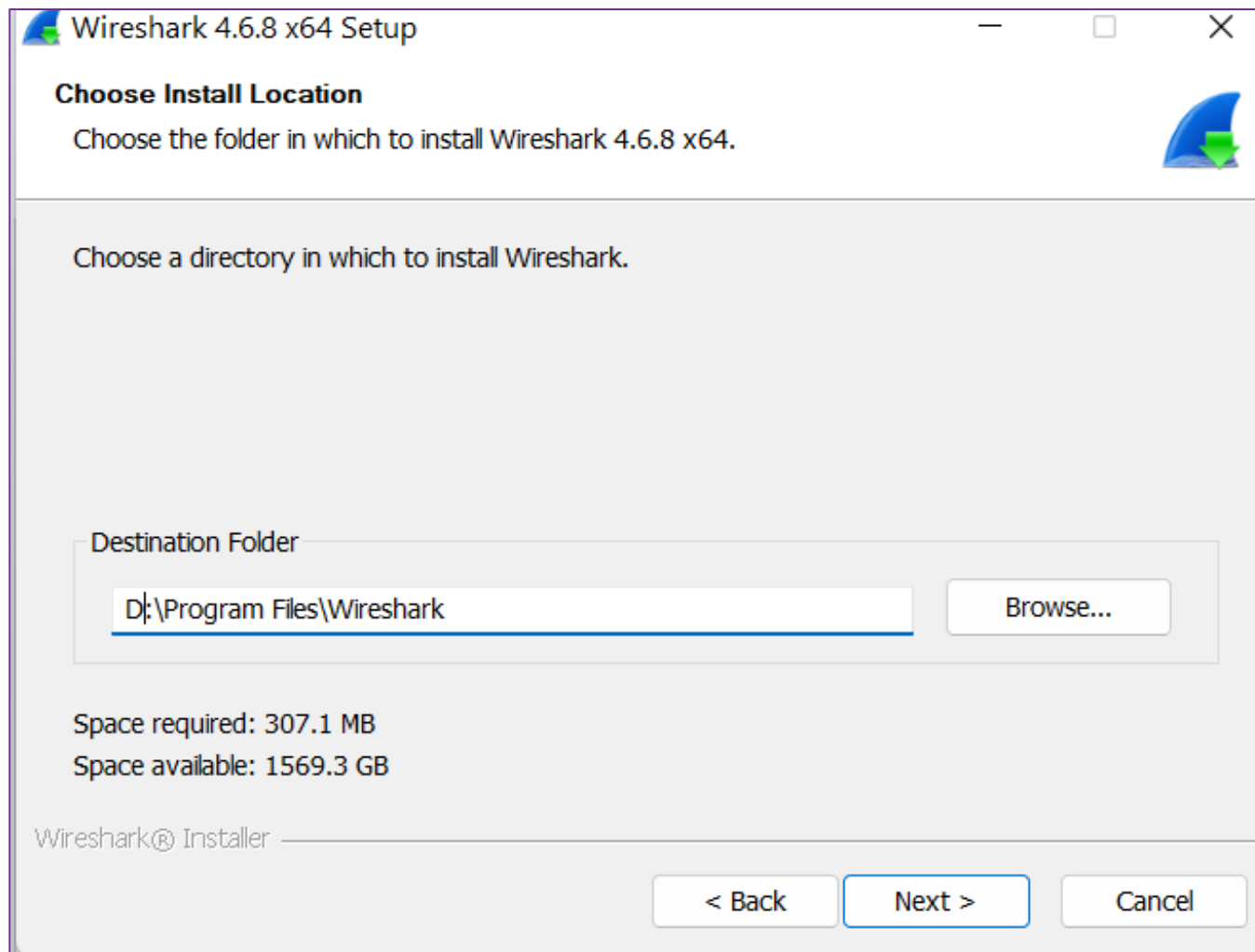
I recommend leaving the default selections as they are and clicking Next >.

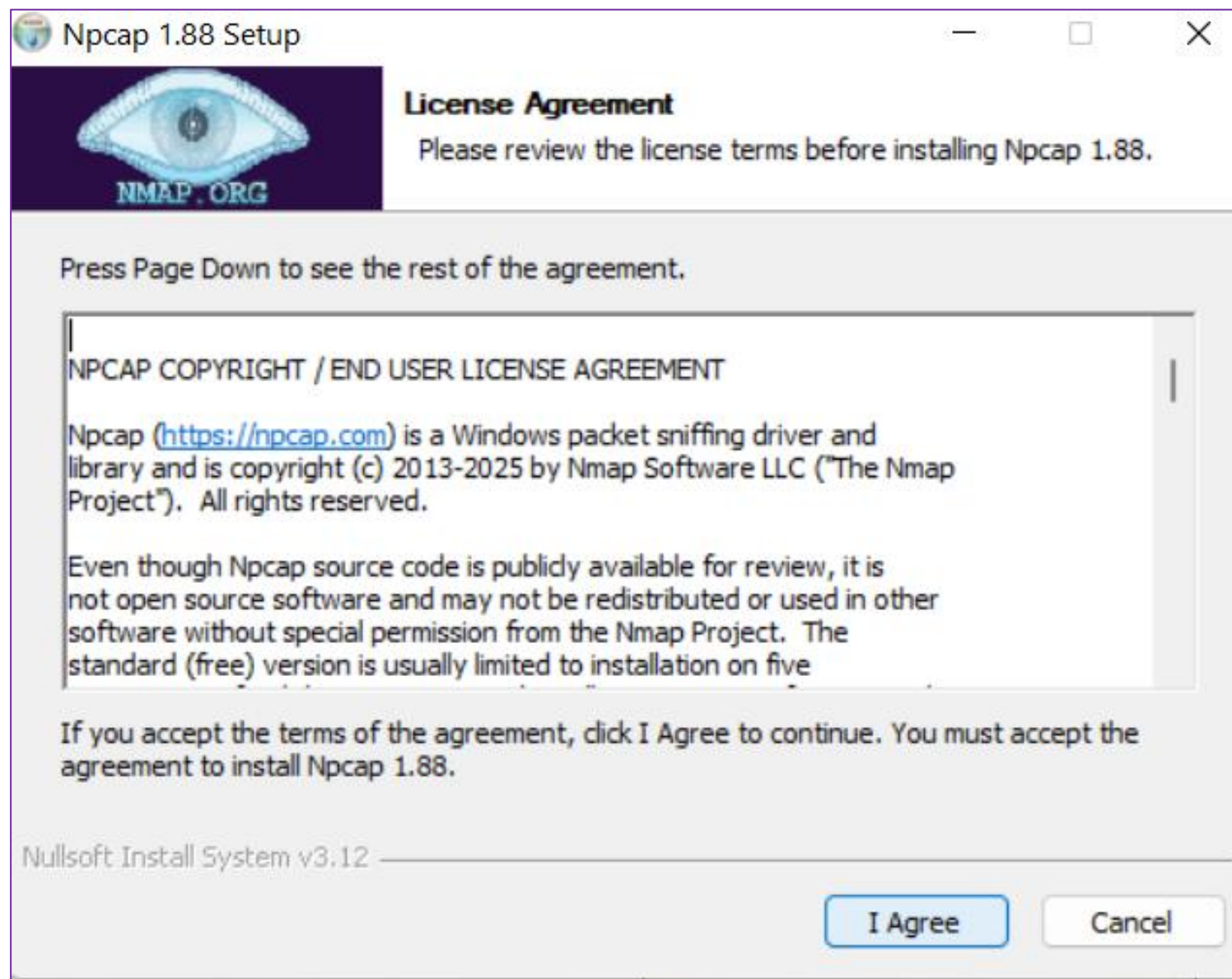
The optional items shown—Androiddump, Etwdump, Randpktdump, SSHdump/Ciscodump/Wifidump, and UDPdump—are mainly for specialised or remote capture situations. Lab 3 does not require them.

What matters more is a later installation step: when Wireshark asks about Npcap, make sure Npcap is installed, because it enables Wireshark to capture traffic from your Windows Wi-Fi/Ethernet interfaces.

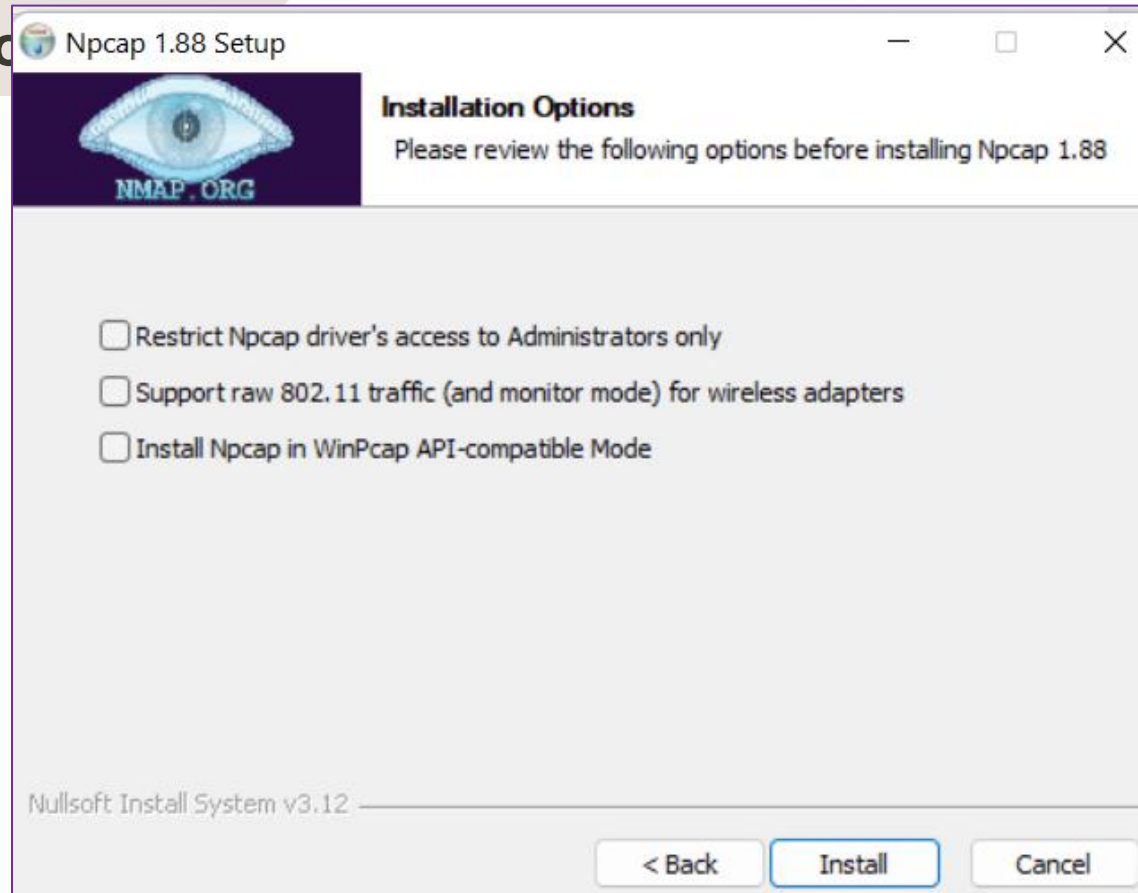
Choose Components: Keep the default settings and click Next. No additional external capture tools are required for this lab. Then, when you reach the Npcap screen, send me a screenshot if you want, and I can tell you exactly which options to select.







Preparation and Guideline for

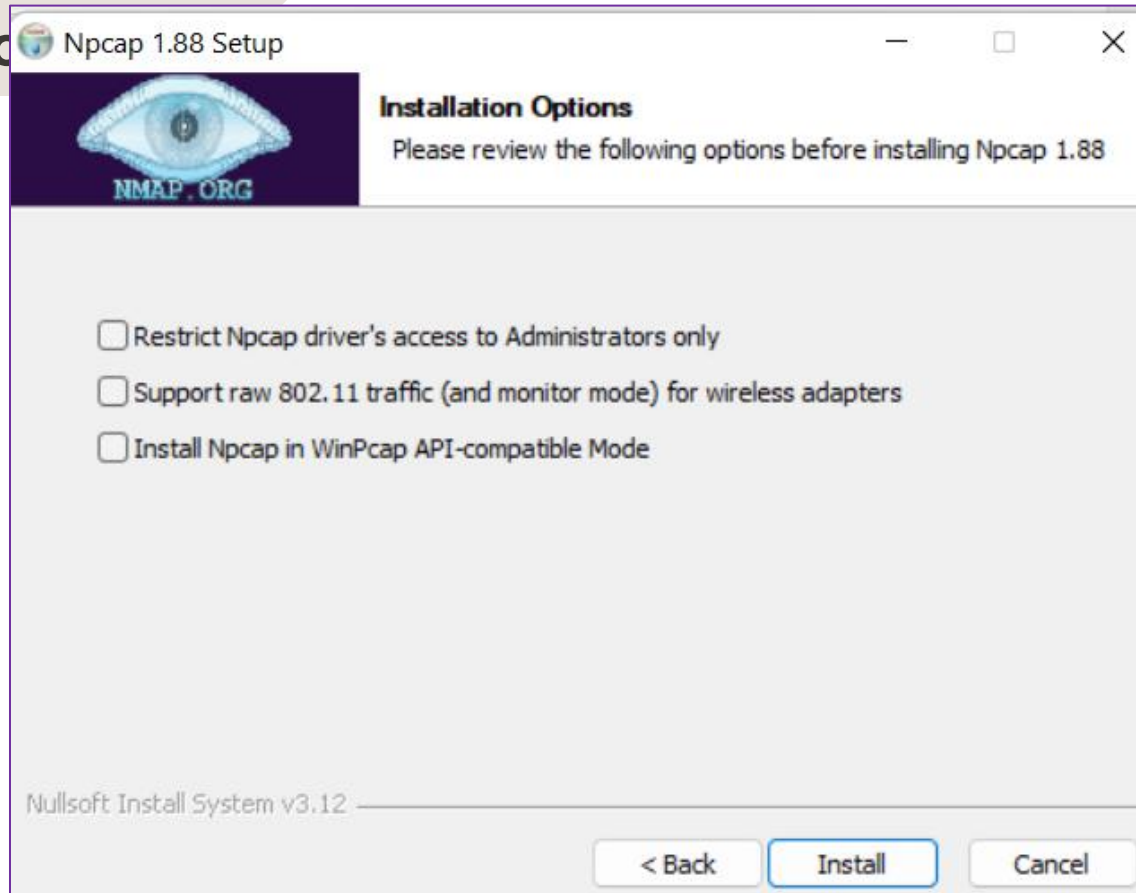


We do not need to tick:

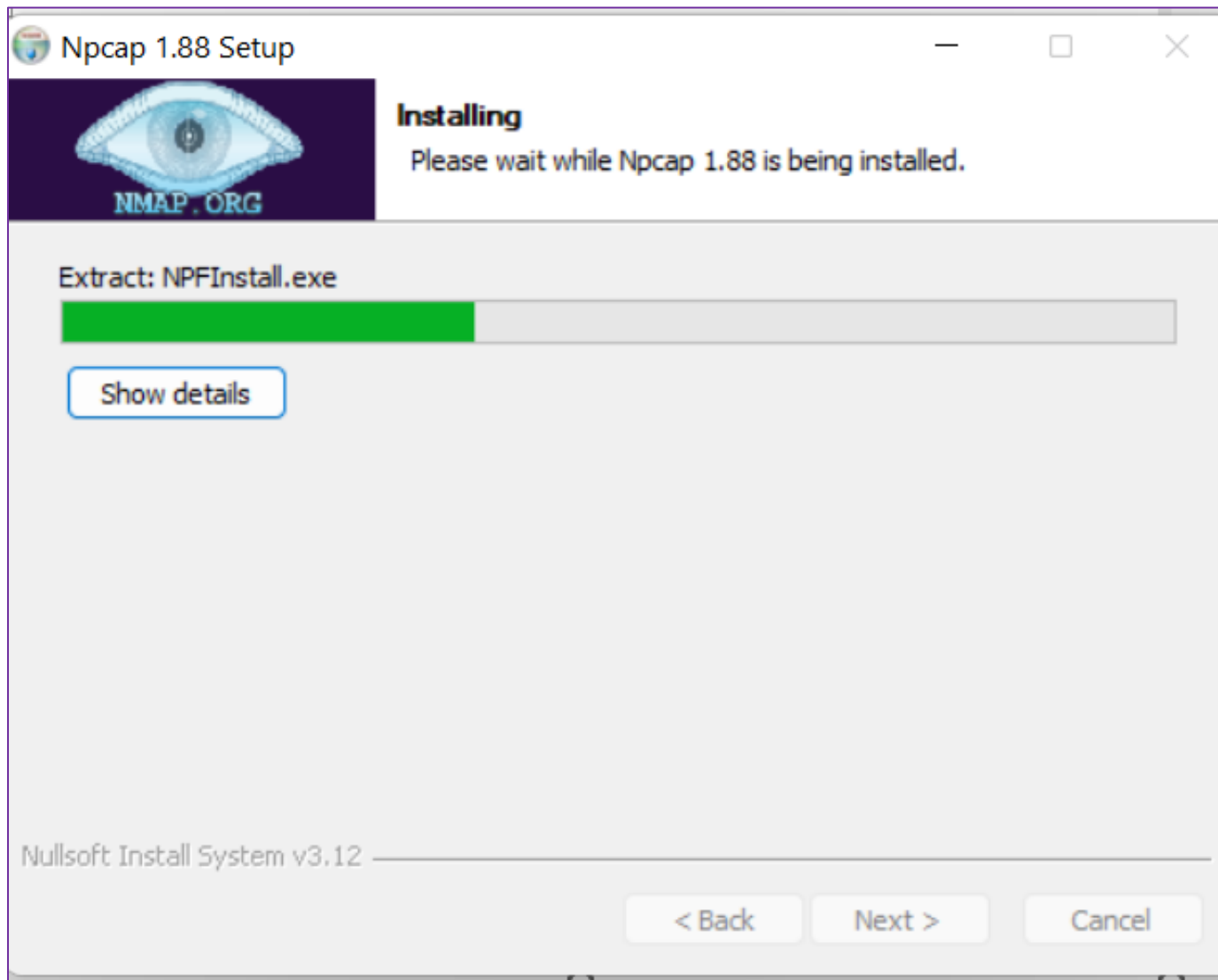
- Restrict Npcap driver's access to Administrators only
- Support raw 802.11 traffic (and monitor mode) for wireless adapters
- Install Npcap in WinPcap API-compatible Mode

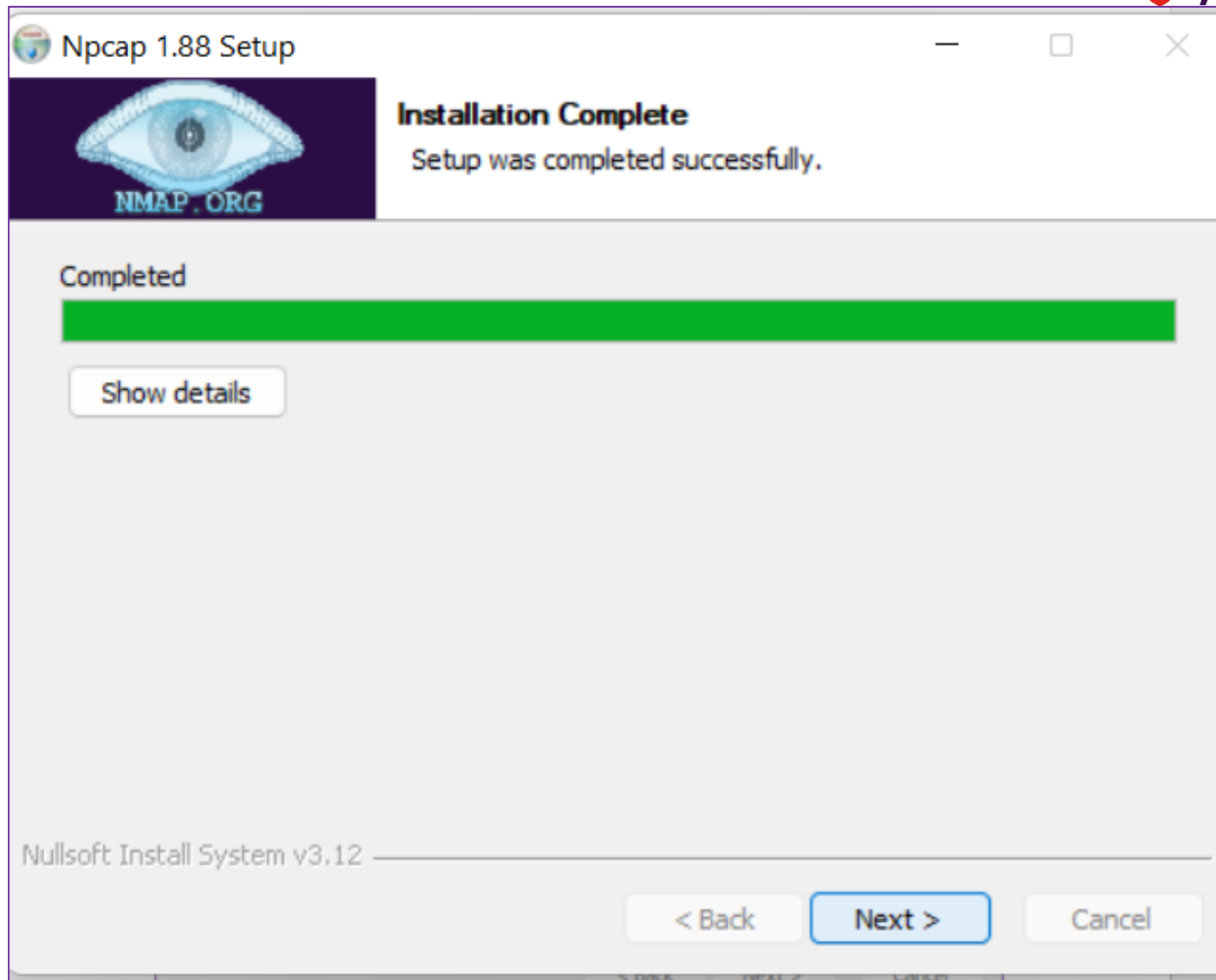
The normal/default Npcap installation is sufficient for capturing your PC's ARP and ICMP traffic in Wireshark.

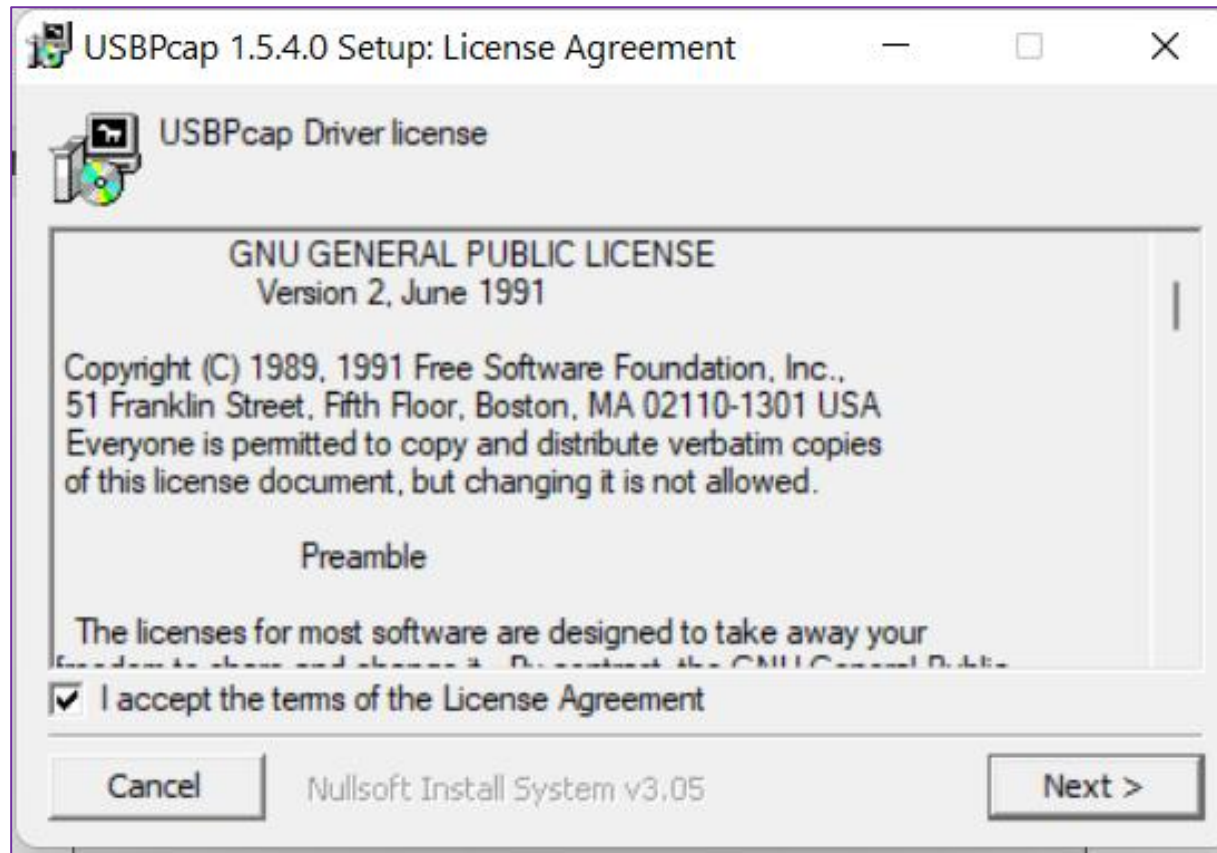
Preparation and Guideline for

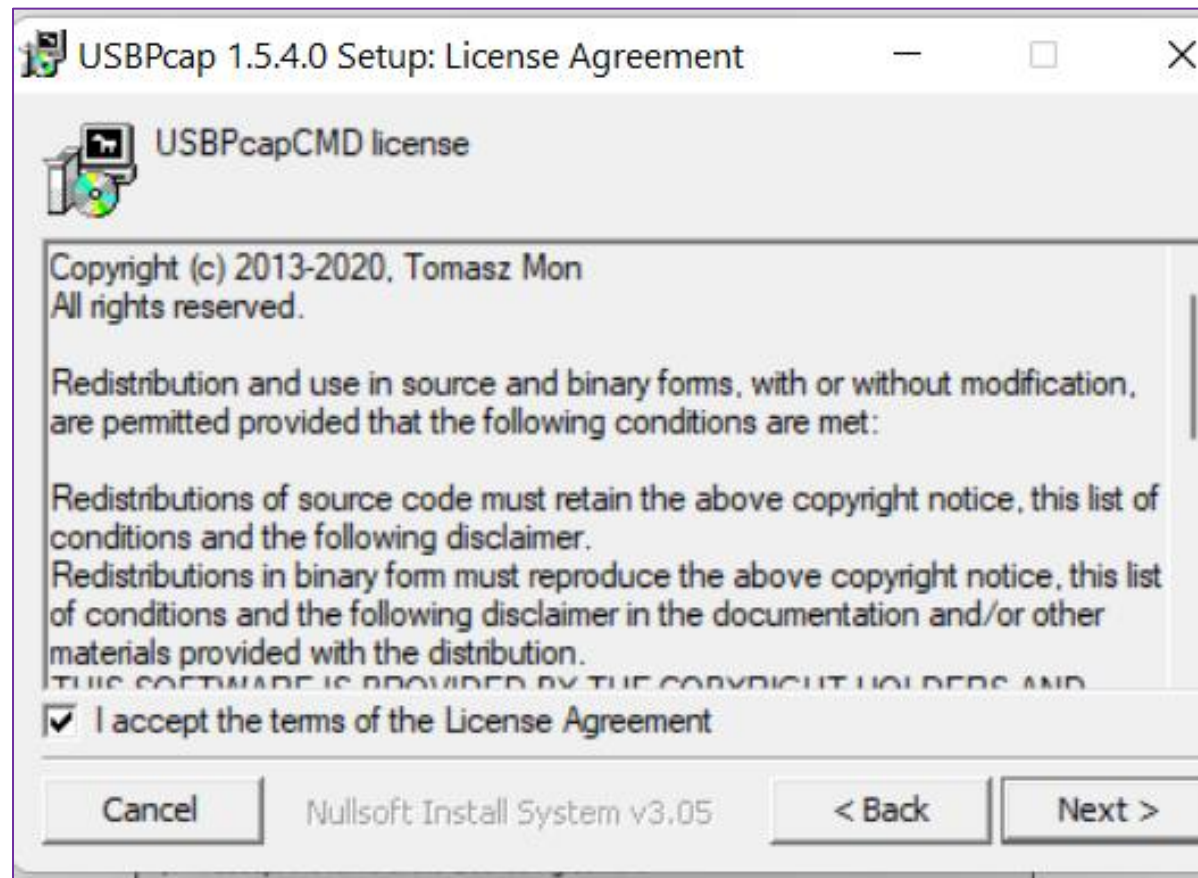


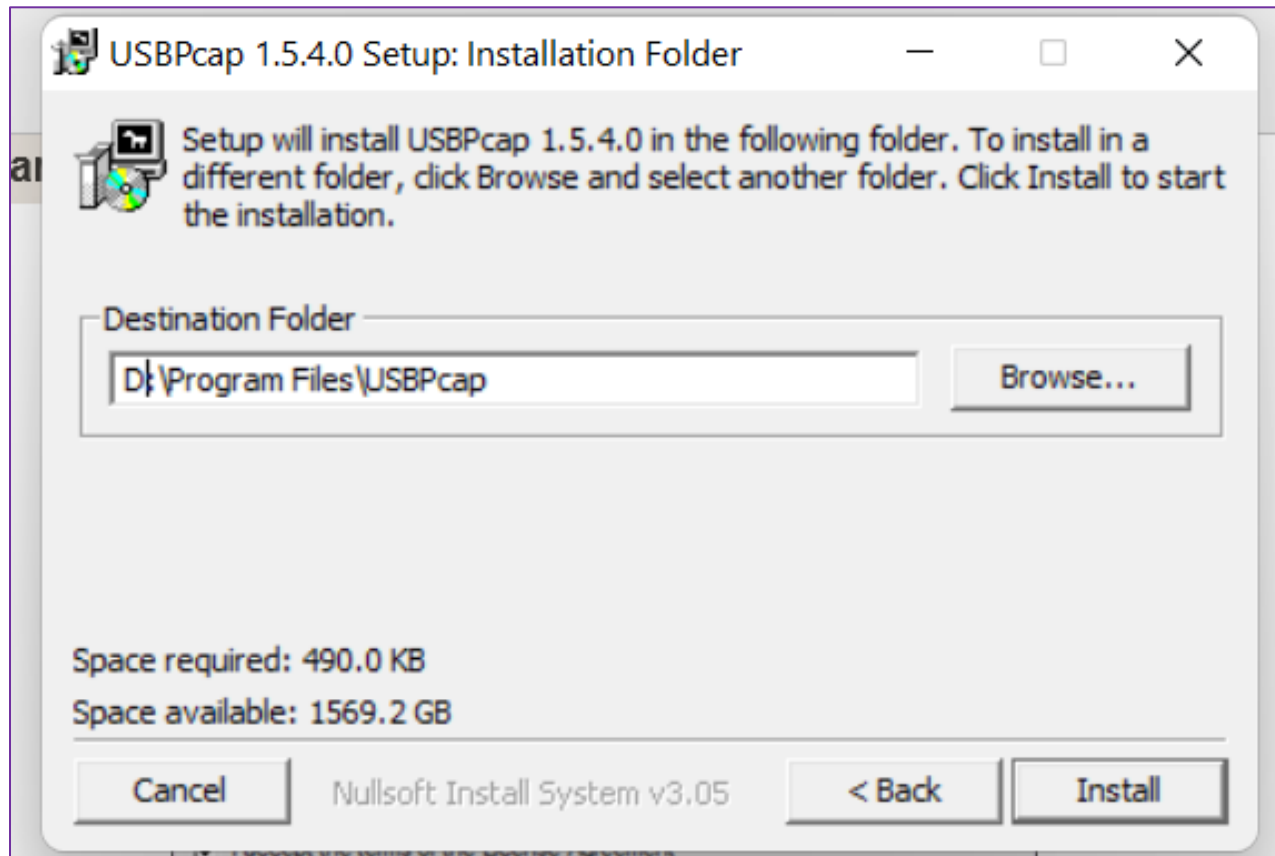
Npcap Installation Options: Keep the default settings (leave all optional boxes unticked), then click Install.



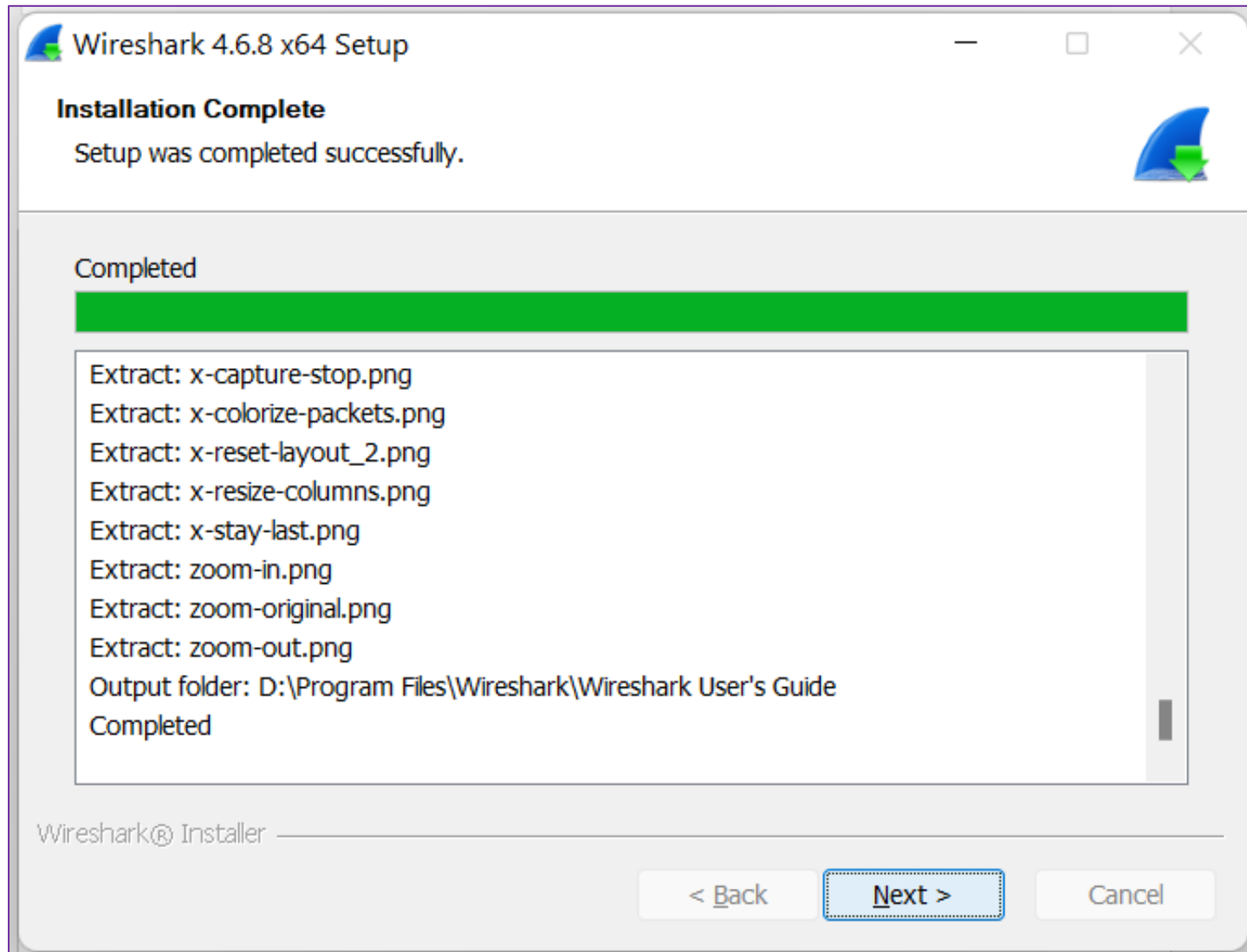




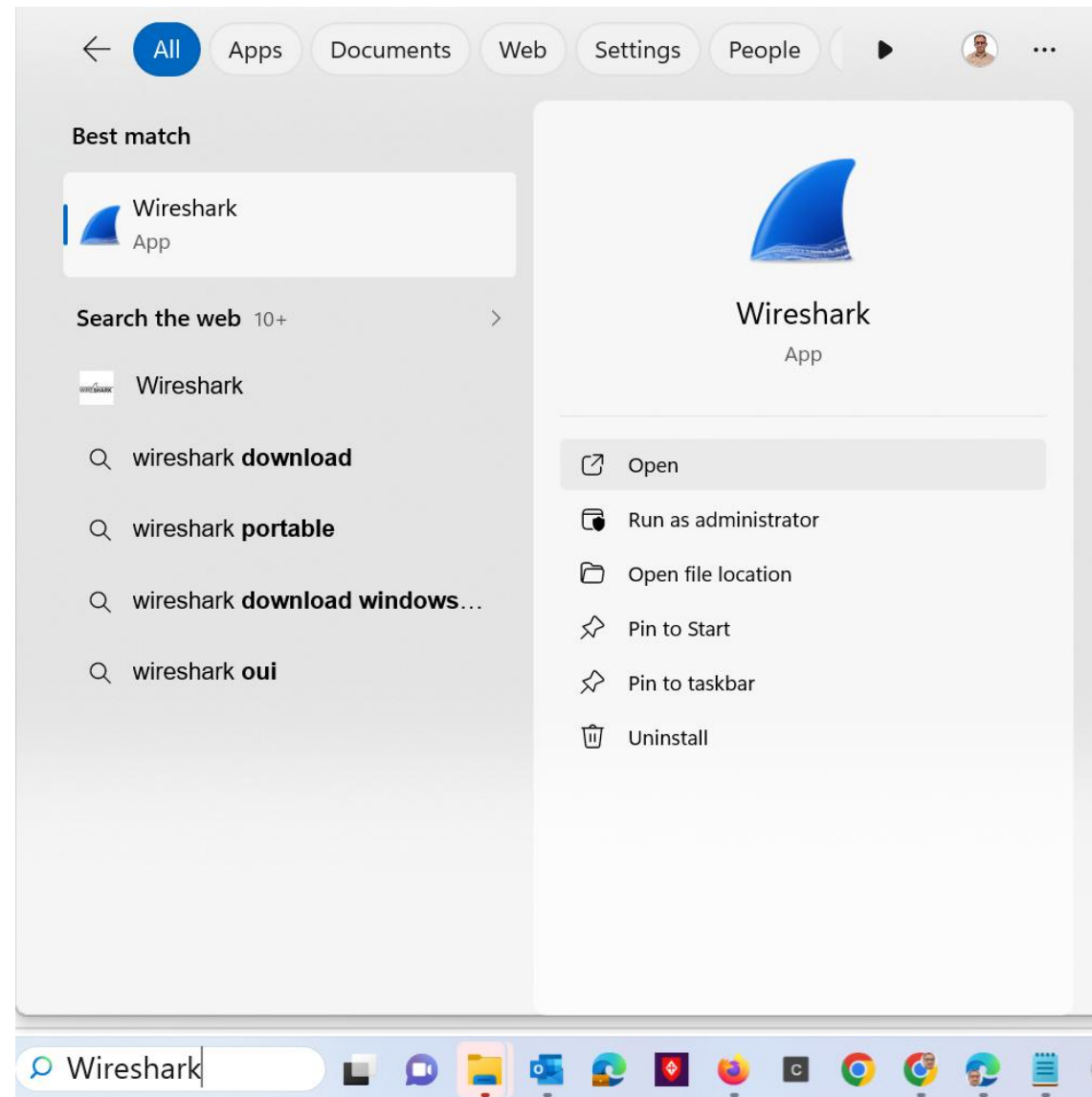




Restart PC

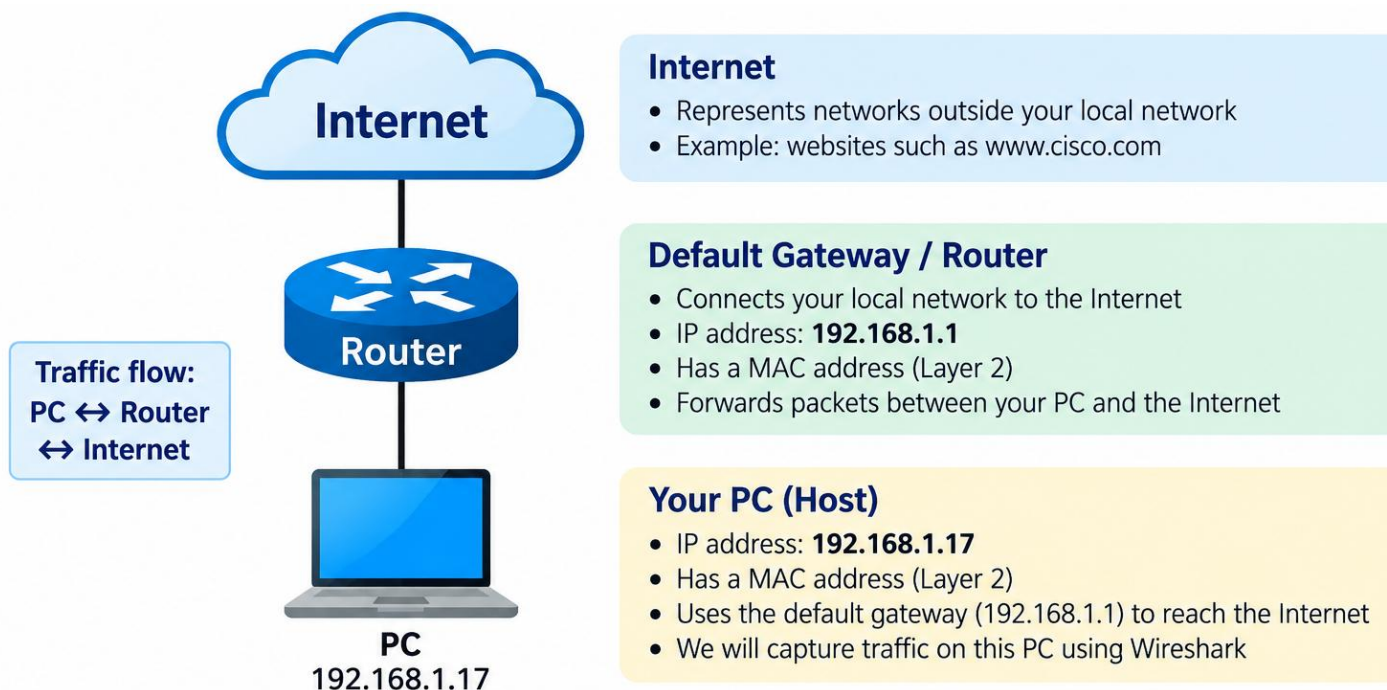


Run Wireshark



Step 1 – Understand the basic network

For this lab, think about this simple network:



In this lab, we will capture and analyse the network traffic on the PC to see how communication happens between your PC, the router and the Internet.

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Instead, use your actual Windows PC.

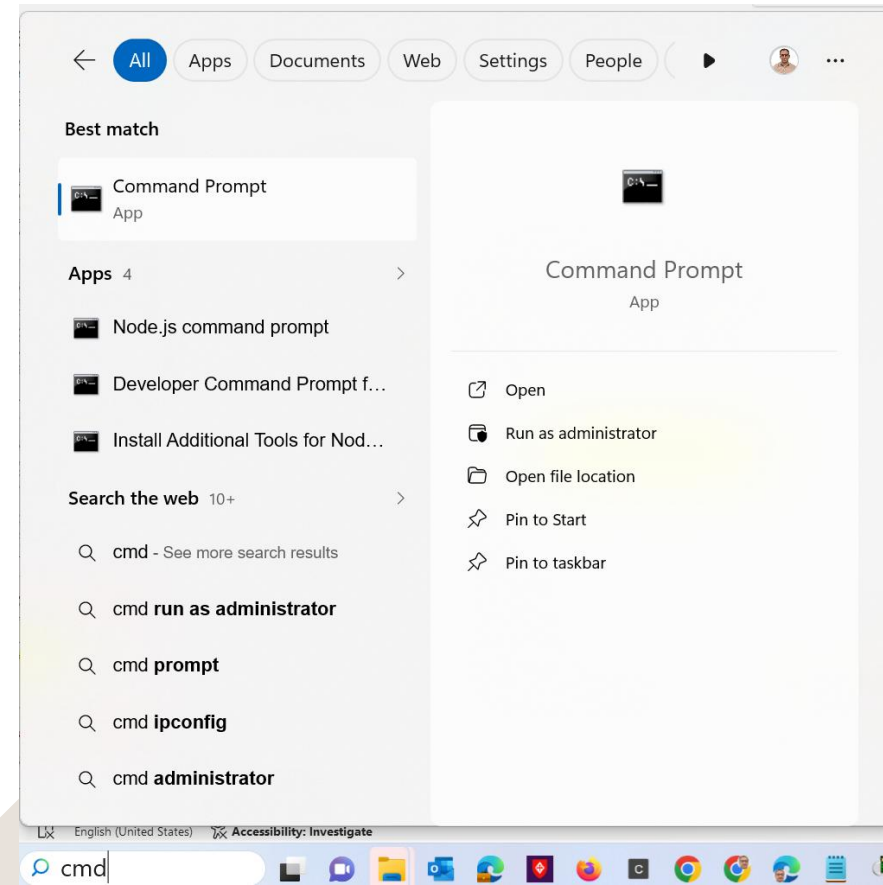
Open: **Start → Command Prompt**

Type: **ipconfig /all**

Press Enter.

Look under the network
connection you are currently
using, normally:

Wireless LAN adapter Wi-Fi





Command Prompt

```
Microsoft Windows [Version 10.0.22000.2538]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\gaming 15>ipconfig /all
```

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```
Command Prompt

Connection-specific DNS Suffix  . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : E0-D4-64-9A-9F-7A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8(Preferred)
Temporary IPv6 Address. . . . . : 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7(Preferred)
Link-local IPv6 Address . . . . . : fe80::5b3b:760e:e528:d3e7%5(Preferred)
IPv4 Address. . . . . : 10.232.47.203(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, 5 September 2026 11:29:51
Lease Expires . . . . . : Saturday, 5 September 2026 20:29:35
Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                             10.232.47.24
DHCP Server . . . . . : 10.232.47.24
DHCPv6 IAID . . . . . : 232838244
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
DNS Servers . . . . . : 10.232.47.24
NetBIOS over Tcpi . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : E0-D4-64-9A-9F-7E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```


macOS — Preparation for Lab 3

Open Terminal:

Command + Space → type Terminal → Enter

Then run:

```
ifconfig
```

Look for the network interface currently being used. For Wi-Fi, this is commonly en0 (although it can differ).

To quickly find the Mac's current Wi-Fi IPv4 address, run:

```
ipconfig getifaddr en0
```

For the MAC (physical) address, run:

```
ifconfig en0 | grep ether
```

For the default gateway/router, run:

```
route -n get default | grep gateway
```

So the Windows-to-Mac equivalents for your slide are:

Information	Windows	macOS
Full network configuration	<code>ipconfig /all</code>	<code>ifconfig</code>
IPv4 address	shown in <code>ipconfig /all</code>	<code>ipconfig getifaddr en0</code>
MAC / Physical Address	Physical Address	<code>ifconfig en0 grep ether</code>
Default Gateway	Default Gateway	<code>route -n get default grep gateway</code>

Mac users: Open Terminal and run `ifconfig`. Locate the active Wi-Fi interface (commonly `en0`) and identify its IP and MAC address. To find the default gateway, run `route -n get default`.

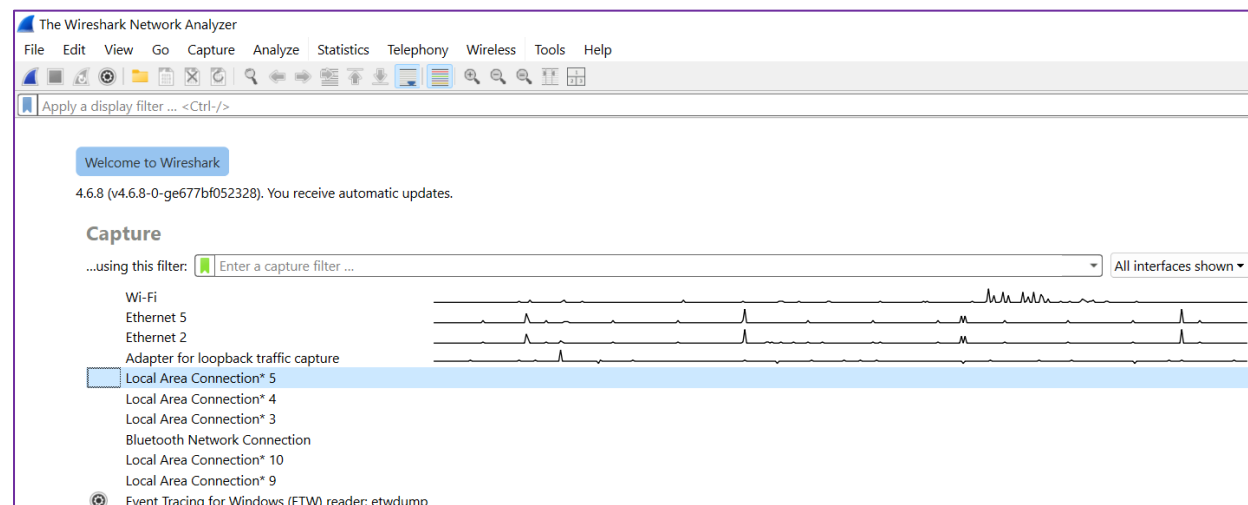
Find Your PC's Network Configuration

Open Windows Command Prompt and type: `ipconfig /all`

Press Enter.

Windows may show many network adapters. Do not automatically use the first adapter listed.

Find the adapter that your PC is currently using to connect to the network/Internet.



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For this computer, the correct adapter is:

Wireless LAN adapter Wi-Fi

We can confirm this because it has an active IPv4 address and a Default Gateway.

From the screenshot, record e.g.:

Adapter: Wi-Fi

IPv4 Address: 10.232.47.203

Subnet Mask: 255.255.255.0

Default Gateway: 10.232.47.24

Physical Address: E0-D4-64-9A-9F-7A

```
Command Prompt
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 5:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-00-27-00-00-11
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::6c22:cef4:65b2:23f1%17(Preferred)
    IPv4 Address. . . . . : 192.168.56.1(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 
    DHCPv6 IAID . . . . . : 805961767
    DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
    NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : E0-D4-64-9A-9F-7B
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
    Physical Address. . . . . : E2-D4-64-9A-9F-7A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
    Physical Address. . . . . : E0-D4-64-9A-9F-7A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8(Preferred)
```

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So, for my actual computer, the basic network is:

My PC

10.232.47.203

E0-D4-64-9A-9F-7A →

Default Gateway

10.232.47.24 →

Other Networks / Internet

```
cmd Command Prompt
Wireless LAN adapter Local Area Connection* 9:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : E0-D4-64-9A-9F-7B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : E2-D4-64-9A-9F-7A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : E0-D4-64-9A-9F-7A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8(Preferred)
Temporary IPv6 Address. . . . . : 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7(Preferred)
Link-local IPv6 Address . . . . . : fe80::5b3b:760e:e528:d3e7%5(Preferred)
IPv4 Address. . . . . : 10.232.47.203(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, 5 September 2026 11:29:51
Lease Expires . . . . . : Saturday, 5 September 2026 20:29:35
Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                          10.232.47.24
DHCP Server . . . . . : 10.232.47.24
DHCPv6 IAID . . . . . : 232838244
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
DNS Servers . . . . . : 10.232.47.24
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Why are we doing this?

Wireshark will soon display many packets and many addresses. Before analysing them, we need to know:

- Which IP address belongs to my PC?
- Which MAC address belongs to my PC?
- What is my default gateway's IP address?

This allows us to recognise our own PC and gateway when they appear in the Wireshark capture.

Important Clarification for Students

Your addresses will probably be different from the lecturer's addresses.

Do not copy:

10.232.47.203

10.232.47.24

These are the addresses on the lecturer's computer/network.

Also, do not copy the original lab examples:

192.168.1.17

192.168.1.1

Instead, record the values displayed by your own ipconfig
/all command.

Be Careful with Virtual Adapters

You may see adapters such as:

Ethernet 2

Ethernet 5

VirtualBox Host-Only Ethernet Adapter

Local Area Connection*

Bluetooth Network Connection

```
Command Prompt
Connection-specific DNS Suffix . : 
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-15
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::6b95:e362:dcdb:78d3%21(Preferred)
Autoconfiguration IPv4 Address. . : 169.254.165.95(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 973733927
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 5:

Connection-specific DNS Suffix . : 
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-11
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::6c22:cef4:65b2:23f1%17(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 805961767
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 9:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : E0-D4-64-9A-9F-7B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : E2-D4-64-9A-9F-7A
DHCP Enabled. . . . . : Yes
```

For example, this screenshot shows 192.168.56.1 under a VirtualBox Host-Only Ethernet Adapter. Do not use that address for this activity.

Use the adapter that provides your actual network connection.

In lecturer's case:

Wireless LAN adapter Wi-Fi →

10.232.47.203

```
Command Prompt
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-15
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::6b95:e362:dadb:78d3%21(Preferred)
Autoconfiguration IPv4 Address. . : 169.254.165.95(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 973733927
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 5:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-11
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::6c22:cef4:65b2:23f1%17(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 805961767
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 9:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : E0-D4-64-9A-9F-7B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : E2-D4-64-9A-9F-7A
DHCP Enabled. . . . . : Yes
```

Preparation and Guideline for Lab 3

Next practical step: in Wireshark, double-click **Wi-Fi** (the first interface) to start capturing traffic. Then we can deliberately generate the ARP request/reply needed for Task 1.

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
5	0.796734900	10.232.47.203	172.188.155.25	TCP	120	[TCP Retransmission] 51052 → 443 [PSH, ACK] Seq=1 Ack=60 Win=513 Len=66
6	0.812120600	172.188.155.25	10.232.47.203	TCP	54	443 → 51052 [ACK] Seq=60 Ack=67 Win=287 Len=0
7	1.129717300	172.188.155.25	10.232.47.203	TCP	66	[TCP Dup ACK 6#1] 443 → 51052 [ACK] Seq=60 Ack=67 Win=287 Len=0 SLE=1 SRE=67
8	1.327005200	10.232.47.203	170.114.2.62	TLSv1.2	82	Application Data
9	1.607429100	170.114.2.62	10.232.47.203	TLSv1.2	78	Application Data
10	1.622552900	2606:4700:4403::ac4...	2001:8004:1f11:e25:...	TLSv1.2	98	Application Data
11	1.622893400	2001:8004:1f11:e25:...	2606:4700:4403::ac4...	TLSv1.2	102	Application Data
12	1.641938700	2606:4700:4403::ac4...	2001:8004:1f11:e25:...	TCP	74	443 → 51919 [ACK] Seq=25 Ack=29 Win=16 Len=0
13	1.651511200	10.232.47.203	170.114.2.62	TCP	54	59290 → 443 [ACK] Seq=29 Ack=25 Win=511 Len=0
14	2.268879800	2001:8004:1f11:e25:...	2606:4700:4408::ac4...	TCP	75	51787 → 443 [ACK] Seq=1 Ack=1 Win=2057 Len=1
15	2.309617500	2606:4700:4408::ac4...	2001:8004:1f11:e25:...	TCP	86	443 → 51787 [ACK] Seq=1 Ack=2 Win=605 Len=0 SLE=1 SRE=2
16	3.076483000	fe80::a484:2eff:fee...	2001:8004:1f11:e25:...	ICMPv6	86	Neighbor Solicitation for 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7 from a6:84:2e:e5:18:52
17	3.076564900	2001:8004:1f11:e25:...	fe80::a484:2eff:fee...	ICMPv6	86	Neighbor Advertisement 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7 (sol, ovr) is at e0:d4:64:9a:9f:7a
18	6.090325300	91.108.56.199	10.232.47.203	SSL	511	Continuation Data
19	6.092979500	10.232.47.203	91.108.56.199	SSL	271	Continuation Data

> Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF...
> Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
> Internet Protocol Version 6, Src: 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7, Dst: 2603:1063:c:11f::365
> Transmission Control Protocol, Src Port: 52021, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 86 dd 60 0b ...R...d...z...
0010 99 76 00 15 06 3f 20 01 80 04 1f 11 0e 25 1c eb ...v...? ...%...
0020 df 60 ef 95 f2 d7 26 03 10 63 00 0c 01 1f 00 00 ...&...c...
0030 00 00 03 65 7e a3 cb 35 01 bb f2 17 c8 69 c0 81 ...e...5 ...i...
0040 4f 4a 50 10 02 01 65 aa 00 00 00 00 00 00 00 00 OJP...e...

These are examples.

Do not change your PC to these addresses.

Use the addresses shown by your own ipconfig command.

Step 2 – Find the Default Gateway's MAC Address Using ARP

We already know from the lecturer's PC:

PC IPv4 Address: 10.232.47.203

PC MAC Address: E0-D4-64-9A-9F-7A

Default Gateway IP: 10.232.47.24

But we do not yet know the gateway's MAC address.

That is exactly the problem ARP solves.

2.1 Start Wireshark Capture

Go back to Wireshark.

You saw several interfaces:

Wi-Fi

Ethernet 5

Ethernet 2

Local Area Connection*

...

Because your active connection is: Wireless LAN adapter Wi-Fi

double-click: **Wi-Fi**

Wireshark should immediately begin capturing packets.

Why?

Wireshark needs to listen on the network interface that your computer is actually using.

Your communication path is approximately:

Your PC

10.232.47.203 → Wi-Fi → Local Network → Default Gateway

10.232.47.24

Step 3 – Display Only ARP Traffic

There will probably be many packets appearing very quickly.

At the top of Wireshark, find:

Apply a display filter...

Type: arp

Press Enter.

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	Time	Source	Destination	Protocol	Length	Info
420	12.542136400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
421	12.542156000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
705	37.380751200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
706	37.380788400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1366	57.087321100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1367	57.087352300	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1908	97.534750700	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1909	97.534767200	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
2082	121.088478600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
2083	121.088500600	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
2315	161.798182900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
2316	161.798212400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
2515	184.325924100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
2516	184.325959300	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
3121	204.295593300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
3122	204.295607000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a

> Frame 420: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
> Address Resolution Protocol (request)

```
0000  e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01  ..d..z.. ...R...
0010  08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18  ....R.../..
0020  00 00 00 00 00 00 0a e8 2f cb  ....//..
```

Why?

Lecturer's PC is generating many different types of network traffic.

We are currently interested only in:

ARP – Address Resolution Protocol

The display filter hides unrelated packets and makes the ARP communication easier to analyse.

Important: the filter does not stop other traffic from being captured. It only changes what Wireshark displays.

After entering: **arp**

Wireshark now displays only ARP packets.

From your screen, we can identify two important messages.

1. ARP Request

Look at a line such as:

Who has 10.232.47.203? Tell 10.232.47.24

In simple terms, the device at: 10.232.47.24

is asking:

"Which device has IP address 10.232.47.203? Please tell me your MAC address."

We already know:

10.232.47.203 = Your PC

10.232.47.24 = Default Gateway

So in this particular packet, the gateway is asking for your PC's MAC address.

This is important: ARP can work in either direction. It is not always the PC asking for the router.

ARP Reply

Immediately after the request, you can see:

10.232.47.203 is at e0:d4:64:9a:9f:7a

This means your PC responds:

"I am 10.232.47.203, and my MAC address is

E0:D4:64:9A:9F:7A."

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	Time	Source	Destination	Protocol	Length	Info
18689	787.981291800	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
18690	787.981335400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
19260	811.783795900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
19261	811.783815300	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
23468	838.539938100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
23469	838.539950100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
24767	864.079795600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
24769	864.079824000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26025	888.073719300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26026	888.073740800	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26289	907.527727300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26290	907.527758000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26695	945.415285100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26696	945.415297000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
27617	973.578465200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24

> Frame 420: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
> Address Resolution Protocol (request)

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 ..d..z...R....
0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R../..
0020 00 00 00 00 00 00 0a e8 2f cb/..

That matches the Physical Address you found earlier using ipconfig /all.

So we have confirmed using two different tools:

Command Prompt

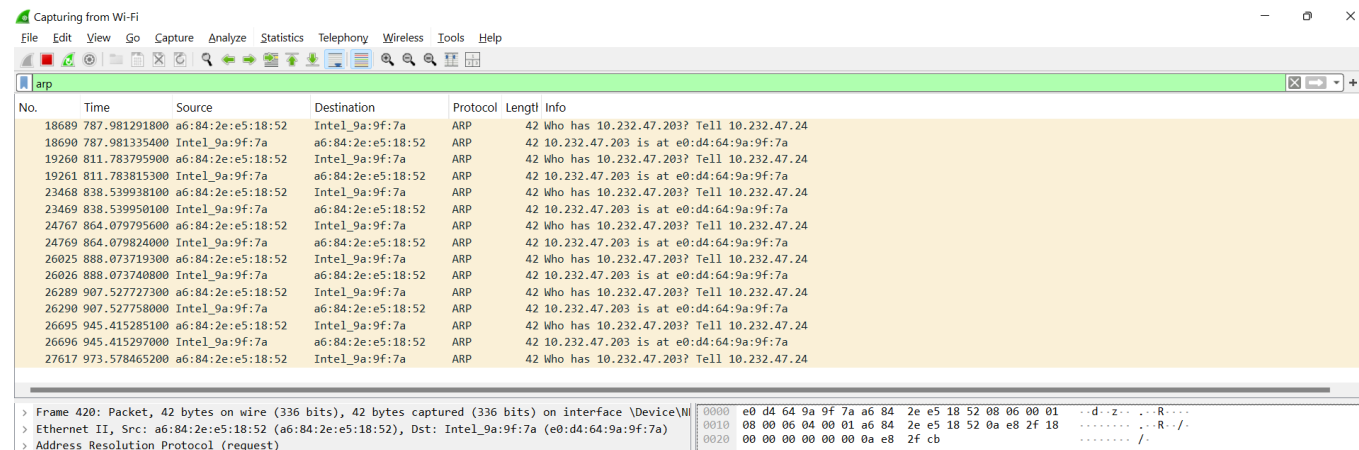
ipconfig /all → PC MAC: E0-D4-64-9A-9F-7A

AND

Wireshark

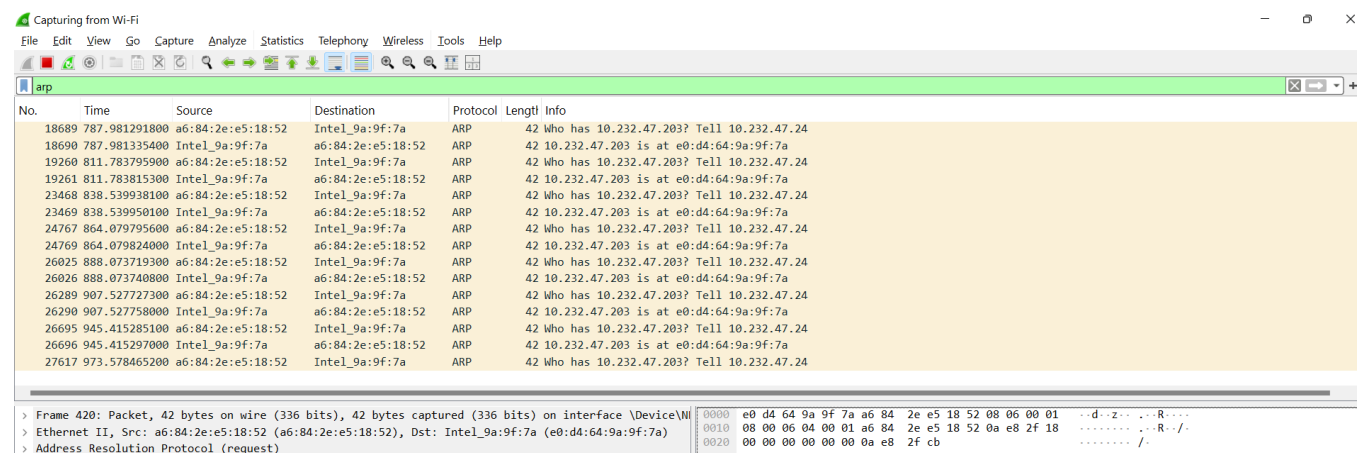
ARP Reply → 10.232.47.203 is at

e0:d4:64:9a:9f:7a



What Is Happening?

The communication you captured shows an ARP address-resolution exchange between the default gateway and your PC:



Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

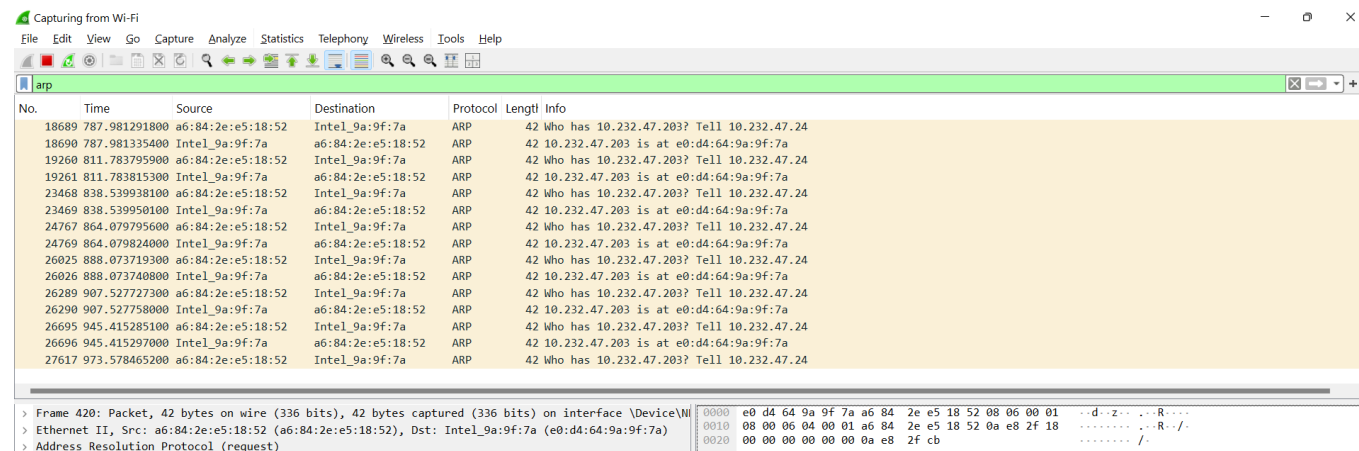
No.	Time	Source	Destination	Protocol	Length	Info
18689	787.981291800	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
18690	787.981335400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
19260	811.783795900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
19261	811.783815300	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
23468	838.539938100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
23469	838.539950100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
24767	864.079795600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
24769	864.079824000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26025	888.073719300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26026	888.073740800	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26289	907.527727300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26290	907.527758000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26695	945.415285100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26696	945.415297000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
27617	973.578465200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24

> Frame 420: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
> Address Resolution Protocol (request)

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 ...d...z...R....
0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R-./..
0020 00 00 00 00 00 0a e8 2f cb/..

The default gateway already knows your PC's IP address (10.232.47.203), but it needs the PC's MAC address before it can send an Ethernet frame to your PC on the local network. So it asks: “Who has 10.232.47.203?”

Lecturer's PC recognises its own IP address and replies: “10.232.47.203 is at E0:D4:64:9A:9F:7A.”



Capturing from Wi-Fi

No.	Time	Source	Destination	Protocol	Length	Info
18689	787.981291800	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
18690	787.981335400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
19260	811.783795900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
19261	811.783815300	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
23468	838.539938100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
23469	838.539950100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
24767	864.079795600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
24769	864.079824000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26025	888.073719300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26026	888.073740800	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26289	907.527727300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26290	907.527758000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26695	945.415285100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26696	945.415297000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
27617	973.578465200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24

> Frame 420: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
> Address Resolution Protocol (request)

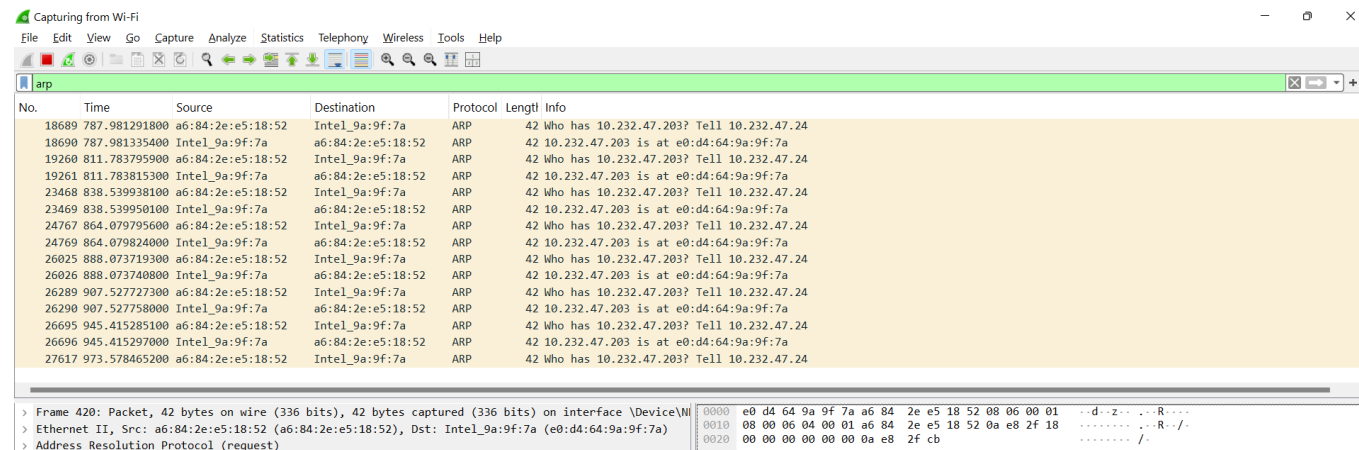
0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 ...d...z...R....
0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R.../
0020 00 00 00 00 00 0a e8 2f cb/..

Key Concept

ARP maps an IPv4 address to a MAC address on the local network.

IP address

10.232.47.203 → ARP → MAC address E0:D4:64:9A:9F:7A



Capturing from Wi-Fi

No.	Time	Source	Destination	Protocol	Length	Info
18689	787.981291800	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
18690	787.981335400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
19260	811.783795900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
19261	811.783815300	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
23468	838.539938100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
23469	838.539950100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
24767	864.079795600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
24769	864.079824000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26025	888.073719300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26026	888.073740800	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26289	907.527727300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26290	907.527758000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26695	945.415285100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26696	945.415297000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
27617	973.578465200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24

> Frame 420: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
> Address Resolution Protocol (request)

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 ...d...z...R....
0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R-.../
0020 00 00 00 00 00 0a e8 2f cb/..

Main idea) An IP address and a MAC address are different.

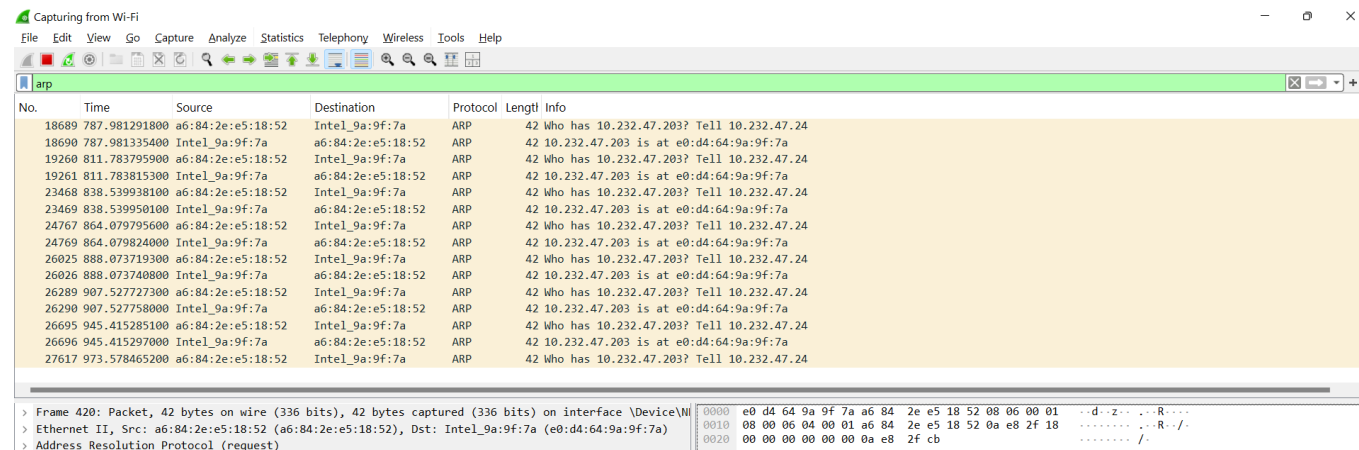
The gateway knows: IP = 10.232.47.203

but needs to know:

MAC = ?

ARP provides the connection:

10.232.47.203 → ARP → E0:D4:64:9A:9F:7A



Capturing from Wi-Fi

No.	Time	Source	Destination	Protocol	Length	Info
18689	787.981291800	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
18690	787.981335400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
19260	811.783795900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
19261	811.783815300	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
23468	838.539938100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
23469	838.539950100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
24767	864.079795600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
24769	864.079824000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26025	888.073719300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26026	888.073740800	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26289	907.527727300	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26290	907.527758000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
26695	945.415285100	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
26696	945.415297000	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
27617	973.578465200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24

arp

> Frame 420: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
> Address Resolution Protocol (request)

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 ...d...z...R...
0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R.../
0020 00 00 00 00 00 00 0a e8 2f cb/..

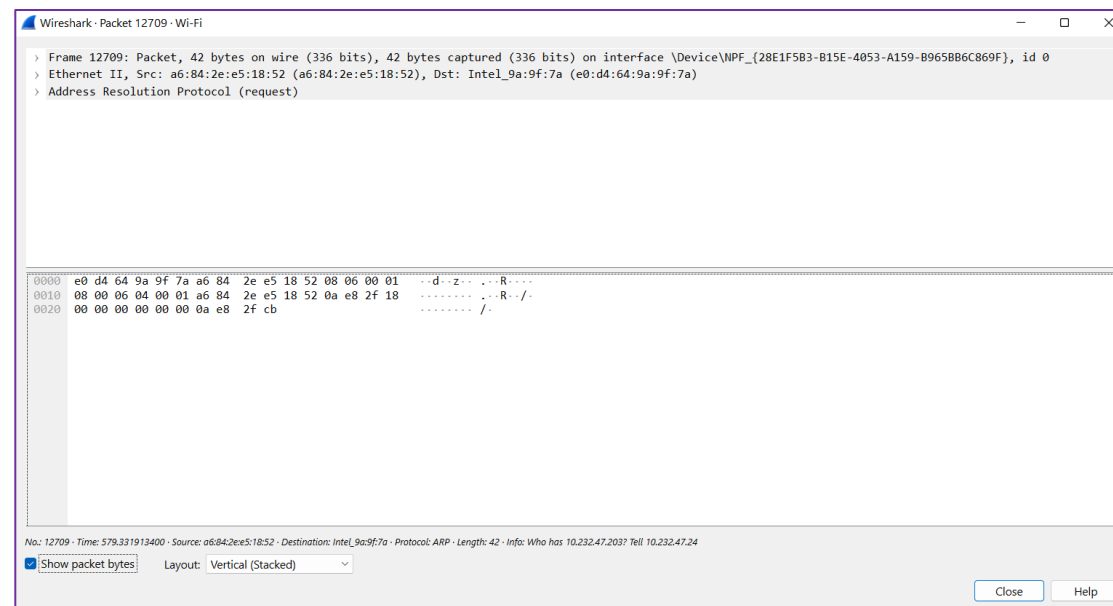
Step 4 – Examine One ARP Request

Now let's actually open one of these packets.

Clicking this line from the screenshot: No. 12709

Who has 10.232.47.203?

Tell 10.232.47.24



What does this mean?

The packet is organised in layers.

Frame 12709

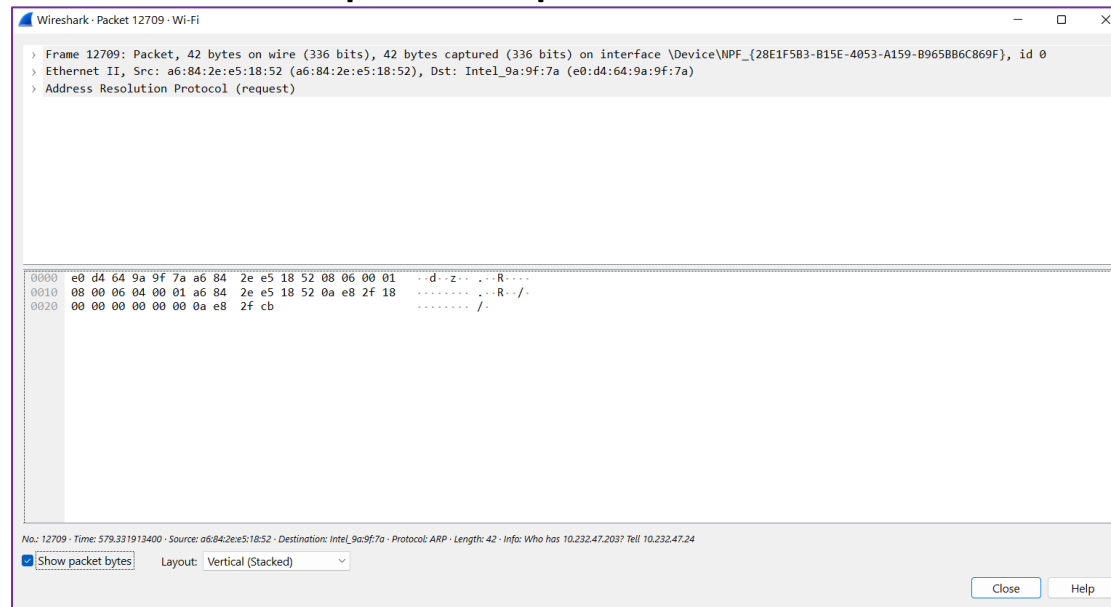
This gives general information about the captured packet, such as its size.

The capture shows:

42 bytes on wire

42 bytes captured

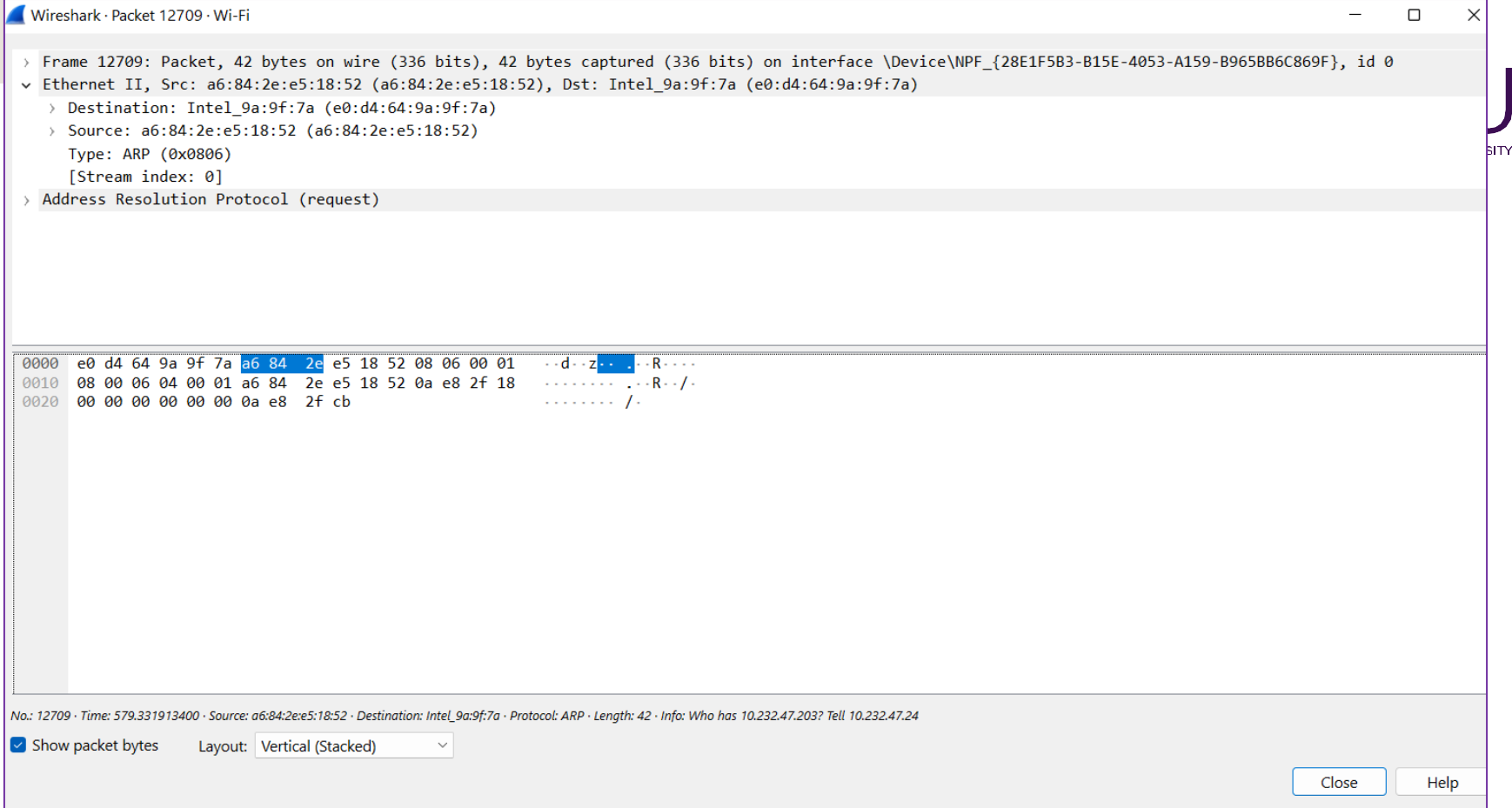
Ethernet II



This contains the Layer 2 information, including the source and destination MAC addresses.

Address Resolution Protocol (request)

This contains the actual ARP request.

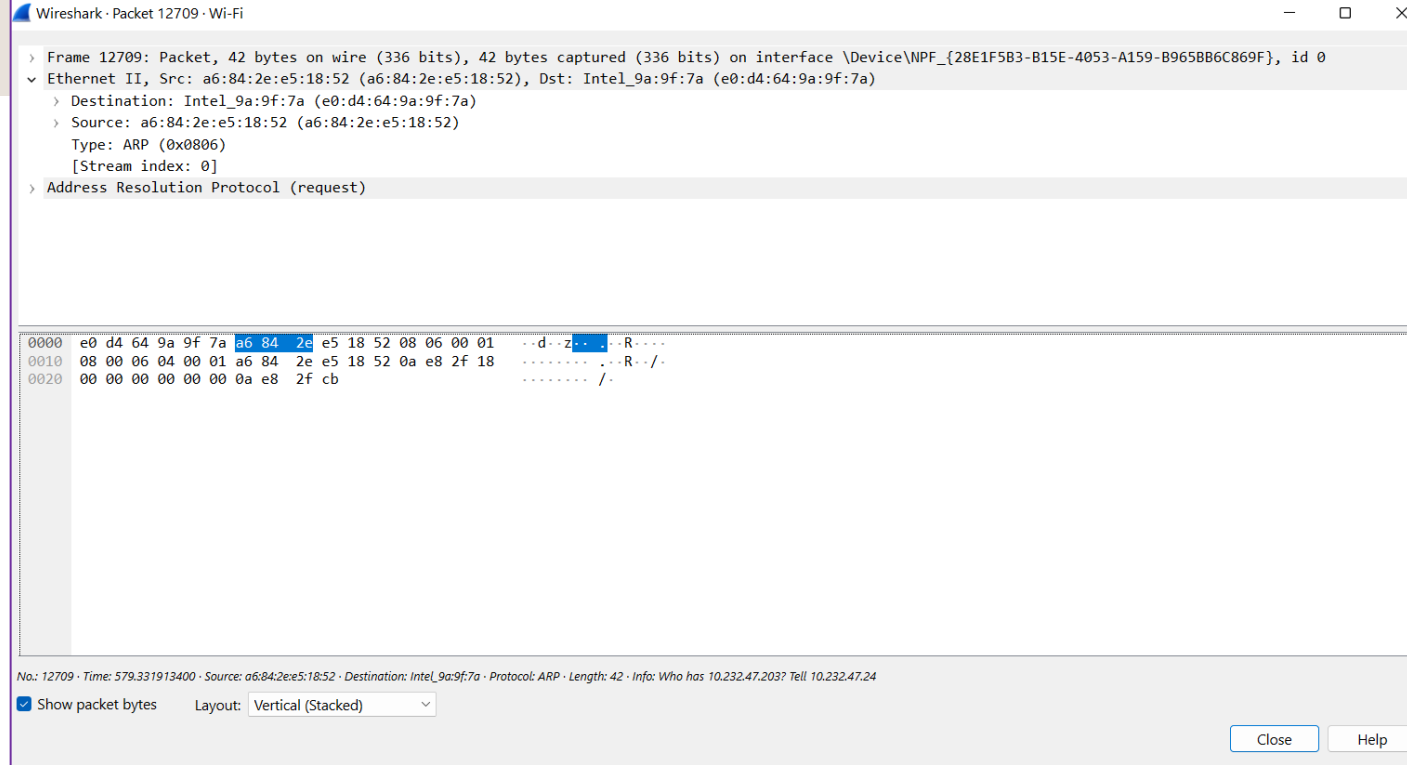


Step 4.1 – Expand Ethernet II

Click the small arrow > beside:

Ethernet II

It will expand and show more information.



Look for:

Destination: ...

Source: ...

Type: ARP (0x0806)

From the line already visible in your screenshot, we can see:

Source MAC: a6:84:2e:e5:18:52

Destination MAC: e0:d4:64:9a:9f:7a

Remember:

e0:d4:64:9a:9f:7a = Your PC's MAC address

So this ARP request is being sent to your PC.

Step 4.2 – Expand the ARP Request

Next, click the arrow > beside:

Address Resolution Protocol (request)

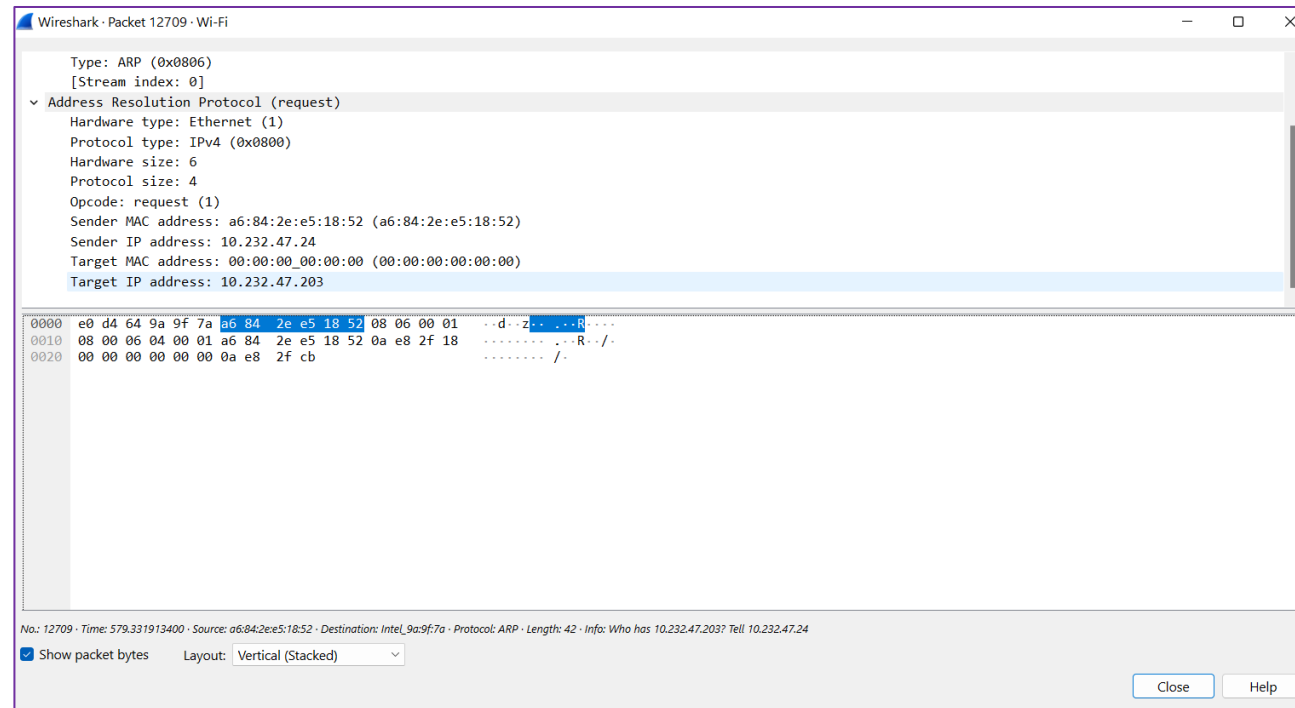
Look for:

Sender MAC address

Sender IP address

Target MAC address

Target IP address



The important question)

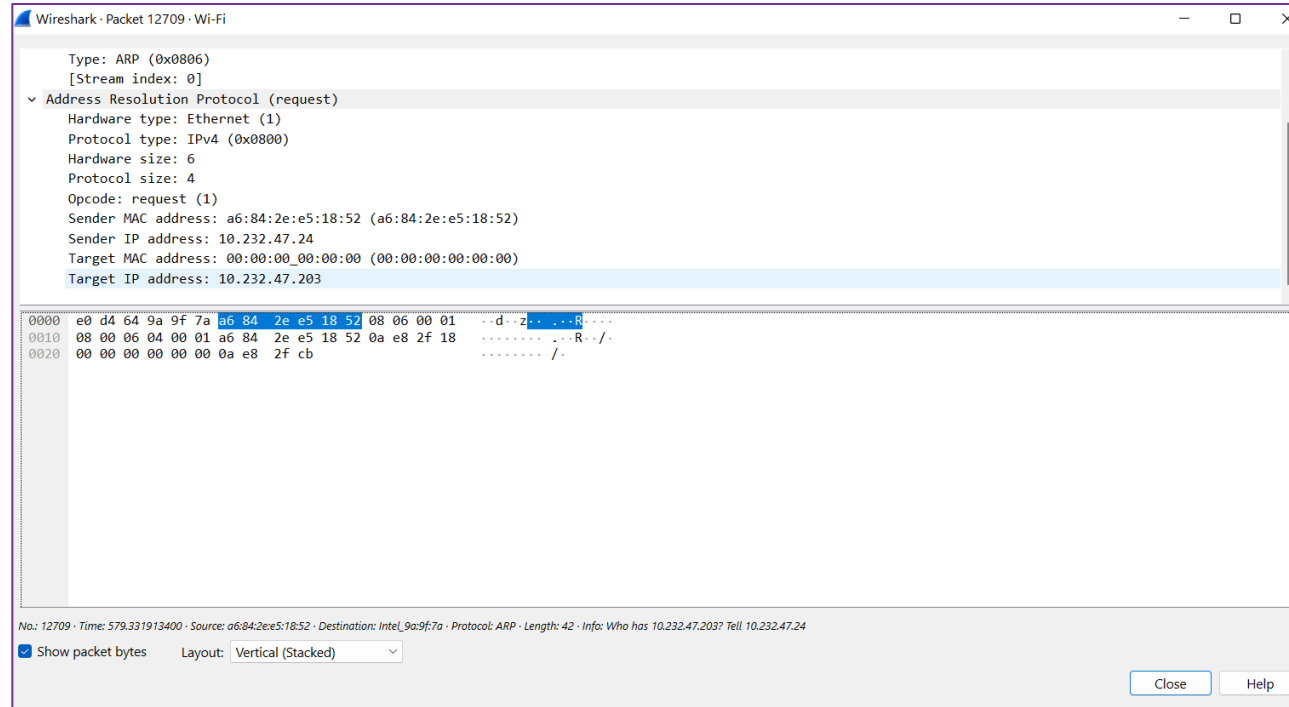
Who has 10.232.47.203?

Tell 10.232.47.24

In simple terms:

Gateway

10.232.47.24



"I need the MAC address

for 10.232.47.203." → Lecturer's PC

10.232.47.203

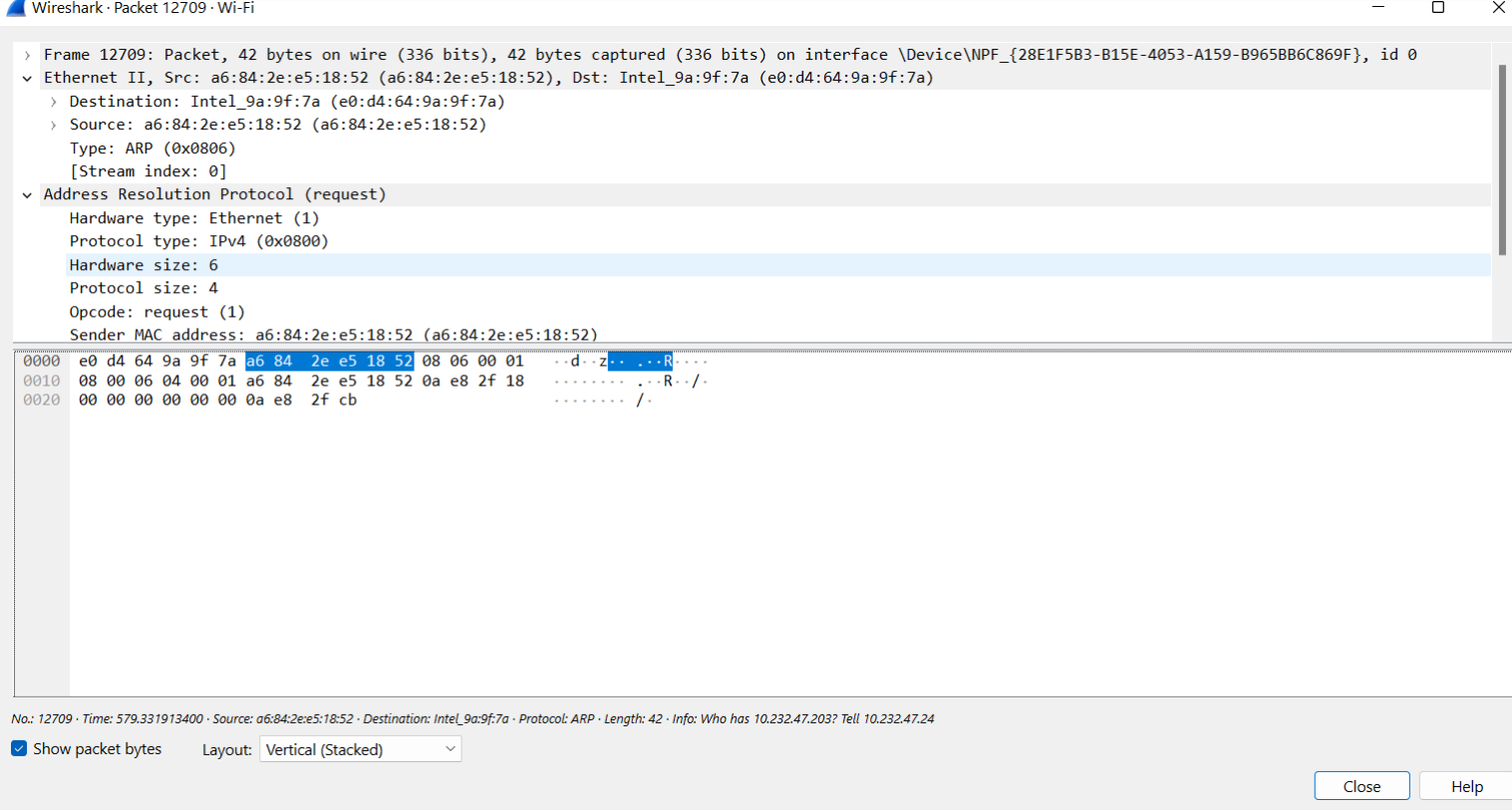
Why are we doing this?

We are no longer just looking at the word ARP in the packet list.

We are opening the packet to see how the information is actually stored inside the Ethernet frame:

Ethernet II Frame

- |
 - Destination MAC
 - Source MAC
 - Type: ARP (0x0806)
- |
 - ARP Request
 - Sender IP/MAC
 - Target IP/MAC



Key concept) 0x0806 means the Ethernet frame is carrying an ARP message.

So we can connect the theory to the real capture:

Ethernet II = the frame

ARP = the message carried inside the frame

Task 1 Completed – What Have We Done?

In Task 1, we learned how Ethernet and ARP work together on a real network.

We have:

- Identified our PC's actual network configuration using `ipconfig /all`
- Identified the active Wi-Fi network adapter
- Recorded our PC's IPv4 address, subnet mask, default gateway and MAC address
- Started a live capture using Wireshark
- Selected the correct network interface (Wi-Fi)

- Used the arp display filter to show only ARP traffic

- Identified ARP Requests and ARP Replies

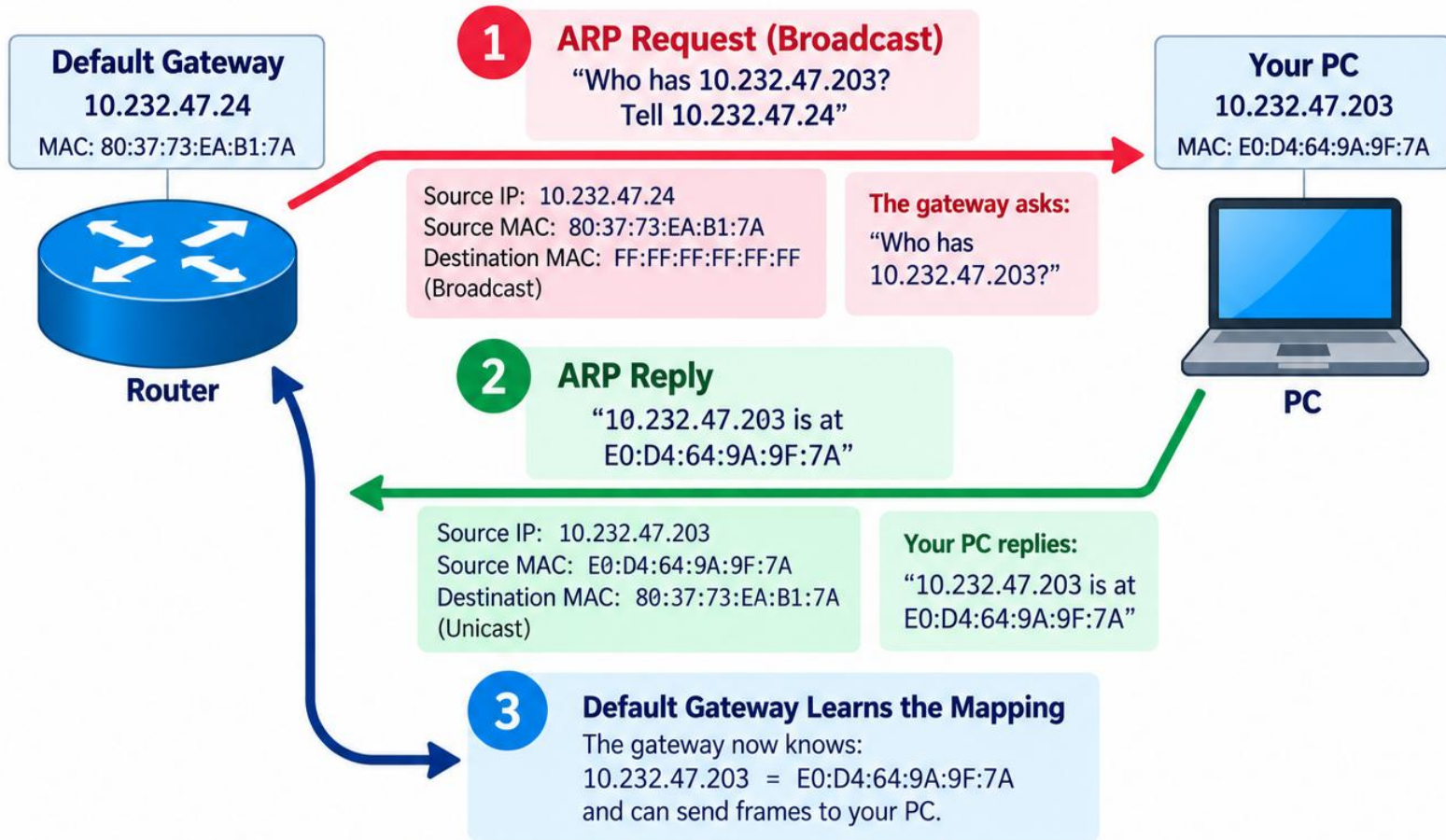
- Examined a real ARP Request:

Who has 10.232.47.203? Tell 10.232.47.24

- Opened the captured packet and examined its Ethernet II frame
- Identified the source MAC address and destination MAC address
- Identified the Ethernet Type field: ARP (0x0806)
- Expanded the Address Resolution Protocol (ARP) section and examined the ARP information
- Observed how Wireshark displays the actual frame bytes in hexadecimal

What Is Happening?

The communication you captured shows an ARP address-resolution exchange between the default gateway and your PC.



Key Point

ARP maps an IPv4 address to a MAC address on the local network.

In this capture, the router is asking for the PC's MAC address.

ARP "Who has ...?" Format

Who has 10.232.47.203?
Tell 10.232.47.24

This is a broadcast message, so all devices on the local network see it, but only the device with that IP address will reply.

Result: Gateway ARP Table

IP Address	MAC Address
10.232.47.203	E0:D4:64:9A:9F:7A

After the ARP reply, the gateway adds this entry to its ARP table.

What Have We Learned?

The key relationship is:

IP Address

|

| ARP resolves



MAC Address

And inside the captured traffic:

Ethernet II Frame

|

— Destination MAC

— Source MAC

— Type = ARP (0x0806)

|

└— ARP Message

— Sender MAC

— Sender IP

— Target MAC

— Target IP

Key Message

IP addresses identify devices logically.

MAC addresses are used for local Ethernet delivery.

ARP provides the connection between an IPv4 address and its corresponding MAC address.

For your particular capture, students have also seen something valuable that is slightly different from the old textbook example: the gateway/network-side device was asking for your PC's MAC address:

Who has 10.232.47.203?

Tell 10.232.47.24

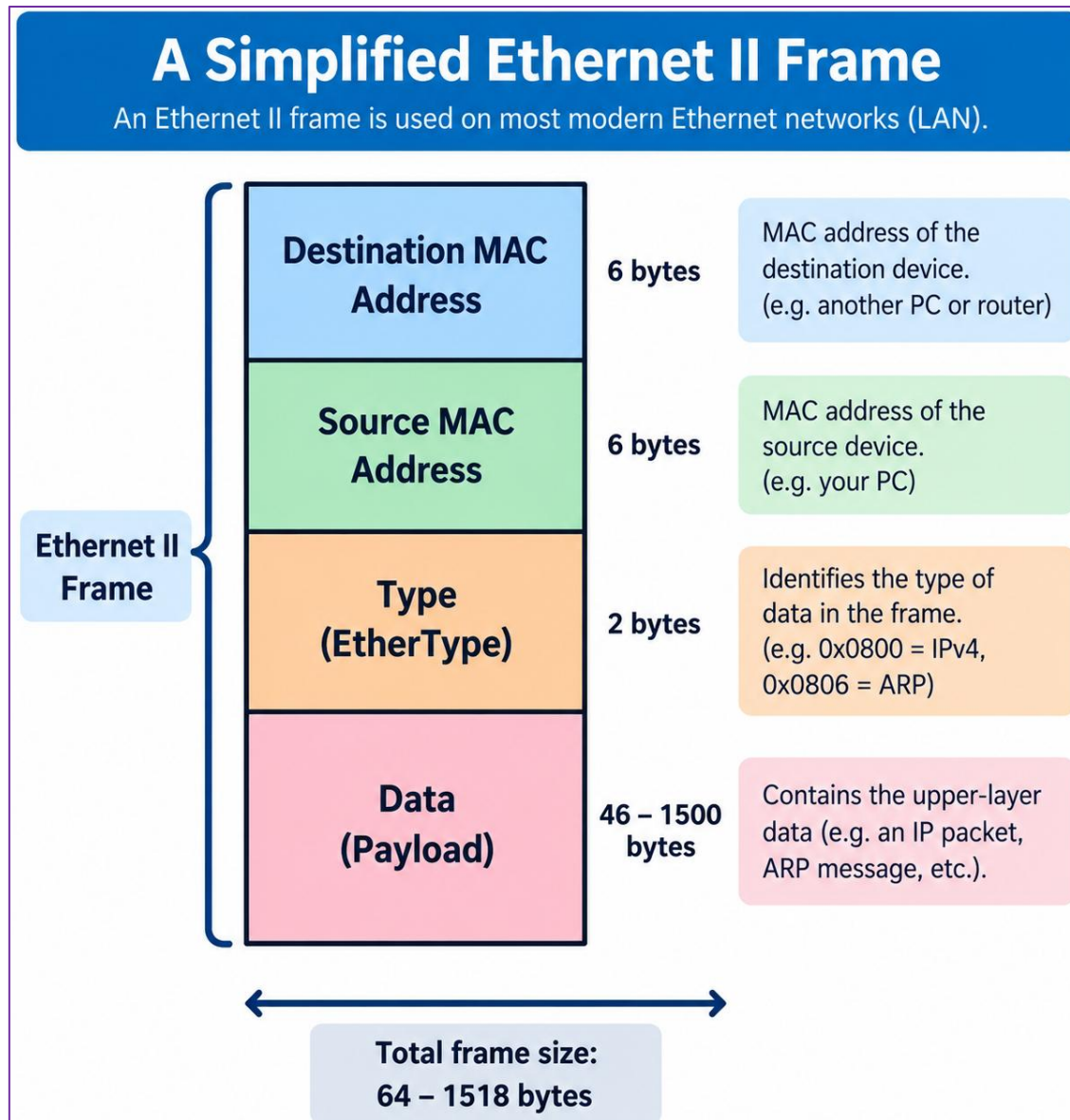
So they are seeing real network behaviour rather than simply reproducing the old 192.168.1.17 / 192.168.1.1 example.

A review of Task 1)

Understand the Ethernet II frame

Data sent across an Ethernet network is placed inside an Ethernet frame.

A simplified Ethernet II frame looks like this:



The complete frame contains:

Field	Size	Simple meaning
Preamble	8 bytes	Helps devices synchronise
Destination MAC	6 bytes	Who should receive the frame
Source MAC	6 bytes	Who sent the frame
Type	2 bytes	What protocol is inside
Data	46–1500 bytes	The information being carried
FCS	4 bytes	Helps detect transmission errors

A review of Task 1)

Important idea

The two fields we look at most often are:

Source MAC

Destination MAC

Check the PC network configuration

Open: **Windows Command Prompt**

Type: **ipconfig**

Look for:

IPv4 Address

Subnet Mask

Default Gateway

Example:

IPv4 Address: 192.168.1.17

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Why are we doing this?

Later, Wireshark will show many addresses.

We need to know:

Which address belongs to my PC?

Which address belongs to my router?

Otherwise, the packet capture may look confusing.

Important clarification

Your IP addresses will probably not be exactly the same as the example.

Students should record the addresses shown on their own PC.

Understand the problem before the first ping

Suppose the PC wants to ping: 192.168.1.1

The PC knows the router's IP address.

But Ethernet uses MAC addresses to deliver frames on the local network.

So the PC has a problem:

I know the router's IP address.

But what is its MAC address?

This is where ARP is used.

Understand the ARP Request

ARP stands for:

Address Resolution Protocol

The PC sends a message like:

Who has 192.168.1.1?

Tell 192.168.1.17

This is an:

ARP Request

The destination MAC address is: ff:ff:ff:ff:ff:ff

This means: Broadcast

What is a broadcast?

A broadcast is sent to all devices on the local network.

Why?

Because the PC does not yet know which MAC address belongs to 192.168.1.1.

So it asks everyone:

Who owns 192.168.1.1?

A review of Task 1)

Understand the ARP Reply

The router recognises its IP address and responds.

Example: 192.168.1.1 is at 80:37:73:ea:b1:7a

This is the:

ARP Reply

Now the PC knows:

IP Address

192.168.1.1 → ARP resolves this

MAC Address

80:37:73:ea:b1:7a

Now the ping can happen

Once the PC knows the router's MAC address, it can send the ping: **ping 10.232.47.24**

```
C:\> Command Prompt - ping 10.232.47.24

Lease Obtained. . . . . : Saturday, 5 September 2026 11:29:51
Lease Expires . . . . . : Saturday, 5 September 2026 20:29:35
Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                             10.232.47.24
DHCP Server . . . . . : 10.232.47.24
DHCPv6 IAID . . . . . : 232838244
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-2F-3D-4E-B0-22-7A-87-DD-C0
DNS Servers . . . . . : 10.232.47.24
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : E0-D4-64-9A-9F-7E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\gaming 15>ping 10.232.47.24

Pinging 10.232.47.24 with 32 bytes of data:
Reply from 10.232.47.24: bytes=32 time=4ms TTL=64
Reply from 10.232.47.24: bytes=32 time=10ms TTL=64
Reply from 10.232.47.24: bytes=32 time=3ms TTL=64
```

Why are we doing this?

A ping creates ICMP Echo Request and ICMP Echo Reply packets.

Before your PC can send an Ethernet frame to the gateway, it needs to know the gateway's MAC address.

We then ran: ping 10.232.47.24

and received:

Sent = 4

Received = 4

Lost = 0 (0% loss)

Test Communication with the Default Gateway

In Command Prompt, we typed: `ping 10.232.47.24`

Pressed Enter.

What happened?

Our PC sent four ICMP Echo Requests to the default gateway.

The gateway returned four ICMP Echo Replies.

This confirms that our PC can communicate successfully with the default gateway.

But what happened at Layer 2?

This is the important connection to Task 1.

To send the ping across the local network, the PC needs a MAC address for the next-hop device:

Destination IP

10.232.47.24 → ARP → Destination MAC → Ethernet frame
→ ICMP Echo Request

Next practical step

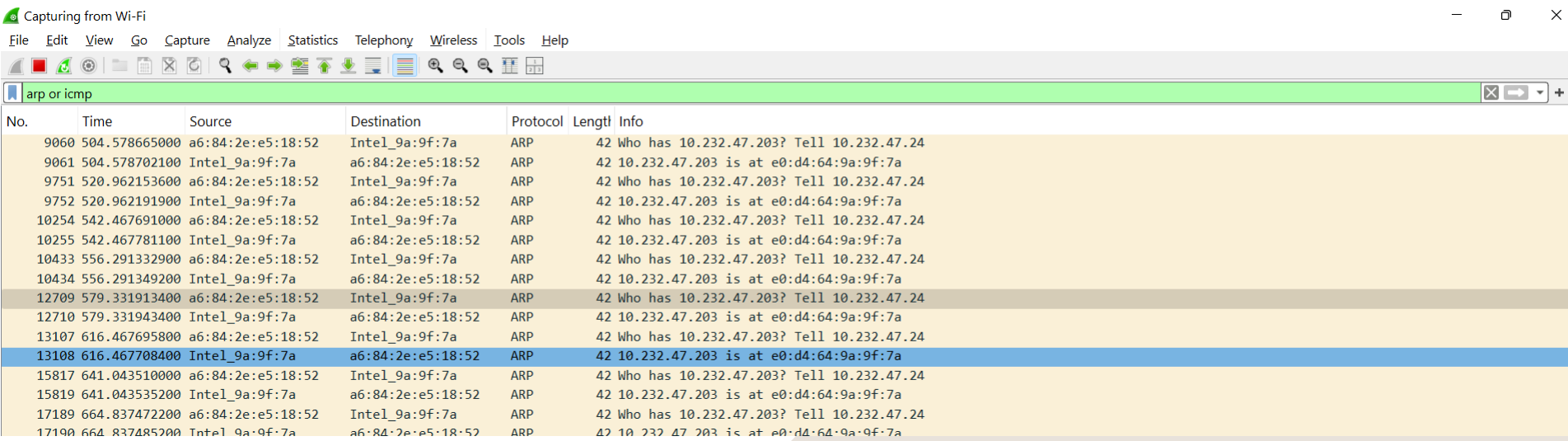
Now go back to Wireshark.

Instead of filtering only:

arp

change the display filter to:

arp or icmp



Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
9060	504.578665000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
9061	504.578702100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
9751	520.962153600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
9752	520.962191900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
10254	542.467691000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
10255	542.467781100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
10433	556.291332900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
10434	556.291349200	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
12709	579.331913400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
12710	579.331943400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
13107	616.467695800	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
13108	616.467708400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
15817	641.043510000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
15819	641.043535200	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
17189	664.837472200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
17190	664.837485200	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a

Preparation and Guideline for Lab 3

The Command Prompt confirms that the ping is working, but our current Wireshark screenshot still does not show the ICMP packets from that ping. It is showing older ARP traffic.

1. Stop the current Wireshark capture.
2. Start a new capture on Wi-Fi.
3. Keep the display filter: **arp or icmp**

Capturing from Wi-Fi

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arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
9060	504.578665000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
9061	504.578702100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
9751	520.962153600	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
9752	520.962191900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
10254	542.467691000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
10255	542.467781100	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
10433	556.291332900	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
10434	556.291349200	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
12709	579.331913400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
12710	579.331943400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
13107	616.467695800	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
13108	616.467708400	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
15817	641.043510000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
15819	641.043535200	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
17189	664.837472200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
17190	664.837485200	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a

Preparation and Guideline for Lab 3

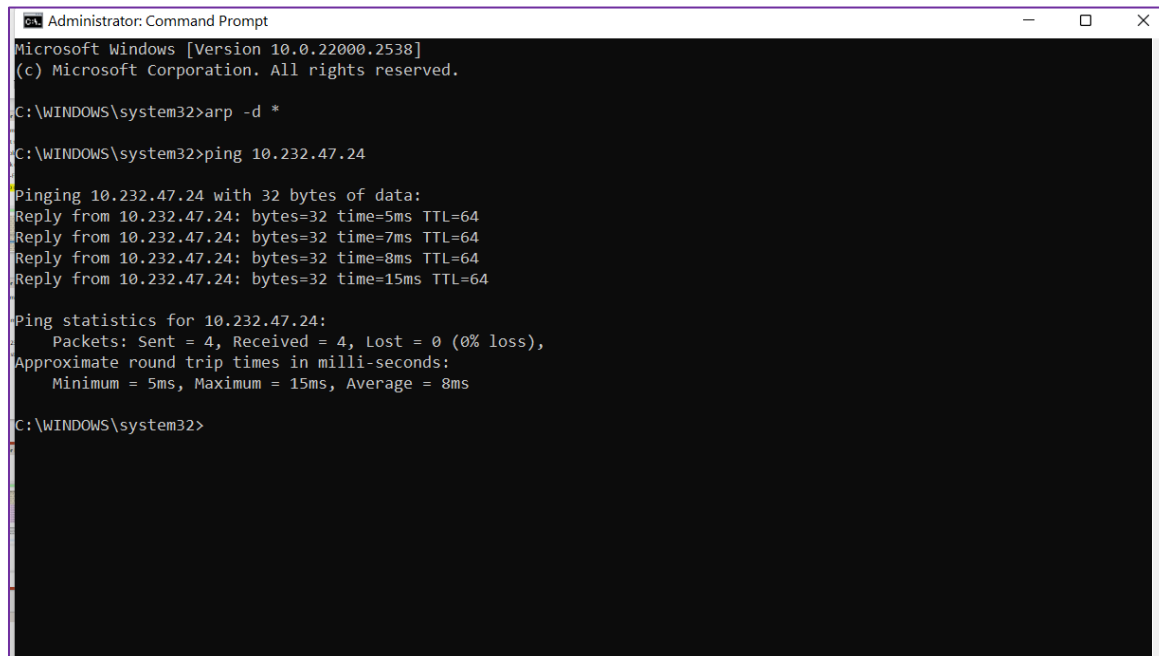
4. Before pinging, open Command Prompt as Administrator

and clear the ARP cache: `arp -d *`

This is useful because the PC may already remember the gateway's MAC address.

5. Immediately run: `ping 10.232.47.24`

6. Go back to Wireshark and stop the capture.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>arp -d *

C:\WINDOWS\system32>ping 10.232.47.24

Pinging 10.232.47.24 with 32 bytes of data:
Reply from 10.232.47.24: bytes=32 time=5ms TTL=64
Reply from 10.232.47.24: bytes=32 time=7ms TTL=64
Reply from 10.232.47.24: bytes=32 time=8ms TTL=64
Reply from 10.232.47.24: bytes=32 time=15ms TTL=64

Ping statistics for 10.232.47.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 15ms, Average = 8ms

C:\WINDOWS\system32>
```

Wi-Fi

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arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
128	9.625060200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
129	9.625071700	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1181	39.928760800	Intel_9a:9f:7a	Broadcast	ARP	42	Who has 10.232.47.24? Tell 10.232.47.203
1182	39.939215000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	10.232.47.24 is at a6:84:2e:e5:18:52
1238	45.207228400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1239	45.207242900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1378	52.346672000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 1379)
1379	52.352226500	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=64 (request in 1378)
1393	53.352239900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 1394)
1394	53.359439000	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=64 (request in 1393)
1403	54.370460400	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 1404)
1404	54.378384100	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=29/7424, ttl=64 (request in 1403)
1407	55.381422900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=30/7680, ttl=128 (reply in 1408)
1408	55.397156300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=30/7680, ttl=64 (request in 1407)

> Frame 128: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...

✓ Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Type: ARP (0x0806)

[Stream index: 0]

✓ Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Sender IP address: 10.232.47.24

Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)

Target IP address: 10.232.47.203

```

0000  e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01  ..d..z...R...
0010  08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18  ....R../.
0020  00 00 00 00 00 00 0a e8 2f cb  ....../.
  
```

See How ARP and Ping Work Together

We are now going to generate a ping and observe both ARP and ICMP in Wireshark.

Use the display filter: arp or icmp

Wi-Fi

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arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
128	9.625060200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
129	9.625071700	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1181	39.928760800	Intel_9a:9f:7a	Broadcast	ARP	42	Who has 10.232.47.24? Tell 10.232.47.203
1182	39.939215000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	10.232.47.24 is at a6:84:2e:e5:18:52
1238	45.207228400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1239	45.207242900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1378	52.346672000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 1379)
1379	52.352226500	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=64 (request in 1378)
1393	53.352239900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 1394)
1394	53.359439000	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=64 (request in 1393)
1403	54.370460400	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 1404)
1404	54.378384100	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=29/7424, ttl=64 (request in 1403)
1407	55.381422900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=30/7680, ttl=128 (reply in 1408)
1408	55.397156300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=30/7680, ttl=64 (request in 1407)

> Frame 128: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...

✓ Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Type: ARP (0x0806)

[Stream index: 0]

✓ Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Sender IP address: 10.232.47.24

Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)

Target IP address: 10.232.47.203

```

0000  e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01  ..d..z...R...
0010  08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18  ....R../.
0020  00 00 00 00 00 00 0a e8 2f cb  ....../.

```

Then ping the default gateway: ping 10.232.47.24

Look for this sequence:

PC

|
— ARP Request

| "What is the MAC address of 10.232.47.24?"

|
— ARP Reply

| "10.232.47.24 is at xx:xx:xx:xx:xx:xx"

|
— ICMP Echo Request

| "Are you there?"

|
— ICMP Echo Reply

"Yes, I am."

What does this show?

ARP and ping have different jobs:

ARP

IP address → MAC address



PC now knows where to send
the Ethernet frame



ICMP

Echo Request → Echo Reply



Connectivity confirmed

Key concept: ARP helps the PC find the correct Layer 2 MAC address. ICMP then performs the actual ping.

Note: if clearing the ARP cache is blocked by the university/network configuration, don't worry. You can still demonstrate ICMP separately and explain that the ARP entry was already cached.

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arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
128	9.625060200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
129	9.625071700	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1181	39.928760800	Intel_9a:9f:7a	Broadcast	ARP	42	Who has 10.232.47.24? Tell 10.232.47.203
1182	39.939215000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	10.232.47.24 is at a6:84:2e:e5:18:52
1238	45.207228400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1239	45.207242900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1378	52.346672000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 1379)
1379	52.352226500	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=64 (request in 1378)
1393	53.352239900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 1394)
1394	53.359439000	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=64 (request in 1393)
1403	54.370460400	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 1404)
1404	54.378384100	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=29/7424, ttl=64 (request in 1403)
1407	55.381422900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=30/7680, ttl=128 (reply in 1408)
1408	55.397156300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=30/7680, ttl=64 (request in 1407)

> Frame 128: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...

> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Type: ARP (0x0806)

[Stream index: 0]

> Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Sender IP address: 10.232.47.24

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 10.232.47.203

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 --d--z--..R----

0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R-./-

0020 00 00 00 00 00 00 0a e8 2f cb/:

Our addresses are:

Your PC

IP: 10.232.47.203

MAC: E0:D4:64:9A:9F:7A

Default Gateway

IP: 10.232.47.24

MAC: A6:84:2E:E5:18:52

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No.	Time	Source	Destination	Protocol	Length	Info
128	9.625060200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
129	9.625071700	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1181	39.928760800	Intel_9a:9f:7a	Broadcast	ARP	42	Who has 10.232.47.24? Tell 10.232.47.203
1182	39.939215000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	10.232.47.24 is at a6:84:2e:e5:18:52
1238	45.207228400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1239	45.207242900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1378	52.346672800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 1379)
1379	52.352226500	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=64 (request in 1378)
1393	53.352239900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 1394)
1394	53.359439000	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=64 (request in 1393)
1403	54.370460400	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 1404)
1404	54.378384100	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=29/7424, ttl=64 (request in 1403)
1407	55.381422900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=30/7680, ttl=128 (reply in 1408)
1408	55.397156300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=30/7680, ttl=64 (request in 1407)

> Frame 128: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...

> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Type: ARP (0x0806)

[Stream index: 0]

> Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Sender IP address: 10.232.47.24

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 10.232.47.203

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 --d--z-- --R----

0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18 --R-/-

0020 00 00 00 00 00 0a e8 2f cb /:

The clearest sequence for teaching is 1181 → 1182 → 1378 → 1379.

1. PC asks for the gateway's MAC address

Packet 1181 says: Who has 10.232.47.24?

Tell 10.232.47.203

In simple words: "I am PC 10.232.47.203. I want to communicate with 10.232.47.24, but I need to know its MAC address."

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No.	Time	Source	Destination	Protocol	Length	Info
128	9.625060200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
129	9.625071700	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1181	39.928760800	Intel_9a:9f:7a	Broadcast	ARP	42	Who has 10.232.47.24? Tell 10.232.47.203
1182	39.939215000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	10.232.47.24 is at a6:84:2e:e5:18:52
1238	45.207228400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1239	45.207242900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1378	52.346672000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 1379)
1379	52.352226500	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=64 (request in 1378)
1393	53.352239900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 1394)
1394	53.359439000	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=64 (request in 1393)
1403	54.370460400	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 1404)
1404	54.378384100	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=29/7424, ttl=64 (request in 1403)
1407	55.381422900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=30/7680, ttl=128 (reply in 1408)
1408	55.397156300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=30/7680, ttl=64 (request in 1407)

> Frame 128: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...

> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Type: ARP (0x0806)

[Stream index: 0]

> Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Sender IP address: 10.232.47.24

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 10.232.47.203

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 --d-z--..R....

0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R../-

0020 00 00 00 00 00 00 0a e8 2f cb/:

This is the ARP Request.

2. Gateway answers

Packet 1182 says: 10.232.47.24 is at A6:84:2E:E5:18:52

In simple words: "I am 10.232.47.24. My MAC address is

A6:84:2E:E5:18:52."

This is the ARP Reply.

Wi-Fi

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arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
128	9.625060200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
129	9.625071700	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1181	39.928760800	Intel_9a:9f:7a	Broadcast	ARP	42	Who has 10.232.47.24? Tell 10.232.47.203
1182	39.939215000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	10.232.47.24 is at a6:84:2e:e5:18:52
1238	45.207228400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1239	45.207242900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1378	52.346672800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 1379)
1379	52.352226500	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=64 (request in 1378)
1393	53.352239900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 1394)
1394	53.359439000	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=64 (request in 1393)
1403	54.370460400	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 1404)
1404	54.378384100	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=29/7424, ttl=64 (request in 1403)
1407	55.381422900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=30/7680, ttl=128 (reply in 1408)
1408	55.397156300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=30/7680, ttl=64 (request in 1407)

> Frame 128: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...

> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

> Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Type: ARP (0x0806)

[Stream index: 0]

> Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Sender IP address: 10.232.47.24

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 10.232.47.203

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 --d-z--..R....

0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18R../.

0020 00 00 00 00 00 00 0a e8 2f cb/:

Now your PC knows:

10.232.47.24

↓ ARP

A6:84:2E:E5:18:52

3. Now the ping can be sent

Look at packet 1378: 10.232.47.203 → 10.232.47.24

Wi-Fi

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arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
128	9.625060200	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
129	9.625071700	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1181	39.928760800	Intel_9a:9f:7a	Broadcast	ARP	42	Who has 10.232.47.24? Tell 10.232.47.203
1182	39.939215000	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	10.232.47.24 is at a6:84:2e:e5:18:52
1238	45.207228400	a6:84:2e:e5:18:52	Intel_9a:9f:7a	ARP	42	Who has 10.232.47.203? Tell 10.232.47.24
1239	45.207242900	Intel_9a:9f:7a	a6:84:2e:e5:18:52	ARP	42	10.232.47.203 is at e0:d4:64:9a:9f:7a
1378	52.346672800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=27/6912, ttl=128 (reply in 1379)
1379	52.352226500	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=27/6912, ttl=64 (request in 1378)
1393	53.352239900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=28/7168, ttl=128 (reply in 1394)
1394	53.359439000	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=28/7168, ttl=64 (request in 1393)
1403	54.370460400	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=29/7424, ttl=128 (reply in 1404)
1404	54.378384100	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=29/7424, ttl=64 (request in 1403)
1407	55.381422900	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=30/7680, ttl=128 (reply in 1408)
1408	55.397156300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=30/7680, ttl=64 (request in 1407)

Frame 128: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...

Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Type: ARP (0x0806)

[Stream index: 0]

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Sender IP address: 10.232.47.24

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 10.232.47.203

0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01 --d-z-- --R--

0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18 --R-/-

0020 00 00 00 00 00 00 0a e8 2f cb /:

ICMP Echo (ping) request

Your PC sends: "Are you there?"

Then packet 1379: 10.232.47.24 → 10.232.47.203

ICMP Echo (ping) reply The gateway effectively answers: "Yes, I'm here."

And this repeats four times:

1378 Echo Request

1379 Echo Reply

1393 Echo Request

1394 Echo Reply

1403 Echo Request

1404 Echo Reply

1407 Echo Request

1408 Echo Reply

That corresponds nicely to the four Windows ping replies.

ARP → Ping relationship?

Four packets:

1181 ARP Who has 10.232.47.24?

Tell 10.232.47.203



1182 ARP 10.232.47.24 is at

A6:84:2E:E5:18:52



1378 ICMP 10.232.47.203 → 10.232.47.24

Echo (ping) Request



1379 ICMP 10.232.47.24 → 10.232.47.203

Echo (ping) Reply

STEP 1 — ARP

"What is the gateway's MAC address?"



STEP 2 — ARP

"Here is my MAC address."



STEP 3 — ICMP

PC sends the ping



STEP 4 — ICMP

Gateway replies



Connectivity confirmed

The main idea for students

ARP does not perform the ping.

ARP and ICMP have different jobs:

ARP

Find the MAC address



Ethernet can deliver the frame



ICMP

Send the ping



Echo Reply received



Connection works ✓

ARP finds the MAC address needed for local Ethernet delivery; ICMP then performs the ping.

Packet 128 says:

Who has 10.232.47.203?

Tell 10.232.47.24

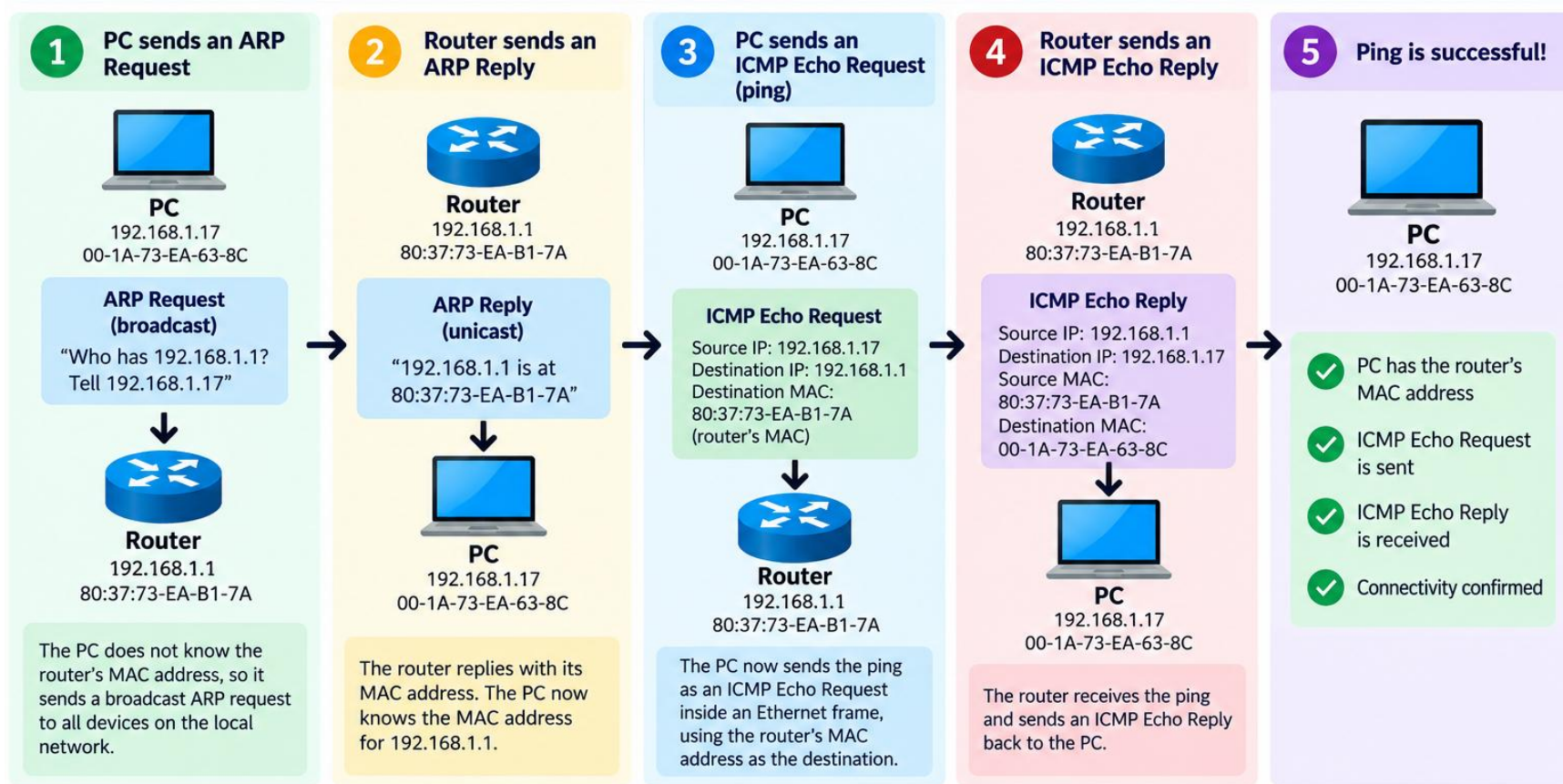
That's the opposite direction: the gateway (10.232.47.24) is asking for lecturer's PC's MAC address.

1181 → 1182 → 1378 → 1379

PC needs gateway MAC → ARP → MAC learned → Ping → Reply.

Ping can happen

Once the PC knows the router's MAC address, it can send the ping:



Main Idea

ARP usually comes before the first ping because the PC first needs the destination MAC address.

Main idea

ARP usually comes before the first ping because the PC first needs the destination MAC address.

Step 8 – Examine an ARP frame

In the example capture:

Source MAC: 00:1a:73:ea:63:8c

Destination MAC: ff:ff:ff:ff:ff:ff

Type:

ARP (0x0806)

What does 0x0806 mean?

It tells us that the Ethernet frame contains: ARP

You may also see: 0x0800

which means: IPv4

Task 1 – Questions to Discuss Together

Question 1

Why is the destination MAC address: **ff:ff:ff:ff:ff:ff**

Because the sender does not yet know the required MAC address, so the ARP request is sent as a broadcast to all devices on the local network.

Question 2

Why does the PC send ARP before the first ping?

The PC knows the destination IP address but needs the destination MAC address to send the Ethernet frame.

Question 3

What is the source MAC address in the example?

00:1a:73:ea:63:8c

Question 4

What part of the MAC address is the OUI?

For:

00:1a:73:ea:63:8c

the first three bytes are:

00:1a:73

This is the OUI.

OUI stands for:

Organisationally Unique Identifier

It normally identifies the manufacturer of the network interface.

Task 1 Key Message

IP address tells us which logical device we want.

MAC address helps deliver the Ethernet frame
on the local network.

ARP connects the two.

TASK 2 – Capture and Analyse a Ping Using Wireshark

We want to generate some network traffic and then use Wireshark to answer: Who sent the packet? Where is it going? Which protocol is being used? What information is inside the frame?

1

Generate network traffic

Use the ping command to create traffic.

```
C:\> ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
```

2

Capture the traffic in Wireshark

Start a capture on the correct network interface.



Wireshark

3

Filter and inspect

Use filters (e.g. arp or icmp) and examine the packets.

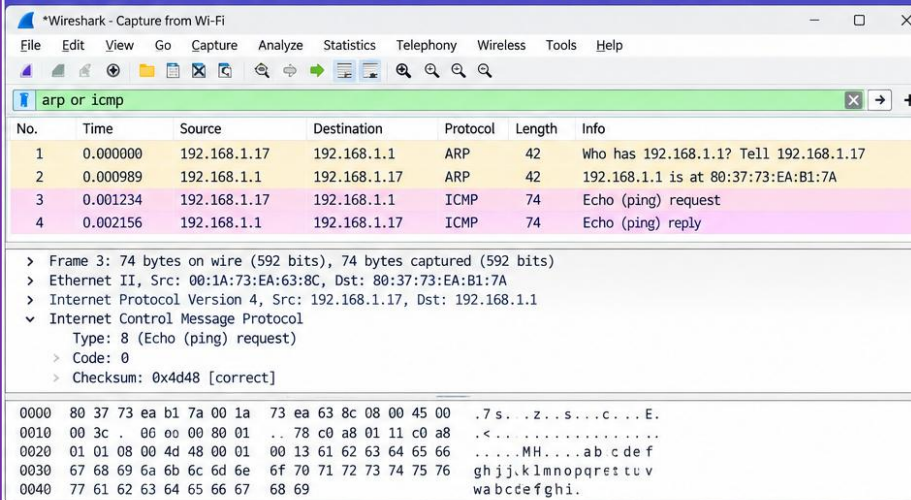
Example filters:

- arp (ARP packets only)
- icmp (ICMP packets only)
- arp or icmp (both ARP and ICMP)



Look at the details of each packet.

Example: Packets captured in Wireshark (filtered with arp or icmp)



Wireshark - Capture from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.17	192.168.1.1	ARP	42	Who has 192.168.1.1? Tell 192.168.1.17
2	0.000989	192.168.1.1	192.168.1.17	ARP	42	192.168.1.1 is at 80:37:73:EA:B1:7A
3	0.001234	192.168.1.17	192.168.1.1	ICMP	74	Echo (ping) request
4	0.002156	192.168.1.1	192.168.1.17	ICMP	74	Echo (ping) reply

> Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 bits)
> Ethernet II, Src: 00:1A:73:EA:63:8C, Dst: 80:37:73:EA:B1:7A
> Internet Protocol Version 4, Src: 192.168.1.17, Dst: 192.168.1.1
v Internet Control Message Protocol
Type: 8 (Echo (ping) request)
> Code: 0
> Checksum: 0x4d48 [correct]

0000 80 37 73 ea b1 7a 00 1a 73 ea 63 8c 08 00 45 00 .7 s . . z . . s . . . c . . . E .
0010 00 3c . 06 00 00 80 01 .. 78 c0 a8 01 11 c0 a8 . <
0020 01 01 08 00 4d 48 00 01 00 13 61 62 63 64 65 66 MH ab c d e f
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 g h j . k l m n o p q r e s t u v
0040 77 61 62 63 64 65 66 67 68 69 w a b c d e f g h i .

Use Wireshark to Answer These Questions

1



Who sent the packet?

Look at the Source MAC and Source IP address in the packet details.

2



Where is it going?

Look at the Destination MAC and Destination IP address.

3



Which protocol is being used?

Check the Protocol field (e.g. ARP, ICMP) and the frame type (e.g. 0x0800 for IPv4).

4



What information is inside the frame?

Expand the packet details to see the Ethernet, IP and ICMP/ARP information, and look at the packet bytes (hex and ASCII).



Key Point

By capturing and analysing real packets, you can see exactly how devices communicate on the network.

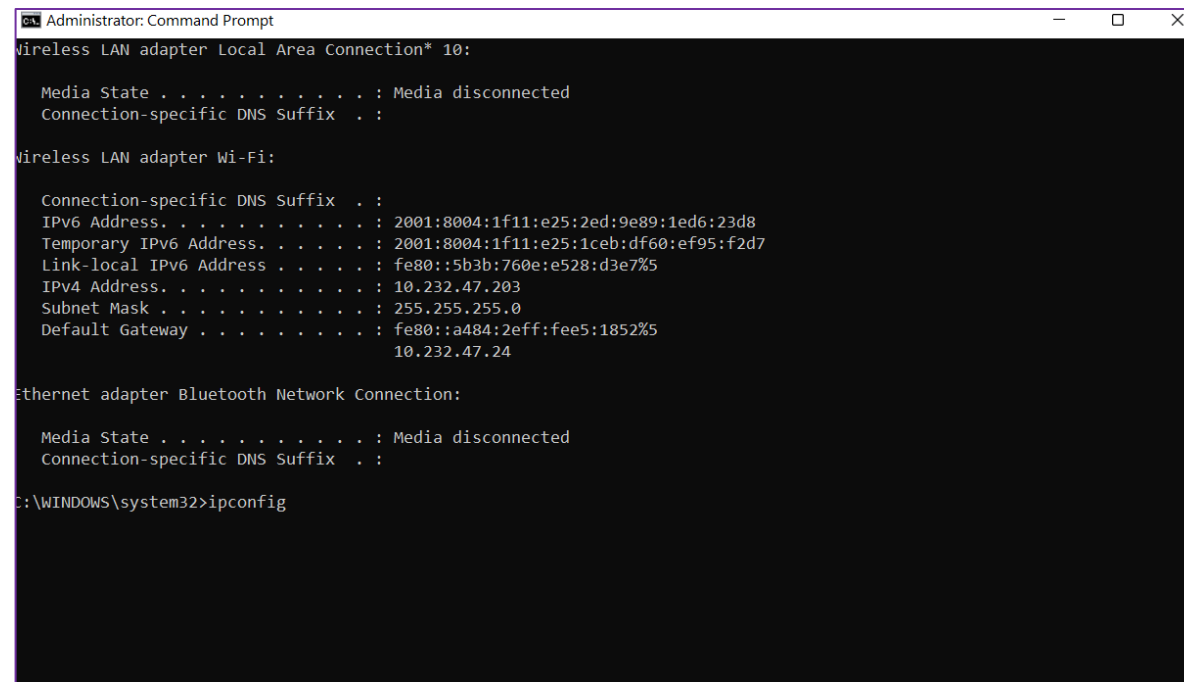
Step 1 – Find Your Default Gateway

Open Command Prompt and type: **ipconfig**

Find the network adapter you are currently using, such as
Wireless LAN adapter Wi-Fi.

Locate: Default Gateway

Write down the
IPv4 default gateway.



```
Administrator: Command Prompt
Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8
    Temporary IPv6 Address. . . . . : 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7
    Link-local IPv6 Address . . . . . : fe80::5b3b:760e:e528:d3e7%5
    IPv4 Address. . . . . : 10.232.47.203
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                              10.232.47.24

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\WINDOWS\system32>ipconfig
```

For example:

Default Gateway: 10.232.47.203

Important: Your address may be different. Use the default gateway shown on your own PC.

```
Administrator: Command Prompt
Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8
    Temporary IPv6 Address. . . . . : 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7
    Link-local IPv6 Address . . . . . : fe80::5b3b:760e:e528:d3e7%5
    IPv4 Address. . . . . : 10.232.47.203
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                                10.232.47.24

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\WINDOWS\system32>ipconfig
```

Why?

We are going to ping your default gateway and capture that communication in Wireshark.

This also matches your actual screenshot much better.

```
Administrator: Command Prompt
Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8
    Temporary IPv6 Address. . . . . : 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7
    Link-local IPv6 Address . . . . . : fe80::5b3b:760e:e528:d3e7%5
    IPv4 Address. . . . . : 10.232.47.203
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                              10.232.47.24

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\WINDOWS\system32>ipconfig
```

Step 2 – Open Wireshark

Open: **Wireshark**

You should see interfaces such as:

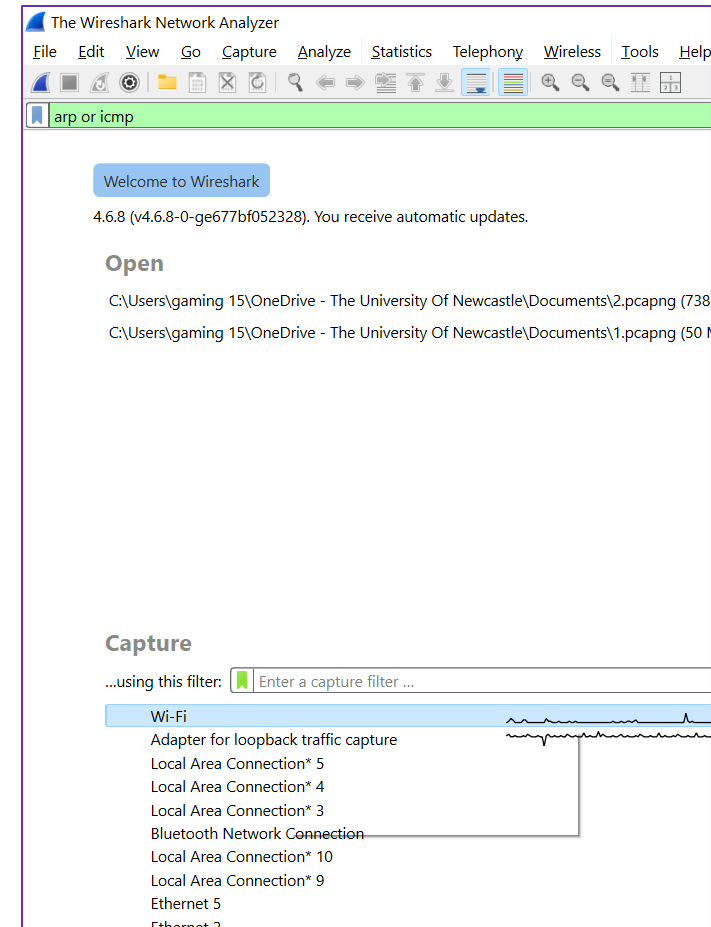
Wi-Fi

Ethernet

Which interface should I choose?

Choose the network interface your computer is currently using.

For example: If you are connected using Wi-Fi: **Choose Wi-Fi**



If you are connected using a cable: **Choose Ethernet**

Helpful clue

Look for the interface showing activity.

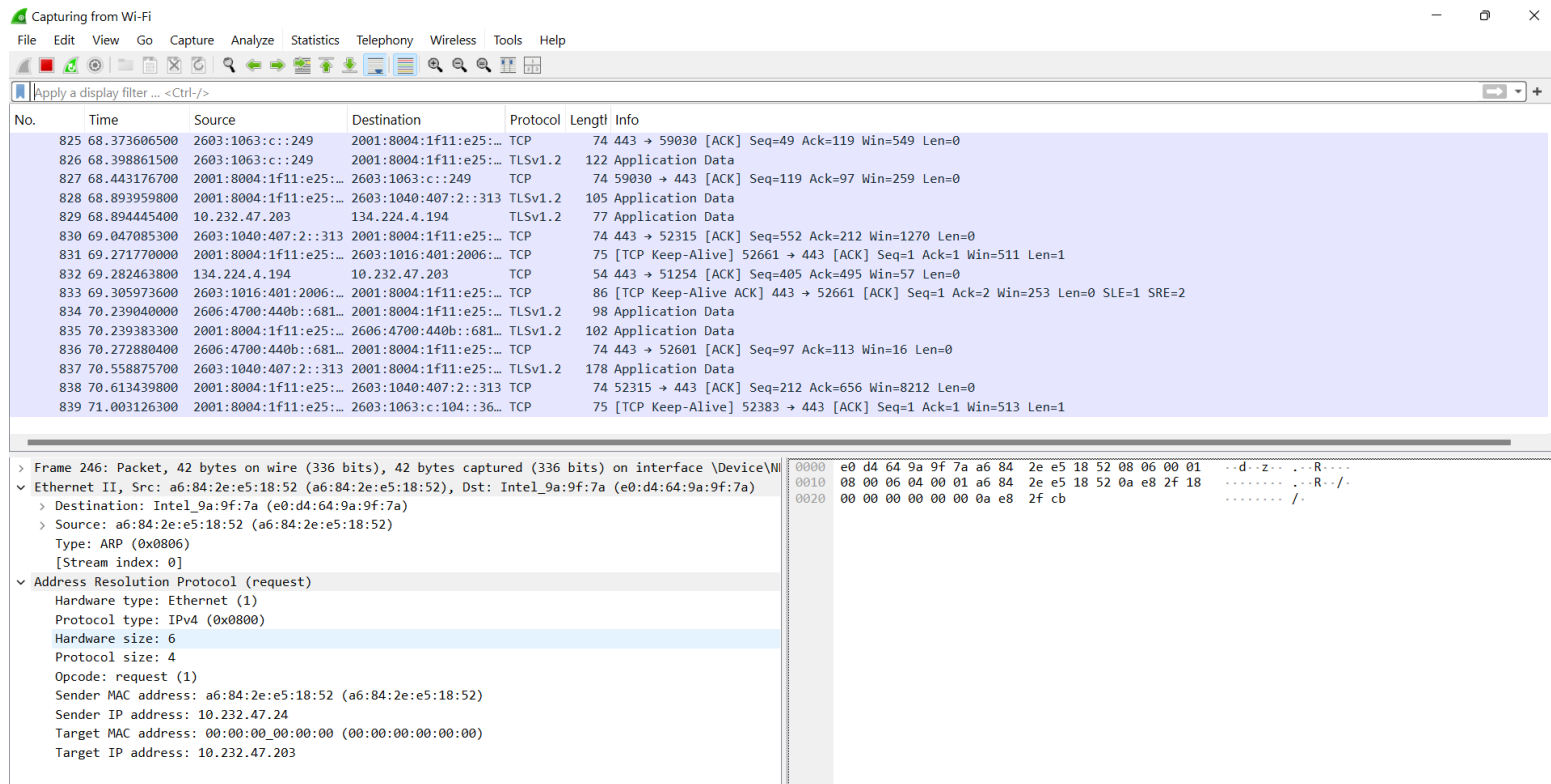
The small graph or packet counter should be moving.

Step 3 – Start capturing

Double-click the correct interface.

Or select it and click: **Start**

You should immediately see packets appearing.



The image shows a Wireshark packet capture window titled "Capturing from Wi-Fi". The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help) and a toolbar. A display filter is applied: "Apply a display filter ... <Ctrl-/>".

The packet list pane shows the following packets:

No.	Time	Source	Destination	Protocol	Length	Info
825	68.373606500	2603:1063:c::249	2001:8004:1f11:e25::...	TCP	74	443 → 59030 [ACK] Seq=49 Ack=119 Win=549 Len=0
826	68.398861500	2603:1063:c::249	2001:8004:1f11:e25::...	TLSv1.2	122	Application Data
827	68.443176700	2001:8004:1f11:e25::...	2603:1063:c::249	TCP	74	59030 → 443 [ACK] Seq=119 Ack=97 Win=259 Len=0
828	68.893959800	2001:8004:1f11:e25::...	2603:1040:407:2::313	TLSv1.2	105	Application Data
829	68.894445400	10.232.47.203	134.224.4.194	TLSv1.2	77	Application Data
830	69.047085300	2603:1040:407:2::313	2001:8004:1f11:e25::...	TCP	74	443 → 52315 [ACK] Seq=552 Ack=212 Win=1270 Len=0
831	69.271770000	2001:8004:1f11:e25::...	2603:1016:401:2006::...	TCP	75	[TCP Keep-Alive] 52661 → 443 [ACK] Seq=1 Ack=1 Win=511 Len=1
832	69.282463800	134.224.4.194	10.232.47.203	TCP	54	443 → 51254 [ACK] Seq=405 Ack=495 Win=57 Len=0
833	69.305973600	2603:1016:401:2006::...	2001:8004:1f11:e25::...	TCP	86	[TCP Keep-Alive ACK] 443 → 52661 [ACK] Seq=1 Ack=2 Win=253 Len=0 SLE=1 SRE=2
834	70.239040000	2606:4700:440b::681...	2001:8004:1f11:e25::...	TLSv1.2	98	Application Data
835	70.239383300	2001:8004:1f11:e25::...	2606:4700:440b::681...	TLSv1.2	102	Application Data
836	70.272880400	2606:4700:440b::681...	2001:8004:1f11:e25::...	TCP	74	443 → 52601 [ACK] Seq=97 Ack=113 Win=16 Len=0
837	70.558875700	2603:1040:407:2::313	2001:8004:1f11:e25::...	TLSv1.2	178	Application Data
838	70.613439800	2001:8004:1f11:e25::...	2603:1040:407:2::313	TCP	74	52315 → 443 [ACK] Seq=212 Ack=656 Win=8212 Len=0
839	71.003126300	2001:8004:1f11:e25::...	2603:1063:c::104::36...	TCP	75	[TCP Keep-Alive] 52383 → 443 [ACK] Seq=1 Ack=1 Win=513 Len=1

The packet details pane shows the selected packet (Frame 246) with the following information:

- Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
- Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
 - Destination: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
 - Source: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
 - Type: ARP (0x0806)
 - [Stream index: 0]
- Address Resolution Protocol (request)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: request (1)
 - Sender MAC address: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
 - Sender IP address: 10.232.47.24
 - Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)
 - Target IP address: 10.232.47.203

The packet bytes pane shows the raw data in hexadecimal and ASCII:

```
0000 e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 06 00 01  ..d..z...R...
0010 08 00 06 04 00 01 a6 84 2e e5 18 52 0a e8 2f 18  ....R.../..
0020 00 00 00 00 00 00 0a e8 2f cb  ....f..
```

Don't worry!

You may see many different protocols:

TCP

DNS

ARP

IPv6

UDP

ICMP

This is normal.

Your computer is communicating with many network services in the background.

Step 4 – First understand the Wireshark screen

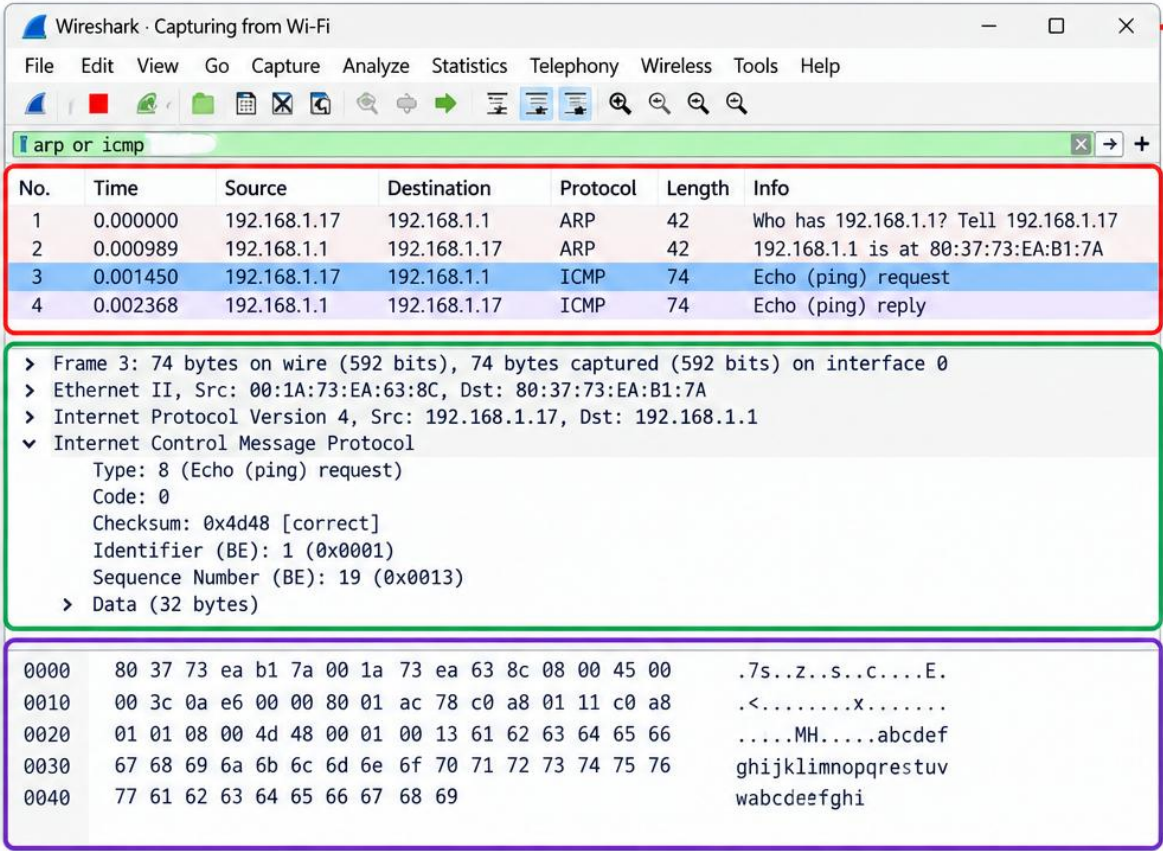
Wireshark has three important sections.



Wireshark – The Three Important Sections

Wireshark shows captured network data in three main sections.

*Capture
Explore
Learn*



The screenshot shows the Wireshark interface with the following sections highlighted:

- TOP – Packet List:** A table of captured packets. The third packet is selected.
- MIDDLE – Packet Details:** A tree view showing the protocol stack of the selected packet.
- BOTTOM – Packet Bytes:** A hex/ASCII view of the selected packet's data.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.17	192.168.1.1	ARP	42	Who has 192.168.1.1? Tell 192.168.1.17
2	0.000989	192.168.1.1	192.168.1.17	ARP	42	192.168.1.1 is at 80:37:73:EA:B1:7A
3	0.001450	192.168.1.17	192.168.1.1	ICMP	74	Echo (ping) request
4	0.002368	192.168.1.1	192.168.1.17	ICMP	74	Echo (ping) reply

Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0

- Ethernet II, Src: 00:1A:73:EA:63:8C, Dst: 80:37:73:EA:B1:7A
- Internet Protocol Version 4, Src: 192.168.1.17, Dst: 192.168.1.1
- Internet Control Message Protocol
 - Type: 8 (Echo (ping) request)
 - Code: 0
 - Checksum: 0x4d48 [correct]
 - Identifier (BE): 1 (0x0001)
 - Sequence Number (BE): 19 (0x0013)
 - Data (32 bytes)

```
0000  80 37 73 ea b1 7a 00 1a 73 ea 63 8c 08 00 45 00  .7s..z...s...C....E.
0010  00 3c 0a e6 00 00 80 01 ac 78 c0 a8 01 11 c0 a8  .<.....X.....
0020  01 01 08 00 4d 48 00 01 00 13 61 62 63 64 65 66  ....MH....abcdef
0030  67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76  ghijklmnopqrestuv
0040  77 61 62 63 64 65 66 67 68 69                    wabcdeefghi
```

1 TOP – Packet List



- Shows the list of packets captured by Wireshark.
- Each row is one packet.
- You can see the source, destination, protocol, length and a short description (Info).

Which packets were captured?

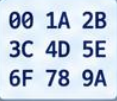
2 MIDDLE – Packet Details



- Shows detailed information about the selected packet.
- You can expand each layer (e.g. Ethernet, IP, ICMP) to see the fields and values.
- Helps you understand what is inside the packet.

What is inside the selected packet?

3 BOTTOM – Packet Bytes



- Shows the raw data of the selected packet in hexadecimal (left) and ASCII (right).
- You can see the exact bytes that were transmitted on the network.
- Useful for low-level analysis and troubleshooting.

Raw data in hexadecimal / ASCII



Simple way to remember:

TOP
= Choose the packet

MIDDLE
= Understand the packet

BOTTOM
= See the raw bytes

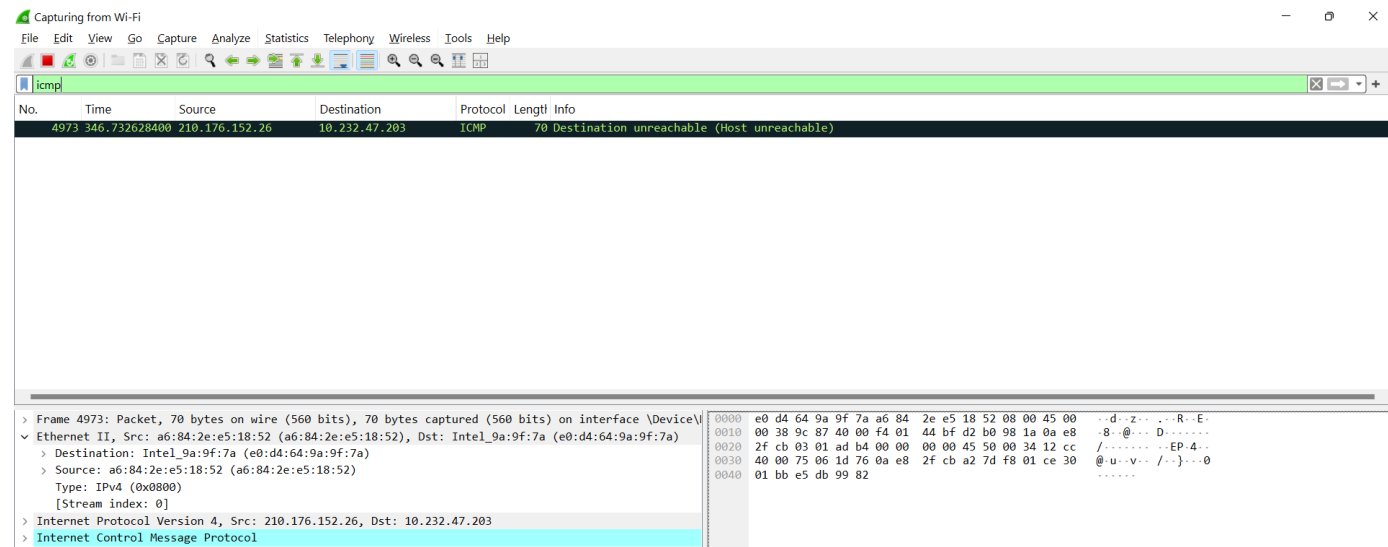
Step 5 – Display Only ICMP Traffic

Keep the capture running.

At the top of Wireshark, click the Display Filter box and type:

icmp

Press Enter. The packet list may now become empty. That is OK. We have not generated our ping yet.



Why?

ICMP is the protocol used by the ping command. The filter makes it easier to see only the ping packets.

Important:

Wireshark is still capturing all traffic.

The display filter only changes what you can see.

This prevents students from thinking something has gone wrong if the screen becomes empty.

Step 6 – Generate the Ping Traffic

While Wireshark is still capturing, return to Command Prompt.

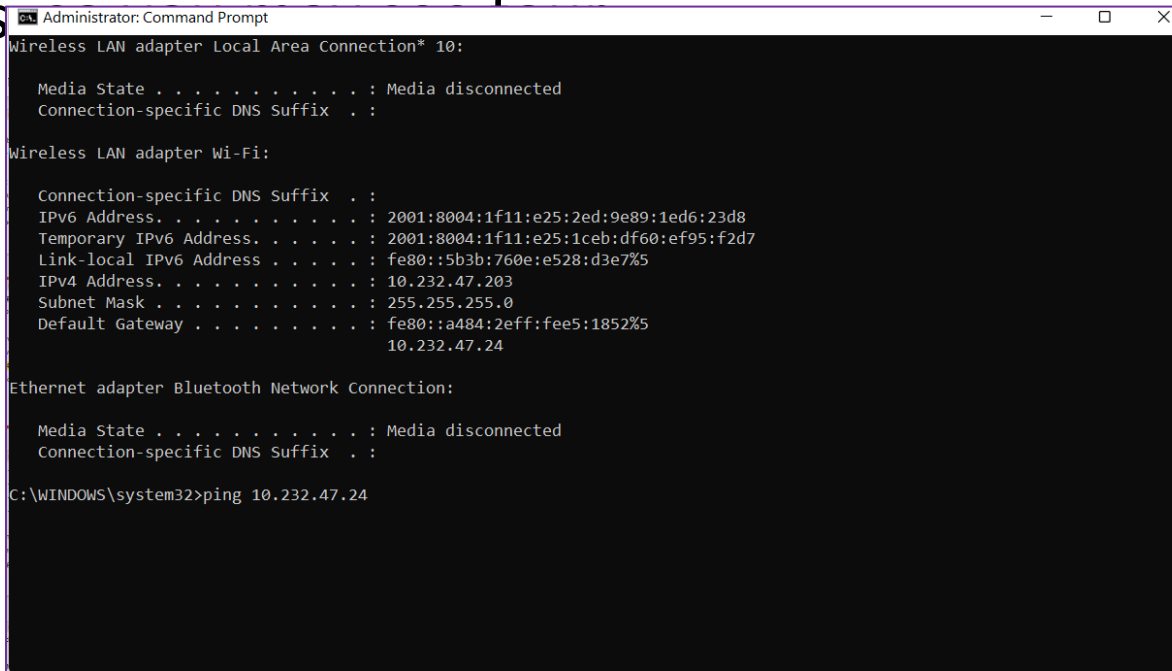
Ping the default gateway you recorded in Step 1:

ping <your-default-gateway> For example: **ping 10.232.47.24**

Windows normally

sends four Echo Requests

replies:



```
Administrator: Command Prompt

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8
    Temporary IPv6 Address. . . . . : 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7
    Link-local IPv6 Address . . . . . : fe80::5b3b:760e:e528:d3e7%5
    IPv4 Address. . . . . : 10.232.47.203
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                                10.232.47.24

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\WINDOWS\system32>ping 10.232.47.24
```

Why are we doing this?

We are deliberately creating known network traffic so that we can capture it and examine exactly what happens when a PC communicates with its gateway.

```
Administrator: Command Prompt

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
IPv6 Address. . . . . : 2001:8004:1f11:e25:2ed:9e89:1ed6:23d8
Temporary IPv6 Address. . . . . : 2001:8004:1f11:e25:1ceb:df60:ef95:f2d7
Link-local IPv6 Address . . . . . : fe80::5b3b:760e:e528:d3e7%5
IPv4 Address. . . . . : 10.232.47.203
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::a484:2eff:fee5:1852%5
                          10.232.47.24

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 

C:\WINDOWS\system32>ping 10.232.47.24

Pinging 10.232.47.24 with 32 bytes of data:
Reply from 10.232.47.24: bytes=32 time=4ms TTL=64
Reply from 10.232.47.24: bytes=32 time=14ms TTL=64
Reply from 10.232.47.24: bytes=32 time=4ms TTL=64
Reply from 10.232.47.24: bytes=32 time=6ms TTL=64

Ping statistics for 10.232.47.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 14ms, Average = 7ms

C:\WINDOWS\system32>
```


Step 7 – Find the Ping Packets in Wireshark

Return to Wireshark.

You should see pairs of ICMP packets:

PC → Gateway Echo (ping) request

Gateway → PC Echo (ping) reply

PC → Gateway Echo (ping) request

Gateway → PC Echo (ping) reply

...

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

Windows normally sends four ping requests, so if all four succeed, you will normally see:

4 Echo Requests

+

4 Echo Replies

=

8 ICMP packets

In simple terms:

PC: Are you there?

Gateway: Yes, I am.

That is very easy for introductory students to understand.

Preparation and Guideline for Lab 3

Step 9 – Select the First Echo Request

In the top Packet List pane, click the first:

Echo (ping) request

Now examine the middle Packet Details pane.

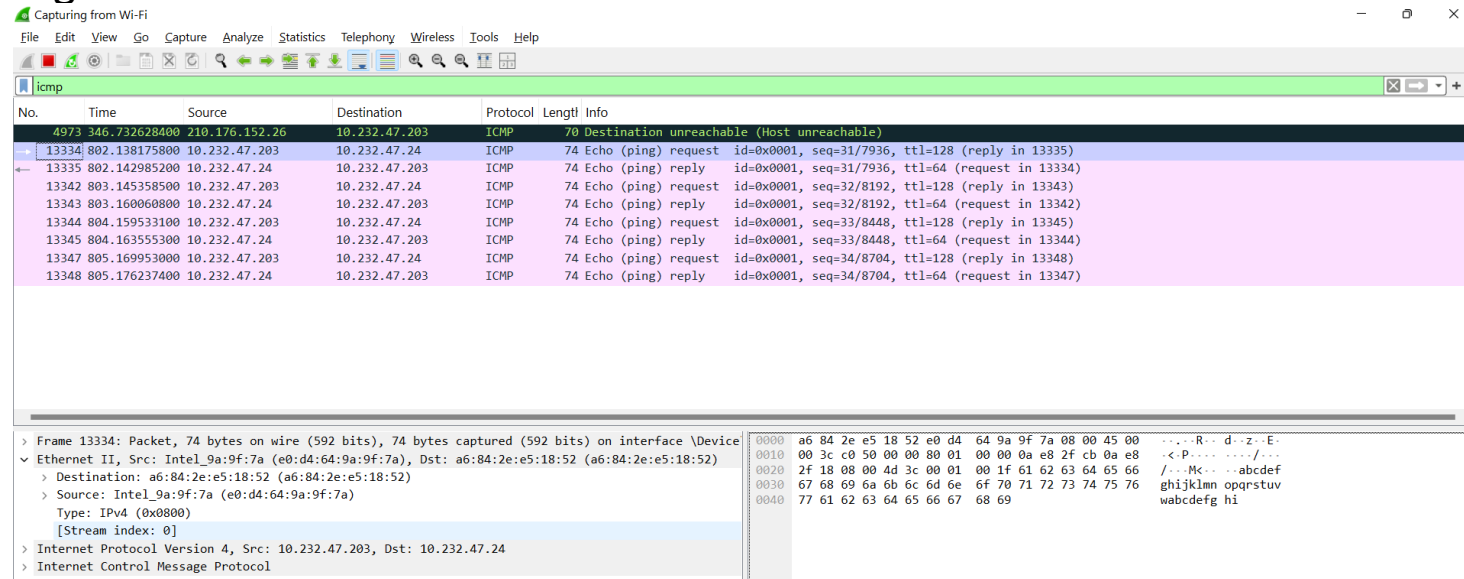
You should see layers similar to:

Frame

Ethernet II

Internet Protocol Version 4

Internet Control Message Protocol



The image shows a Wireshark packet capture interface. The top pane displays a list of captured packets. The middle pane shows the details of the selected packet (Frame 13334), which is an ICMP Echo (ping) request. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{...} [ethernet II]

Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Destination: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

Source: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)

Type: IPv4 (0x0800)

[Stream index: 0]

Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24

Internet Control Message Protocol

0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00 ...R...d...z...E...
0010 00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8 ...<P.../...
0020 2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66 /...M<... ..abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 ..wbcdefg hi

Preparation and Guideline for Lab 3

What are students seeing?

This is a very useful point:

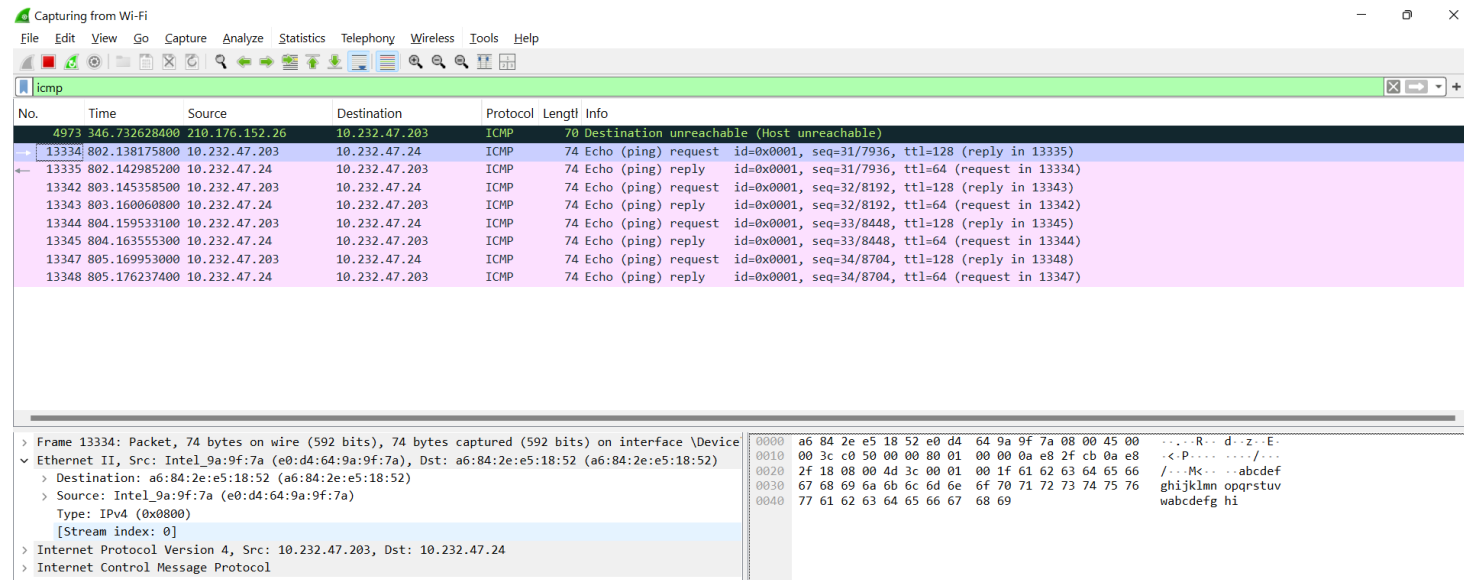
One ping request contains several protocol layers.

Ethernet II → carries the frame using MAC addresses

IPv4 → identifies source and destination using IP addresses

ICMP → contains the actual ping request

“We clicked one packet at the top. The middle pane now lets us open that packet and see what is inside it.”



Wireshark interface showing a packet capture from Wi-Fi. The packet list pane displays several ICMP Echo (ping) request and reply packets. The packet details pane shows the structure of a selected packet (Frame 13334).

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device...
Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
Destination: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
Source: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24
Internet Control Message Protocol

This shows that one captured communication contains information from several protocol layers.

Remember:

TOP → Which packet?

MIDDLE → What is inside it?

BOTTOM → What are the actual bytes?

This directly connects the activity to your figure.

Step 10 – Examine Ethernet II

Expand: Ethernet II

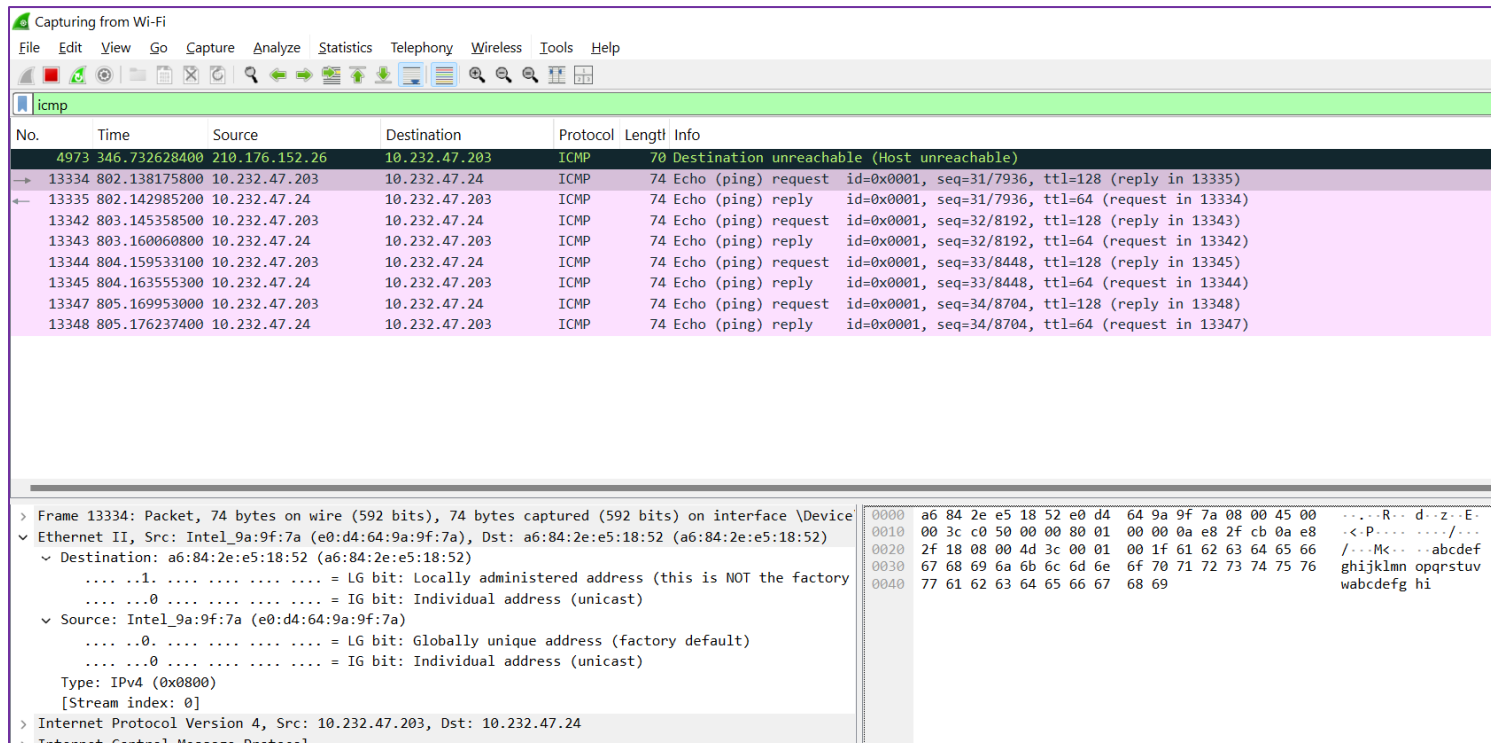
Look for:

Destination

Source

Type

For example:



Capturing from Wi-Fi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
→ 13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
← 13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

> Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device NPF{...}

▼ Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

- ▼ Destination: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
 -1. = LG bit: Locally administered address (this is NOT the factory address)
 -0. = IG bit: Individual address (unicast)
- ▼ Source: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)
- Type: IPv4 (0x0800)
- [Stream index: 0]

> Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24

> Internet Control Message Protocol, Seq: 31, Len: 74

0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00R..d..z..E..

0010 00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8 ..<P...../...

0020 2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66 /...M<...abcdef

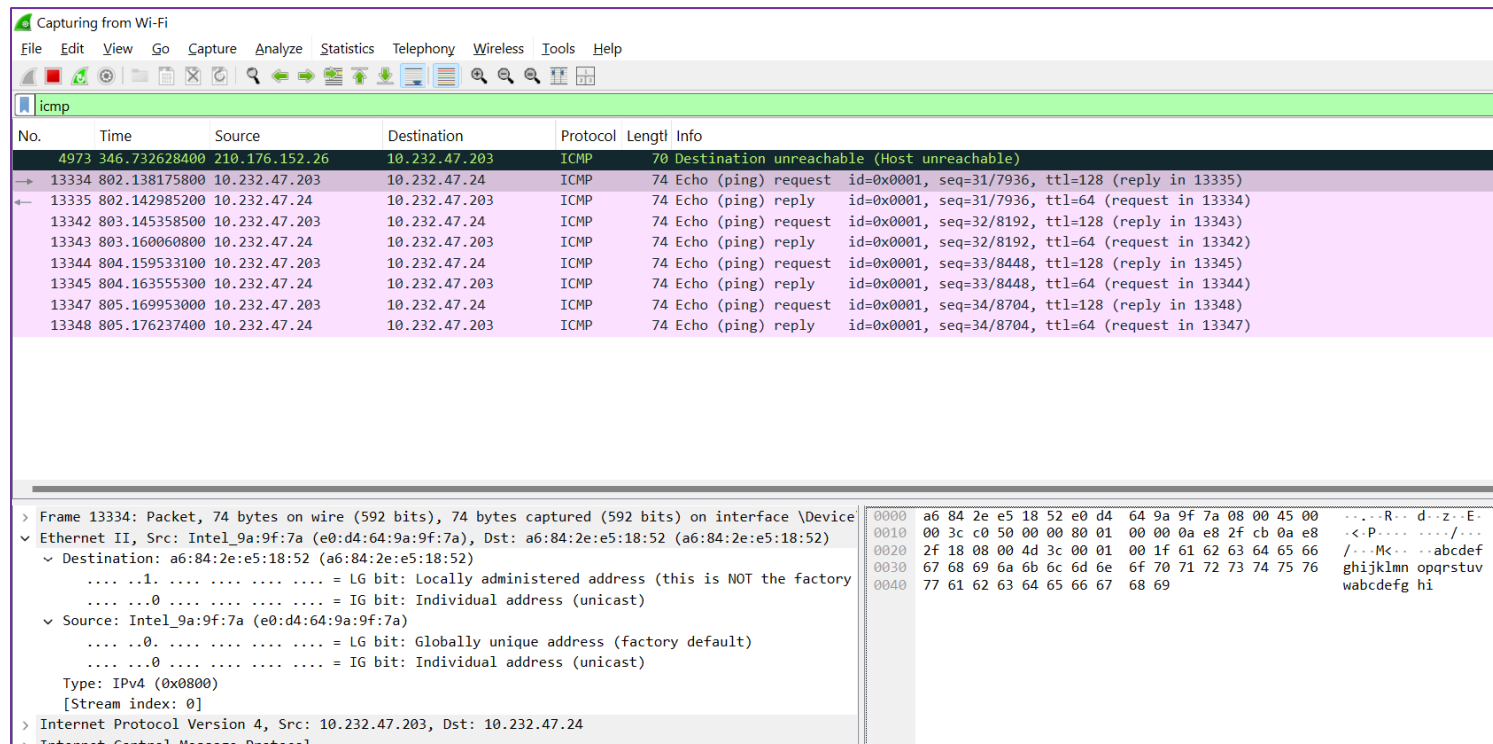
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv

0040 77 61 62 63 64 65 66 67 68 69 ..wabcdfgh i

What does this tell us?

The Ethernet header tells us about Layer 2 communication.

The source MAC is the device sending the frame.



Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
→ 13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
← 13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device...

Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

- Destination: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
 - ...1. = LG bit: Locally administered address (this is NOT the factory)
 - ...0. = IG bit: Individual address (unicast)
- Source: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
 - ...0. = LG bit: Globally unique address (factory default)
 - ...0. = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)
[Stream index: 0]

Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24

0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00 ...R...d...z...E-
0010 00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8 ...<P.../...
0020 2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66 /...M<... ..abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabdefgh hi

Preparation and Guideline for Lab 3

What does this tell us?

The Ethernet II header shows the Layer 2 communication for this ping packet.

Source MAC:

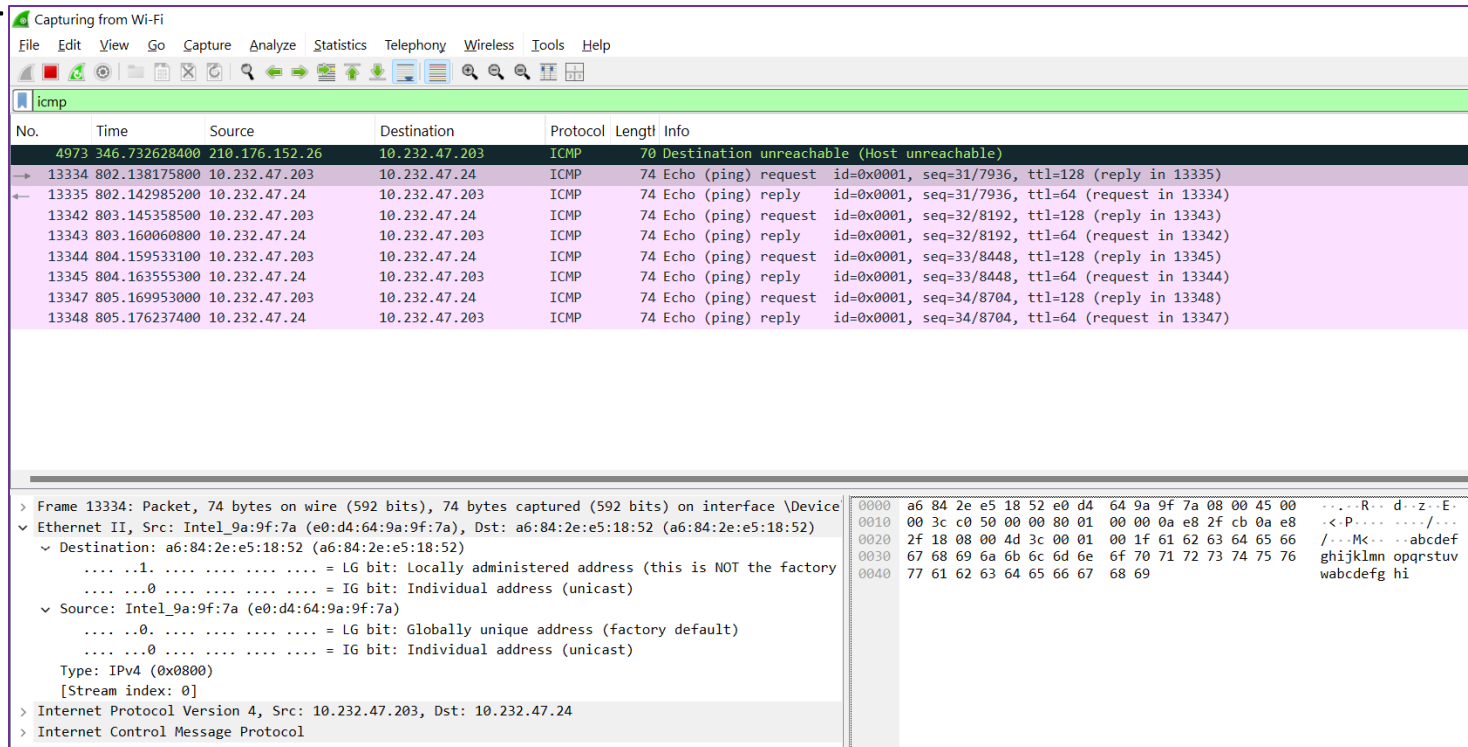
e0:d4:64:9a:9f:7a

Destination MAC:

a6:84:2e:e5:18:52

Type:

IPv4 (0x0800)



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icmp

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
→ 13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
← 13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

> Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device
✓ Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
 Destination: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
 ...1. = LG bit: Locally administered address (this is NOT the factory
 ...0. = IG bit: Individual address (unicast)
 Source: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
 ...0. = LG bit: Globally unique address (factory default)
 ...0. = IG bit: Individual address (unicast)
 Type: IPv4 (0x0800)
 [Stream index: 0]
 Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24
 Internet Control Message Protocol

0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00R..d..z..E.
0010 00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8 -<-P...../...
0020 2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66 /...M<...abcde
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrst
0040 77 61 62 63 64 65 66 67 68 69 wabcdegh i

Preparation and Guideline for Lab 3

What does this tell us?

The Ethernet II header tells us how the frame is being delivered at Layer 2.

- Source MAC → MAC address of the device sending the frame
- Destination MAC → MAC address of the device receiving the frame on the local network
- Type: IPv4 (0x0800) → tells Ethernet that the data carried inside this frame is an IPv4 packet

For this Echo Request:

PC

e0:d4:64:9a:9f:7a

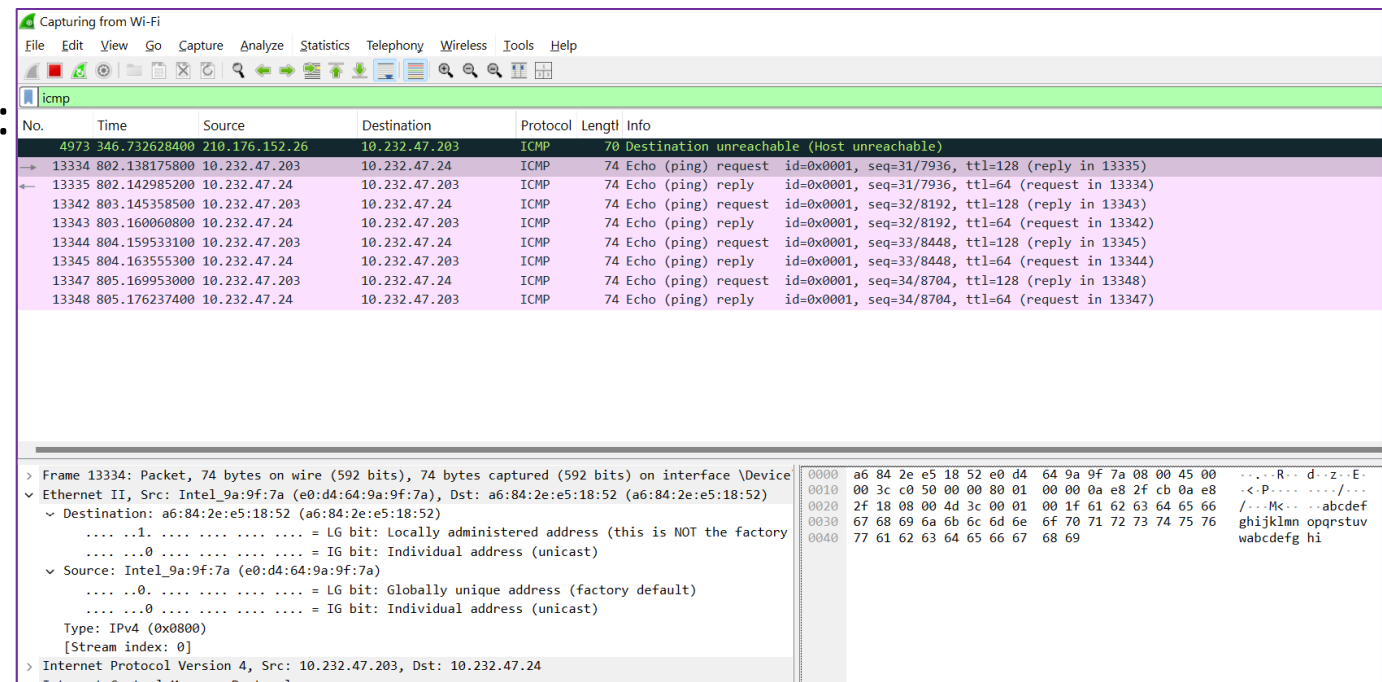
|

| Ethernet frame



Gateway

a6:84:2e:e5:18:52



Capturing from Wi-Fi

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
→ 13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
← 13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

> Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device NPF{...}

✓ Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)

- ✓ Destination: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
 - 01 = LG bit: Locally administered address (this is NOT the factory address)
 - 00 = IG bit: Individual address (unicast)
- ✓ Source: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
 - 00 = LG bit: Globally unique address (factory default)
 - 00 = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)
[Stream index: 0]

> Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24

0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00R..d..z..E..
0010 00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8 ..<P...../...
0020 2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66 /...Mk...abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabcdefg hi

Preparation and Guideline for Lab 3

Ethernet II

Source: e0:d4:64:9a:9f:7a

Destination: a6:84:2e:e5:18:52

Type: IPv4 (0x0800)

↓ contains

IPv4

Source: 10.232.47.203

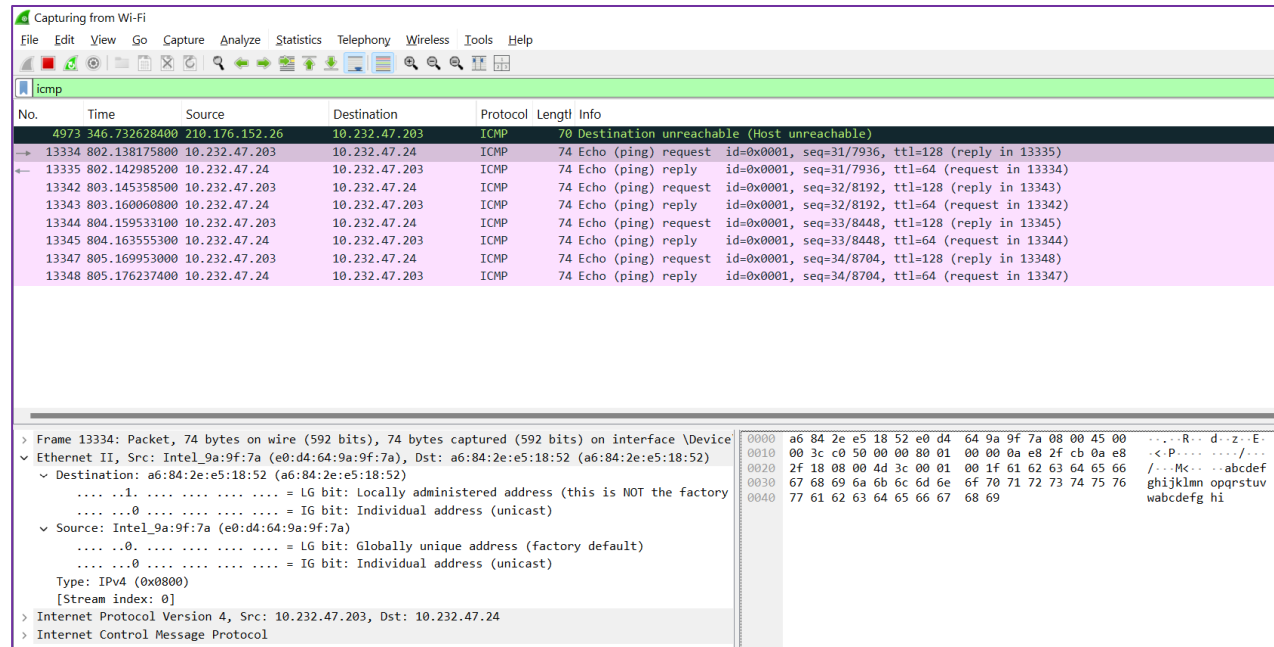
Destination: 10.232.47.24

↓ contains

ICMP

Echo (ping) request

That is an excellent point to emphasize: the ping is ICMP, carried inside an IPv4 packet, which is carried inside an Ethernet frame.



Capturing from Wi-Fi

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

> Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device...
Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
Destination: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
...1. = LG bit: Locally administered address (this is NOT the factory...
...0. = IG bit: Individual address (unicast)
Source: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
...0. = LG bit: Globally unique address (factory default)
...0. = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
> Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24
> Internet Control Message Protocol

Offset	Hex	ASCII
0000	a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00	R...d...E
0010	00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8	...<P... .. /...
0020	2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66	/...M<... ..abcdef
0030	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76	ghijklmn opqrstuv
0040	77 61 62 63 64 65 66 67 68 69	wabcdefg hi

The source MAC identifies the local interface sending the Ethernet frame. The destination MAC identifies the next device receiving the frame on the local network. When the PC sends traffic outside its local network, this will normally be the MAC address of the default gateway, not the MAC address of the final remote server.

Step 11 – Record the Layer 2 Addresses

Using the Ethernet II information from the Echo Request, record:

PC MAC address:

Default gateway MAC address:

PC MAC address:

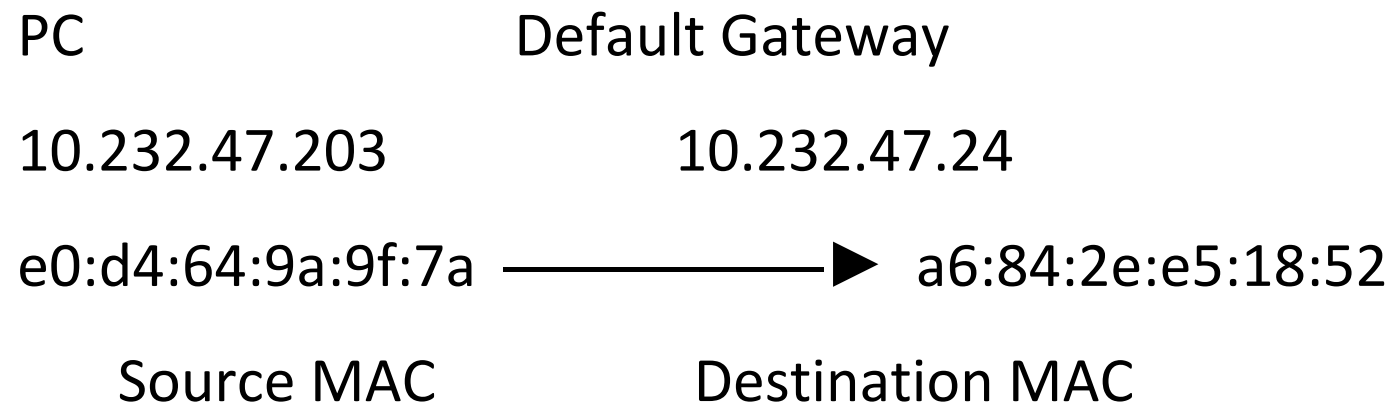
e0:d4:64:9a:9f:7a

Default gateway MAC address:

a6:84:2e:e5:18:52

How do we know?

Look at the selected Echo Request:



Because this is a ping request from your PC to the default gateway:

- Source MAC = your PC's MAC address
- Destination MAC = default gateway's MAC address

Why are we recording these?

I would add this short section to the slide:

Why?

This shows which devices are communicating at Layer 2.

The IP addresses identify the devices at Layer 3, while the MAC addresses are used to deliver the Ethernet frame on the local network.

For an Echo Request sent from your PC to its gateway:

Source MAC = PC

Destination MAC = Default Gateway

Do not copy the example MAC addresses from this guide.

Record the addresses shown in your own capture.

Step 12 – Examine IPv4

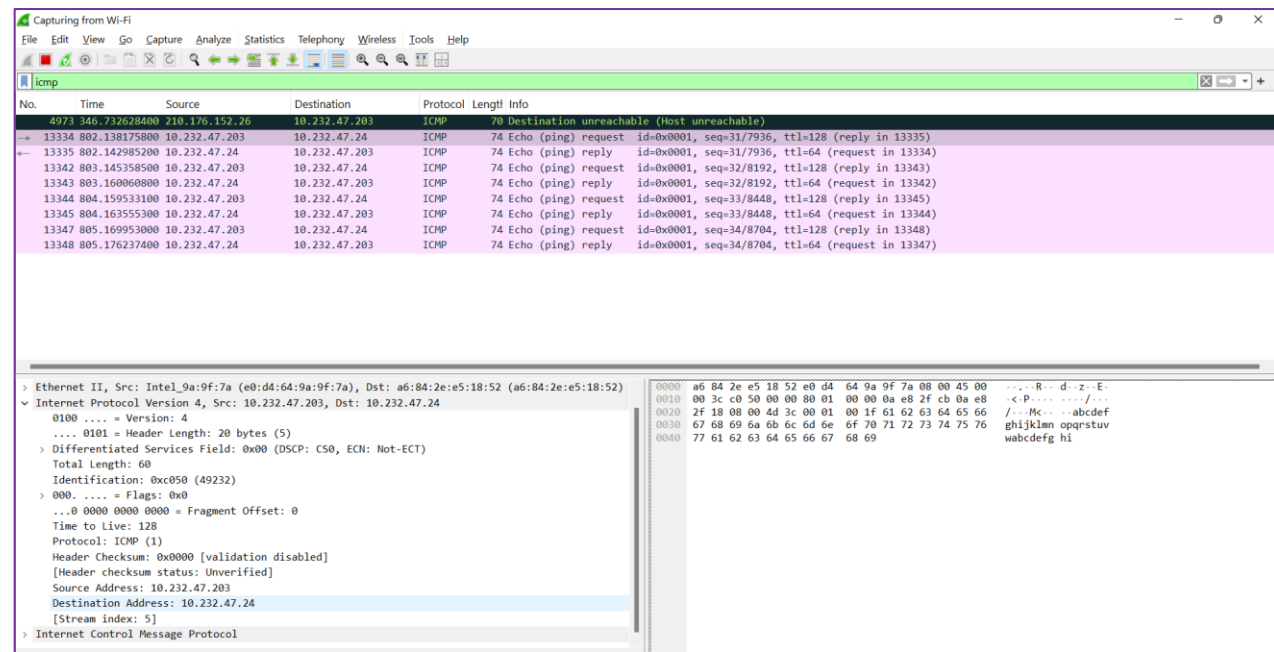
Expand:

Internet Protocol Version 4

Find:

Source Address

Destination Address

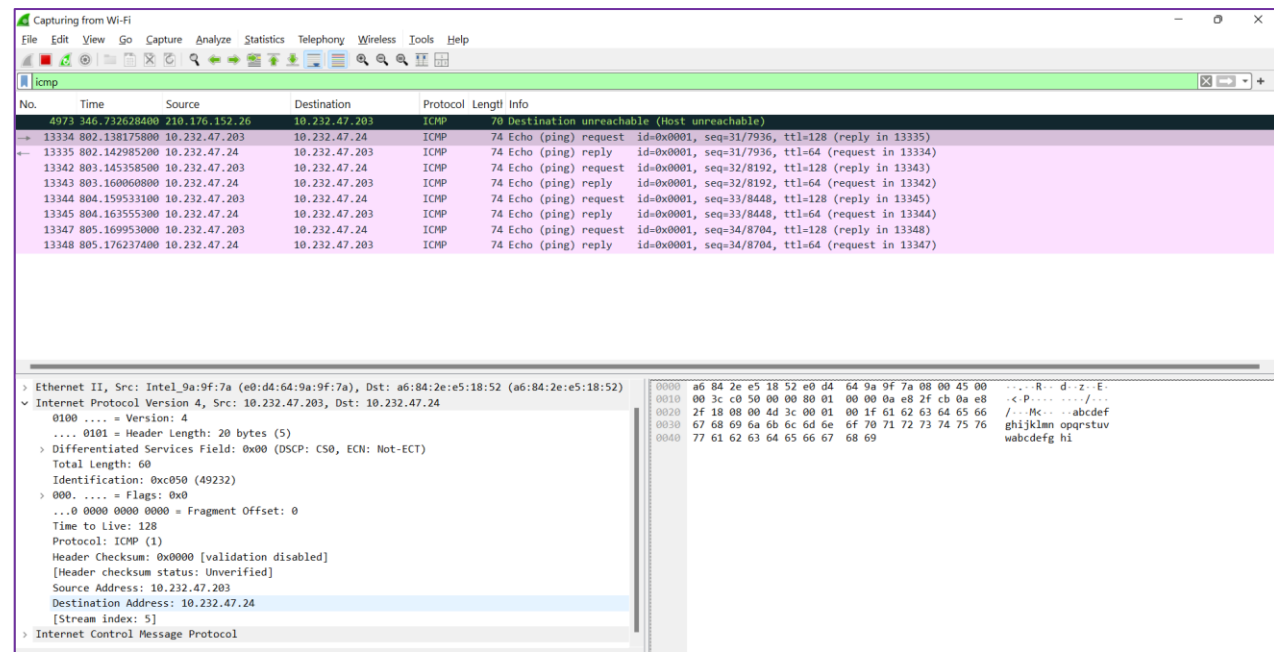


Preparation and Guideline for Lab 3

For a ping from the PC to its gateway:

Source IP = your PC

Destination IP = your default gateway



Now compare Ethernet II and IPv4:

Ethernet II → MAC addresses → Layer 2

Used to deliver the frame on the local network.

IPv4 → IP addresses → Layer 3

Used to identify the source and destination across IP networks.

Key Idea

The same communication contains both MAC addresses and

IP addresses because they have different jobs:

Layer 2 – Ethernet II

PC MAC —————▶ Gateway MAC

e0:d4:64:9a:9f:7a a6:84:2e:e5:18:52

Layer 3 – IPv4

PC IP —————▶ Gateway IP

10.232.47.203 10.232.47.24

Simple way to remember:

MAC = local delivery

IP = network addressing/routing

One small technical wording improvement: rather than saying “*the same packet contains both types*”, I would say:

The same network communication uses both types of addressing because they perform different jobs.

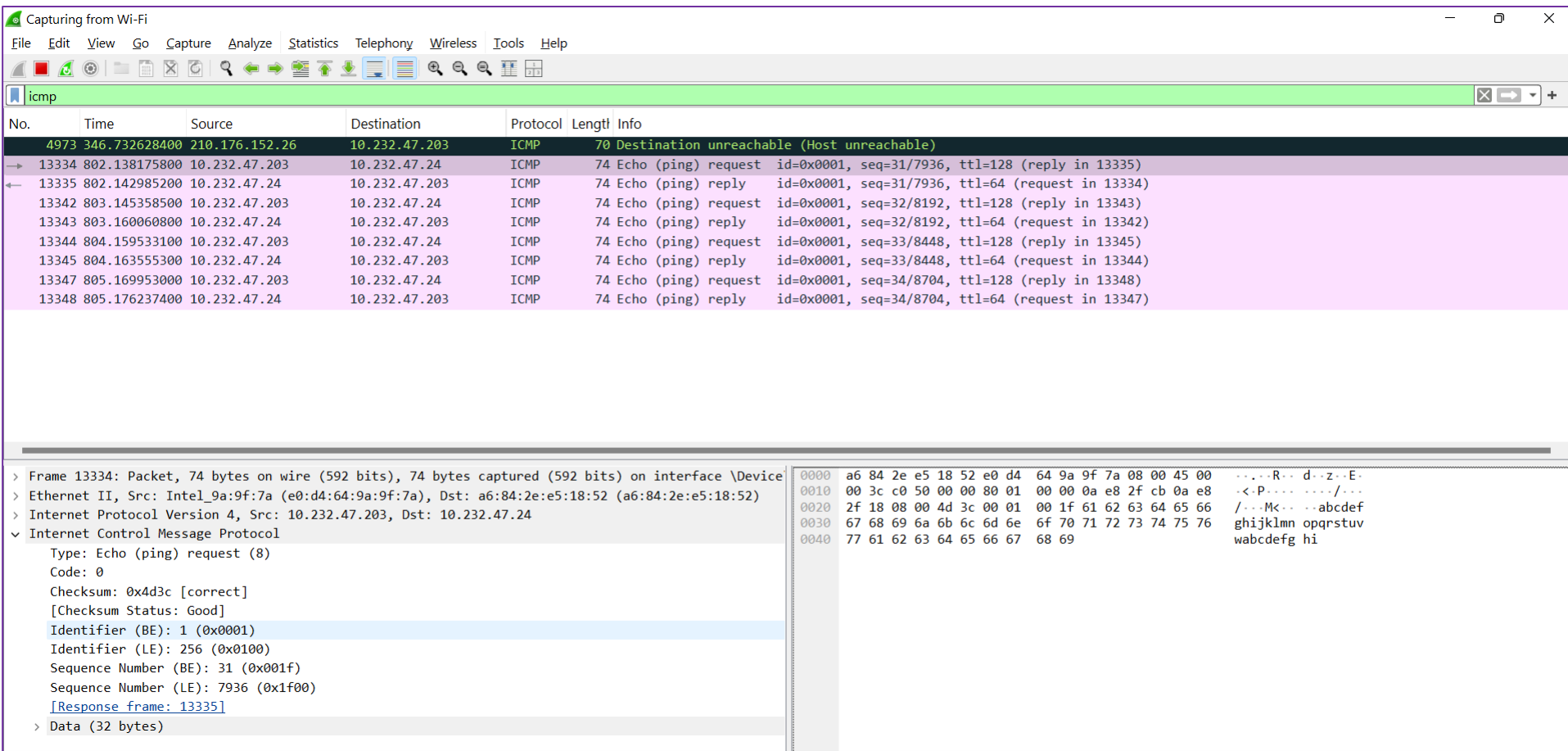
Or, if you want to introduce encapsulation accurately:

The Ethernet frame contains an IPv4 packet, so we can see both MAC addresses and IP addresses in the captured data.

Step 13 – Examine ICMP

Expand: Internet Control Message Protocol

For the request, look for: Type: 8 (Echo (ping) request)



The image shows a Wireshark packet capture window titled "Capturing from Wi-Fi". The packet list on the left shows several ICMP Echo (ping) requests and replies. The selected packet is packet 13334, which is an ICMP Echo (ping) request. The packet details pane on the right shows the structure of the ICMP Echo (ping) request, including the Identifier (BE), Identifier (LE), Sequence Number (BE), and Sequence Number (LE). The packet bytes pane on the right shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and the ICMP Echo (ping) request data.

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

> Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device...
> Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
> Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24
v Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d3c [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 31 (0x001f)
Sequence Number (LE): 7936 (0x1f00)
[Response frame: 13335]
> Data (32 bytes)

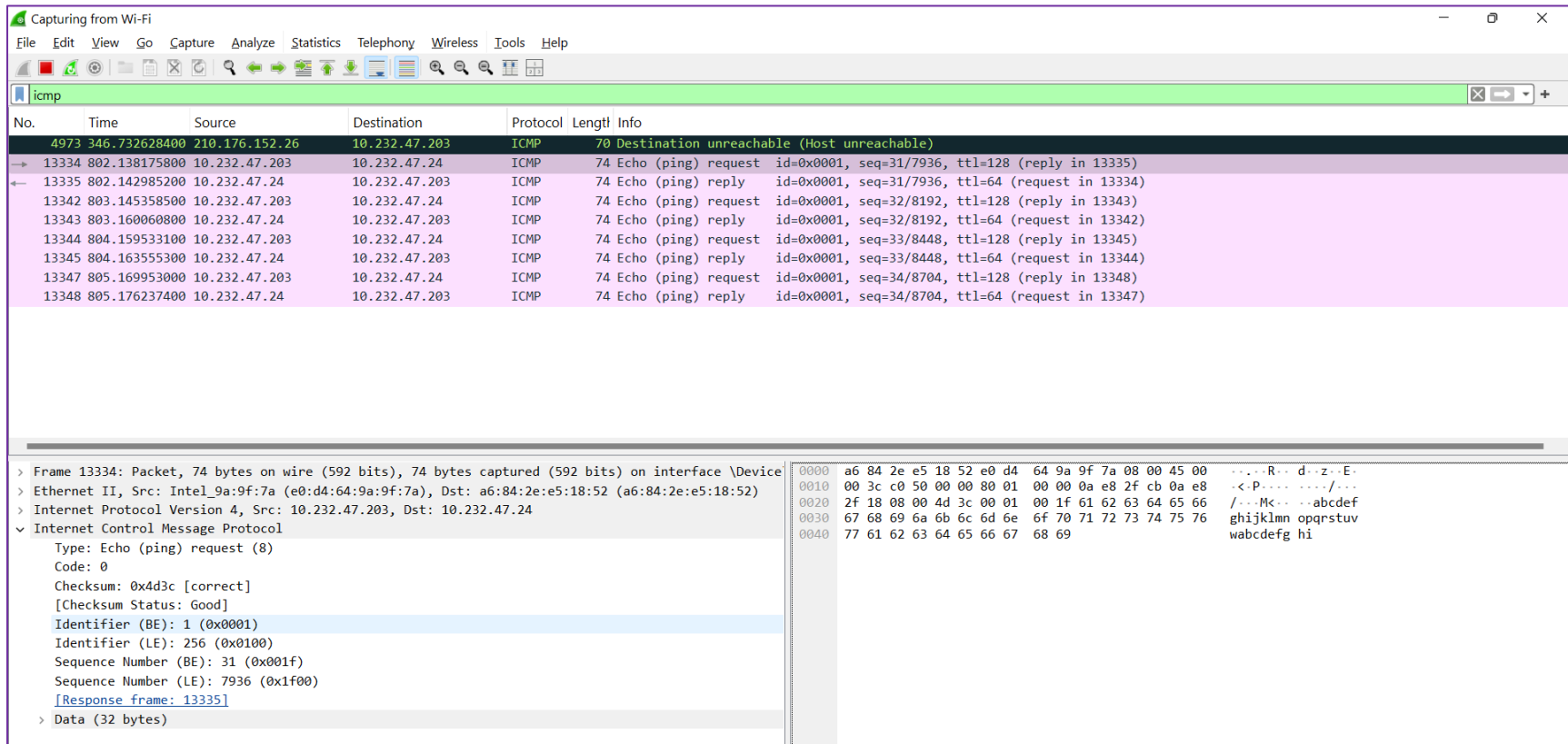
0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00 ...R.. d..z..E..
0010 00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8 ..<.P..... /...
0020 2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66 /...M<... ..abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabcedfg hi

Code: 0

Type 8 means this is an ICMP Echo Request.

In simple terms, the PC is asking the gateway:

“Are you reachable?”



The screenshot shows a Wireshark packet capture window titled "Capturing from Wi-Fi". The packet list on the left shows several ICMP Echo (ping) requests and replies. The selected packet is packet 13334, which is an ICMP Echo (ping) request from 10.232.47.203 to 10.232.47.24. The packet details pane on the right shows the following information:

- Frame 13334: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF...
- Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
- Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24
- Internet Control Message Protocol
 - Type: Echo (ping) request (8)
 - Code: 0
 - Checksum: 0x4d3c [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 1 (0x0001)
 - Identifier (LE): 256 (0x0100)
 - Sequence Number (BE): 31 (0x001f)
 - Sequence Number (LE): 7936 (0x1f00)
 - [\[Response frame: 13335\]](#)
- Data (32 bytes)

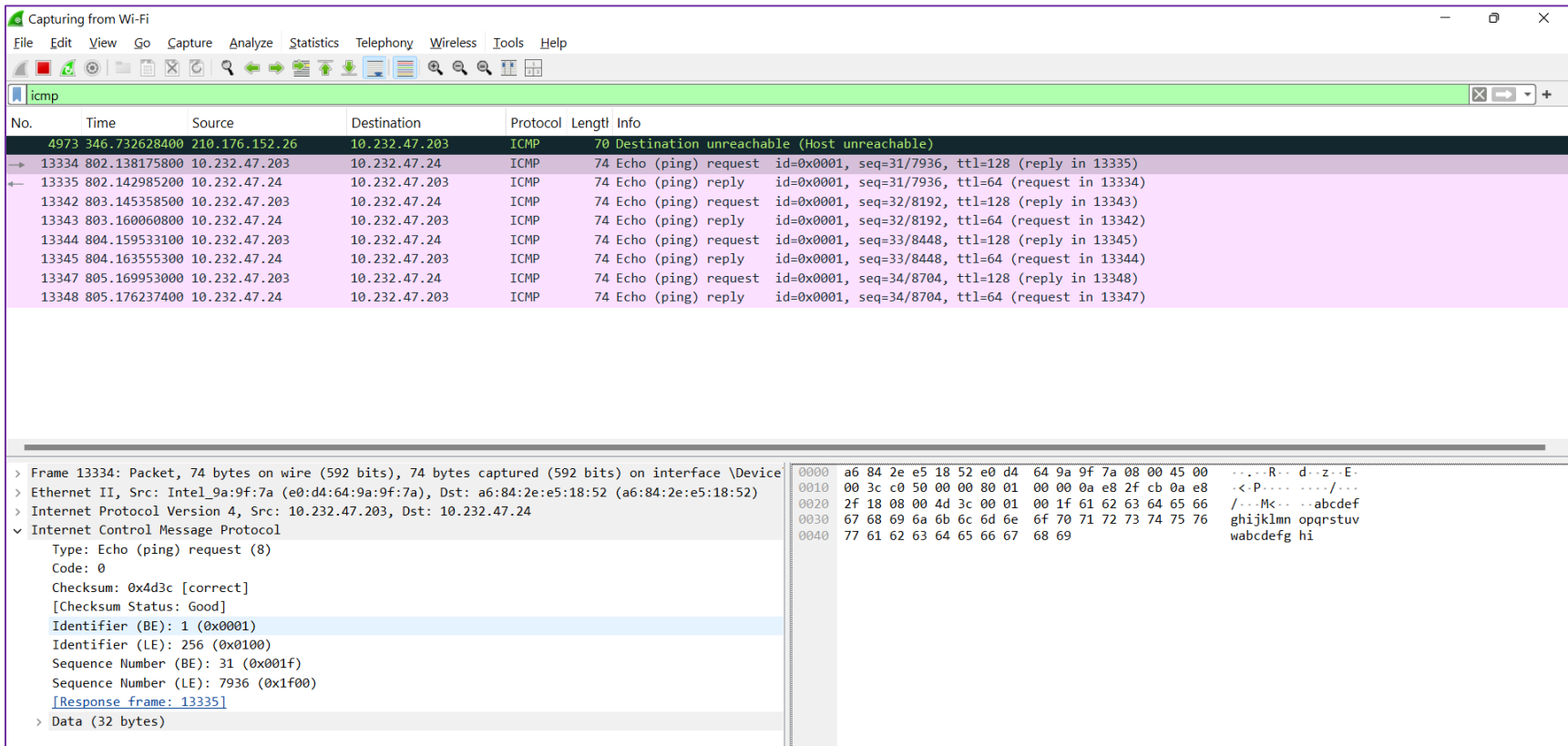
The packet bytes pane on the right shows the raw data of the packet, including the ICMP Echo request header and the data payload.

Step 14 – Examine the Echo Reply

Select the corresponding: Echo (ping) reply

Look at ICMP: Type: 0 (Echo (ping) reply)

Now compare:



Wireshark packet capture showing ICMP Echo (ping) request and reply. The packet list shows a request (No. 13334) and a reply (No. 13335). The packet details for the selected packet (No. 13335) show the ICMP Echo (ping) reply details.

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
→ 13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
← 13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

Frame 13335: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device...
Ethernet II, Src: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a), Dst: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52)
Internet Protocol Version 4, Src: 10.232.47.203, Dst: 10.232.47.24
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d3c [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 31 (0x001f)
Sequence Number (LE): 7936 (0x1f00)
[Response frame: 13335]
Data (32 bytes)

0000 a6 84 2e e5 18 52 e0 d4 64 9a 9f 7a 08 00 45 00 ...R..d..z..E..
0010 00 3c c0 50 00 00 80 01 00 00 0a e8 2f cb 0a e8 ...<P...../...
0020 2f 18 08 00 4d 3c 00 01 00 1f 61 62 63 64 65 66 /...M<...abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabdefgh hi

What does this tell us?

Type 8 means this packet is an ICMP Echo Request.

In simple terms:

Your PC (10.232.47.203)

|

| ICMP Echo Request

| Type = 8



Gateway (10.232.47.24)

"Are you reachable?"

Ping uses ICMP. Type 8 is the question: "Are you reachable?"

Examine the ICMP Echo Reply

Then go back to the top Packet List.

Immediately below packet 13334, you have:

13335 10.232.47.24 → 10.232.47.203

ICMP Echo (ping) reply

Single-click packet 13335. Then expand: > Internet Control Message Protocol

You should find: Type: 0 (Echo (ping) reply) and Code: 0

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

icmp

No.	Time	Source	Destination	Protocol	Length	Info
4973	346.732628400	210.176.152.26	10.232.47.203	ICMP	70	Destination unreachable (Host unreachable)
→ 13334	802.138175800	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=31/7936, ttl=128 (reply in 13335)
← 13335	802.142985200	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=31/7936, ttl=64 (request in 13334)
13342	803.145358500	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=32/8192, ttl=128 (reply in 13343)
13343	803.160060800	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=32/8192, ttl=64 (request in 13342)
13344	804.159533100	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 13345)
13345	804.163555300	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 13344)
13347	805.169953000	10.232.47.203	10.232.47.24	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 13348)
13348	805.176237400	10.232.47.24	10.232.47.203	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 13347)

> Frame 13335: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device
> Ethernet II, Src: a6:84:2e:e5:18:52 (a6:84:2e:e5:18:52), Dst: Intel_9a:9f:7a (e0:d4:64:9a:9f:7a)
> Internet Protocol Version 4, Src: 10.232.47.24, Dst: 10.232.47.203
> Internet Control Message Protocol
Type: Echo (ping) reply (0)
Code: 0
Checksum: 0x553c [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 31 (0x001f)
Sequence Number (LE): 7936 (0x1f00)
[Request frame: 13334]
[Response time: 4.809 ms]
> Data (32 bytes)

0000e0 d4 64 9a 9f 7a a6 84 2e e5 18 52 08 00 45 00 ..d..z...R..E
001000 3c 7a 6b 00 00 40 01 8b a3 0a e8 2f 18 0a e8 -<zk..@..../...
00202f cb 00 00 55 3c 00 01 00 1f 61 62 63 64 65 66 /...U<...abcdef
003067 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
004077 61 62 63 64 65 66 67 68 69 wabcdfgh i

What does this tell us?

Type 0 means this is an ICMP Echo Reply.

In simple terms:

PC Gateway

10.232.47.203 10.232.47.24

ICMP Echo Request

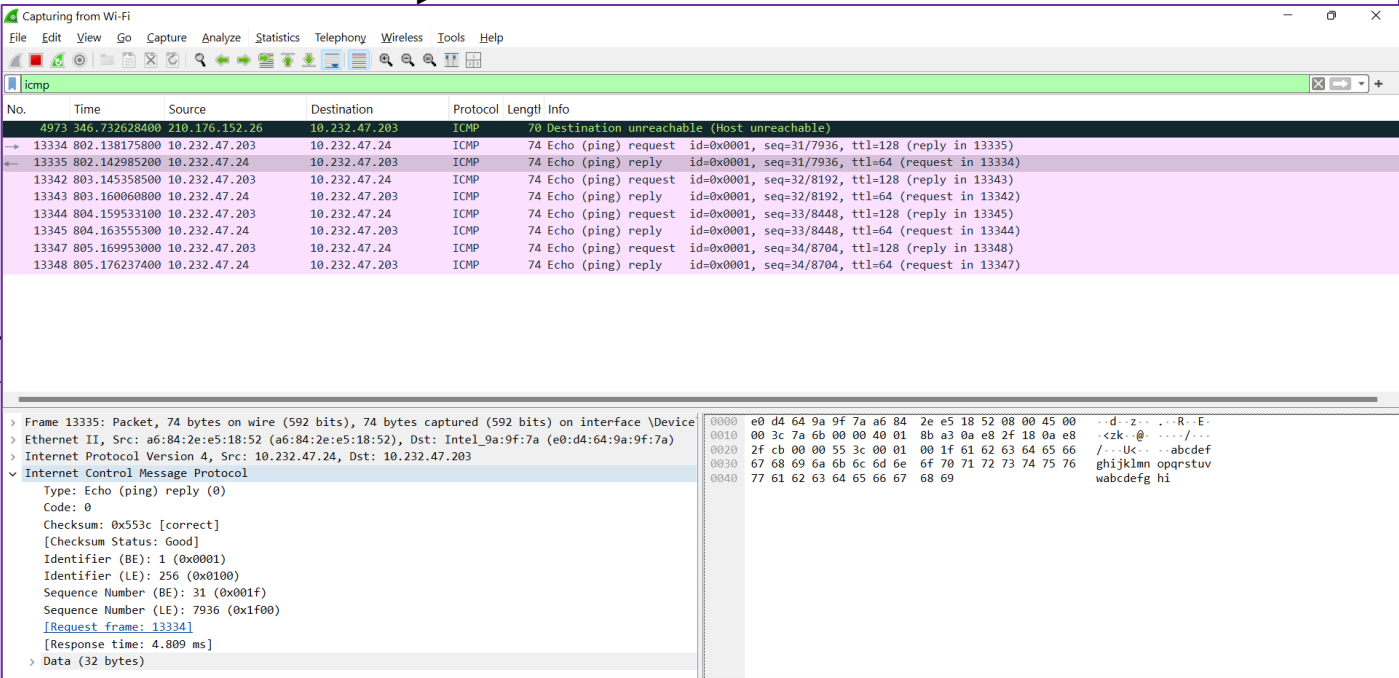
Type 8

"Are you reachable?"

ICMP Echo Reply

Type 0

"Yes, I am."



Now compare the two packets

This would make a very useful slide:

	Echo Request	Echo Reply
ICMP Type	8	0
Source IP	10.232.47.203	10.232.47.24
Destination IP	10.232.47.24	10.232.47.203
Meaning	Are you reachable?	Yes, I am.

Key idea:

Type 8 = Echo Request → PC asks

Type 0 = Echo Reply → Gateway answers

Notice also that the source and destination IP addresses reverse in the reply. That is excellent evidence for students that packet 13334 is going PC → gateway, while packet 13335 is coming gateway → PC.

Step 15 – Understand Encapsulation

Wireshark shows the captured communication as a set of nested protocol layers:

Ethernet II → IPv4 → ICMP → Data

This helps us see the result of encapsulation.

When data is prepared for transmission, each relevant layer adds the information it needs:

ICMP information → IPv4 header added → Ethernet header added → Frame transmitted across the local network

Think of sending a parcel:

Message → Put inside a package → Add addressing information → Deliver it

Main idea:

Encapsulation allows information from different networking layers to travel together in the transmitted frame.

This is more technically accurate than simply saying that Ethernet II → IPv4 → ICMP → Data “is encapsulation.”

Understand Encapsulation

Keep the Echo (ping) reply selected.

In the Packet Details pane, look at the protocol sections:

Frame

Ethernet II

Internet Protocol Version 4

Internet Control Message Protocol

Data (32 bytes)

You can expand or collapse each section using the small arrow beside it.

What are we seeing?

Wireshark is showing the different pieces of information carried in the captured frame:

Ethernet II → IPv4 → ICMP → Data

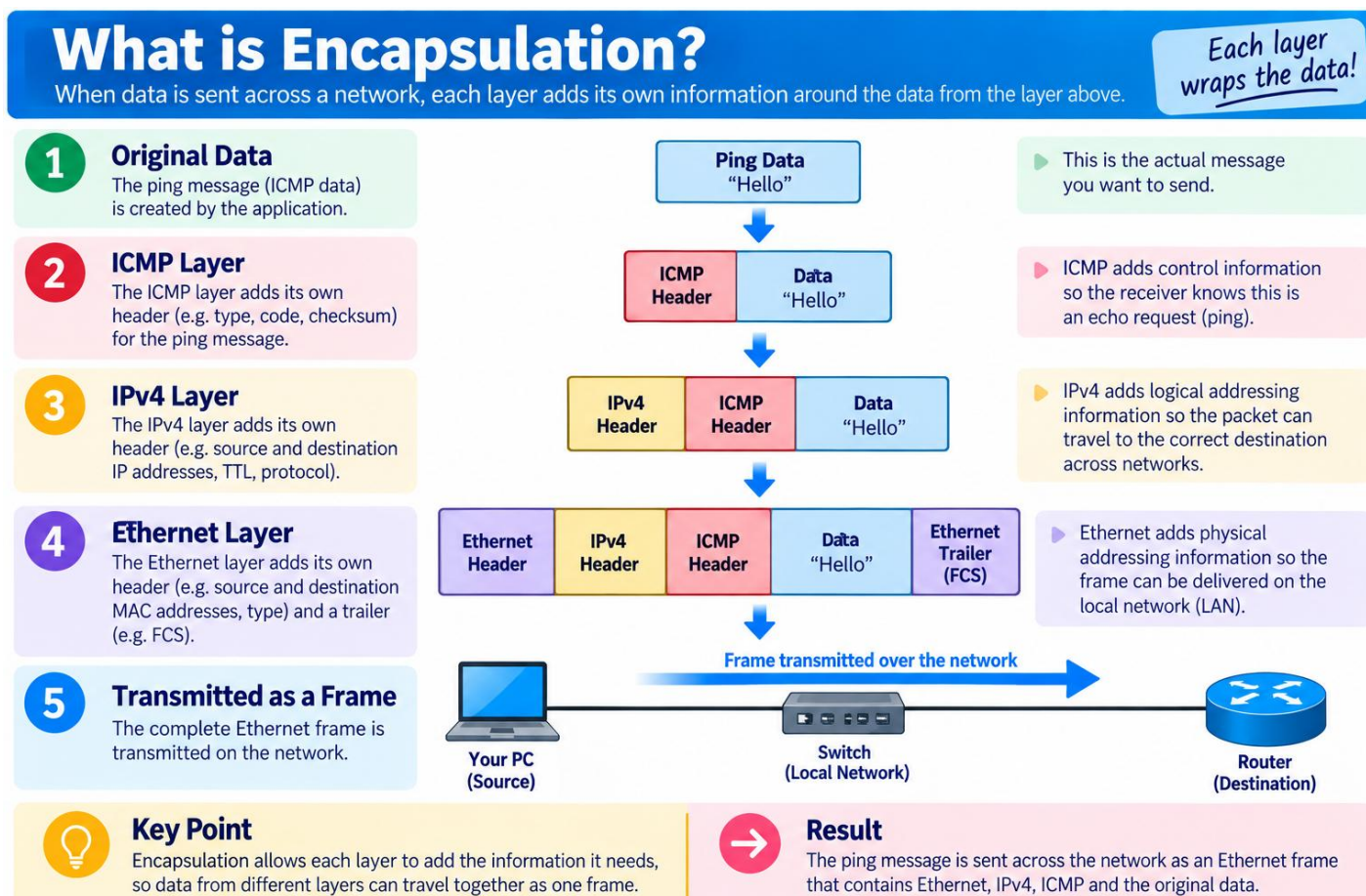
Each protocol has a different job:

Each protocol has a different job:

Layer / Protocol	What it provides in this example
Ethernet II	Source and destination MAC addresses
IPv4	Source and destination IP addresses
ICMP	Ping information such as Echo Request / Echo Reply
Data	The 32 bytes of ping data

So what is encapsulation?

When the ping is prepared for transmission, information is wrapped with the information required by the lower layers:



because the Wireshark hierarchy is showing us the result/structure produced by encapsulation, while encapsulation itself is the process of adding the appropriate protocol information as data moves down the networking stack.

Simple parcel analogy

You can then use your analogy:

Message → Put the message inside a package → Add addressing/delivery information → Send the package

Similarly:

Ping data → ICMP information → IPv4 header → Ethernet header → Frame transmitted

Main idea

I would put this in a highlighted box:

Encapsulation allows information from different networking layers to be carried together in the transmitted frame. Each layer adds information needed for its particular job.

Understand encapsulation

The packet details may show:

Ethernet II → IPv4 → ICMP → Data

This is called: **Encapsulation**

Simple explanation

Each layer adds information needed for communication.

Think of sending a parcel:

Network Encapsulation – Think of Sending a Parcel

When you send data, each layer adds its own header, just like a parcel being packed.

1 ICMP – The message

The actual content you want to send
(e.g. a ping message).

*Ping
message*

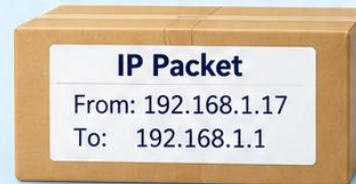
ICMP = the message
e.g. "Echo Request"



wrapped inside

2 IP – Where it must travel logically

Adds the source and destination IP addresses
(e.g. 192.168.1.17 → 192.168.1.1).



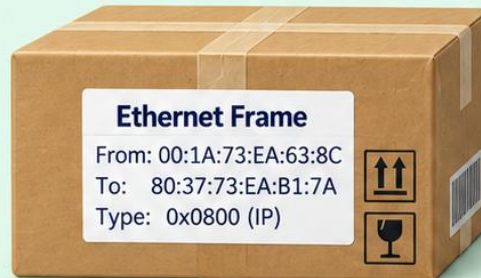
IP = the address
e.g. source IP,
destination IP,
routing information



wrapped inside

3 Ethernet – How it travels across the local network

Adds the source and destination MAC addresses
(e.g. 00:1A:73:EA:63:8C → 80:37:73:EA:B1:7A).



Ethernet = the delivery
e.g. source MAC,
destination MAC,
frame type

The Result: A Complete Frame

ICMP
(The message)

IP
(Where it must
travel logically)

Ethernet
(How it travels across
the local network)

Real-world example



Just like a parcel
needs an envelope,
an address, and
delivery information,
network data is
wrapped in ICMP,
then IP, then
Ethernet to reach
its destination.



Key Takeaway:

Each layer adds its own information, so the data can be delivered
from your PC to the correct destination.

Why did the destination IP address change, but the destination MAC address still belong to the default gateway?

Because the remote server is outside the local network. The PC sends the packet toward the remote server's IP address, but the Ethernet frame must first be delivered to the default gateway, so the destination MAC address is the gateway's MAC address.

Task 2 – What Students Should Record

Students should collect evidence such as:

- Screenshot of the ICMP filter
- Echo Request and Echo Reply
- PC MAC address
- Gateway MAC address
- Source IP address
- Destination IP address
- Ethernet frame type
- ICMP request/reply information

But remind students:

A screenshot alone is not enough.

They should explain what the screenshot shows.

For example:

The captured ICMP Echo Request was sent from my PC to the default gateway. The Ethernet II header shows my PC's MAC address as the source and the gateway's MAC address as the destination.

This is stronger evidence than simply inserting an unexplained screenshot.



End of Guided Work

We have now completed the two main guided activities together.

Well done!



You should now know how to:

Use Wireshark to capture and analyse network traffic.

1 Capture



Start a capture on the correct network interface.

Collect the traffic

2 Filter



Use display filters (e.g. arp, icmp) to show only the packets you need.

Show what is relevant

3 Select



Click on a packet in the list.

Choose the packet

4 Inspect



Look at the packet details and bytes (IPv4, ICMP, ARP, MAC addresses, etc.).

Find the information

5 Explain



Answer the questions using what you see in the capture.

Describe what happened



Next Steps

You will use exactly the same process to complete the remaining Lab 3 activities, including:

- Analysing traffic to a remote destination
- Answering the reflection questions



Key Message

The skills you have practised in Tasks 1 and 2 are the same skills you will use for the rest of the lab and in real network troubleshooting.

Capture → Filter → Select → Inspect → Explain

TASK 3

Continue Independently

Use the same process to analyse traffic to a remote destination.

You can do it!

1



Restart Wireshark capture

Start a new capture on the correct network interface.

2



Ping a remote destination

For example: ping www.cisco.com

3



Stop capture

Stop the capture after you see replies.

4



Find the Echo Request

In Wireshark, locate the first ICMP Echo (ping) request.

5



Compare MAC addresses

Check the source and destination MAC addresses.

6



Compare IP addresses

Check the source and destination IP addresses.

7



Explain what changed

Why is the destination IP different but the destination MAC the same?

8



Complete reflection

Answer the remaining Lab 3 reflection questions.

Scenario: Ping a Remote Destination



Ping to Default Gateway (Local)

Packet Details

- ✓ Ethernet II
 - Source: 00:1A:73:EA:63:8C (Your PC)
 - Destination: 80:37:73:EA:B1:7A (Router)
 - Type: 0x0800 (IPv4)
- ✓ Internet Protocol Version 4
 - Source: 192.168.1.17 (Your PC)
 - Destination: 192.168.1.1 (Default Gateway)
- ✓ Internet Control Message Protocol
 - Type: 8 (Echo request)

Ping to a Remote Destination (e.g. www.cisco.com)

Packet Details

- ✓ Ethernet II
 - Source: 00:1A:73:EA:63:8C (Your PC)
 - Destination: 80:37:73:EA:B1:7A (Router)
 - Type: 0x0800 (IPv4)
- ✓ Internet Protocol Version 4
 - Source: 192.168.1.17 (Your PC)
 - Destination: 72.163.4.161 (Remote IP)
- ✓ Internet Control Message Protocol
 - Type: 8 (Echo request)



Key Concept

When you ping a remote destination:

- The **destination IP address changes** (it is the remote server's IP).
- The **destination MAC address is still normally the local default gateway's MAC address** (the router) because the remote device is on a different network.



Your Task

Use the same capture → filter → select → inspect → explain process to complete the remaining Lab 3 activities and the reflection questions.

Why?

Because the PC cannot directly send an Ethernet frame across the Internet to the remote server.

It first sends the frame to its: **Default Gateway**

How This Lab Connects to Real Jobs

The same skills are used by IT professionals every day.

Network Engineer

Uses packet captures to find connectivity problems.

Cybersecurity Analyst

Looks for suspicious or unusual network traffic.

SOC Analyst

Investigates security alerts and network activity.

System Administrator

Checks why computers and servers cannot communicate.

Cloud / Infrastructure Engineer

Troubleshoots communication between systems.

IT Support Specialist

Uses tools such as ping and Wireshark to investigate user problems.

Workplace Skill

Instead of saying: “The network is not working.”

you can investigate: “Where exactly is the communication failing?”

Preparation and Guideline for Lab 3



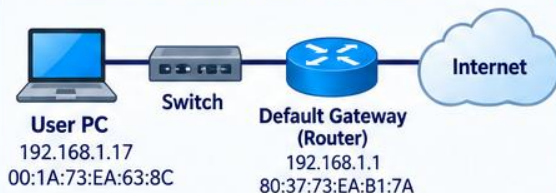
Real-World Example: "I Cannot Access the Internet"

A user reports: "My computer is connected, but the Internet is not working."

Use packet captures to find the problem, not guess!

1 Network Scenario

The user's computer is connected to the local network, but cannot access the Internet.



2 Troubleshooting Steps

we investigate step by step using Wireshark.



1. User PC

Check the PC network configuration (IP address, default gateway).



2. Can it find the gateway MAC?

Check ARP packets.



3. Can it reach the gateway?

Check ICMP ping packets.



4. Did the gateway reply?

Check if an ICMP Echo Reply is received.



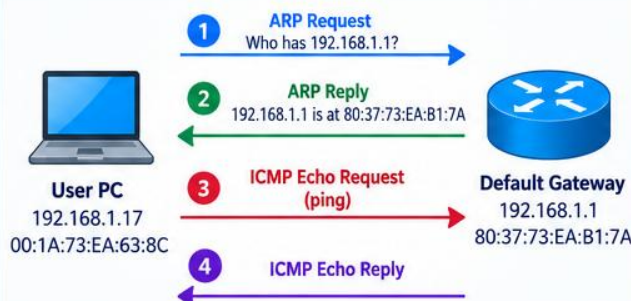
5. Check packet addresses

Look at the source/destination MAC and IP addresses.

3 What Wireshark May Show (Working)

When the PC and gateway are communicating normally, Wireshark may show:

No.	Time	Source	Destination	Protocol	Info
1	0.000000	00:1A:73:EA:63:8C	ff:ff:ff:ff:ff:ff	ARP	Who has 192.168.1.1?
2	0.000512	80:37:73:EA:B1:7A	00:1A:73:EA:63:8C	ARP	192.168.1.1 is at 80:37:73:EA:B1:7A
3	0.001024	192.168.1.17	192.168.1.1	ICMP	Echo (ping) request
4	0.001536	192.168.1.1	192.168.1.17	ICMP	Echo (ping) reply

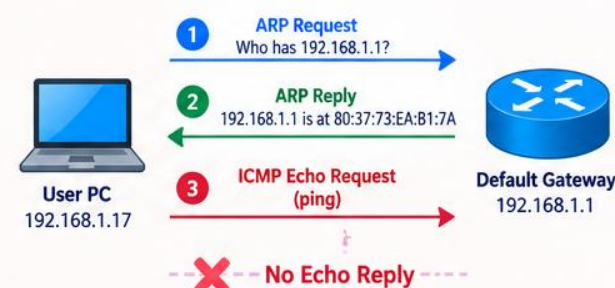


If all four packets appear, communication with the gateway is working.

4 What If There Is a Problem?

If we see an Echo Request but no Echo Reply, it looks like this:

No.	Time	Source	Destination	Protocol	Info
1	0.000000	00:1A:73:EA:63:8C	ff:ff:ff:ff:ff:ff	ARP	Who has 192.168.1.1?
2	0.000512	80:37:73:EA:B1:7A	00:1A:73:EA:63:8C	ARP	192.168.1.1 is at 80:37:73:EA:B1:7A
3	0.001024	192.168.1.17	192.168.1.1	ICMP	Echo (ping) request
-	-	-	-	-	No ICMP Echo Reply



If we see an Echo Request but no Echo Reply, we know that further troubleshooting is needed. The problem may be with the gateway, the network, or beyond.



Real-World Lesson

Packet captures help us locate the problem instead of guessing.



We can see what is happening.



We can identify where the problem is.



We can make informed decisions to fix it.

Capture
Analyse
Troubleshoot
Solve!

We completed Tasks 1 and 2 together. You have now learned how to examine Ethernet II frames, identify MAC and IP addresses, understand ARP requests and replies, capture network traffic in Wireshark, and analyse ICMP ping requests and replies. You have gained the core skills and knowledge required to complete the remaining remote-traffic analysis, evidence collection, and reflection activities independently.

- These guidelines are designed to **support your learning** and help you apply necessary techniques effectively. Please do the **remaining task** and follow the instructions based on **Canvas**. If you have any questions, feel free to ask—I'm happy to help!

You are encouraged to discuss concepts and work alongside classmates, but your final work, screenshots, explanations, and submission must reflect your own understanding and work.

Do not share completed submissions or copy another student's work. Each student must submit individually through Canvas, and submissions may be reviewed using Turnitin and other academic integrity processes.

Make sure you can run your program, explain your work, and demonstrate your understanding.



- 1. Live during class** – Ask directly during sessions.
- 2. Canvas Inbox** – For structured academic queries.
- 3. Email** – For formal communications or attachments.
- 4. Canvas Discussion Forum** – Everyone can see and learn from your question.

Happy Learning and Bright Futures Ahead!

Stay Connected:

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• *Education is the most powerful weapon you can use to change the world. — Nelson Mandela*

